IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Digital Object Identifier

Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]

[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

Corresponding author: Jorge Luis Mayorga Taborda (e-mail: jl.mayorga236@uniandes.edu.co).

ABSTRACT When a load is restored or a line is reclosed, the electrical response can persist and resemble a new intervention. We formulate source attribution from eight phasor measurement units (PMUs) in the IEEE 39-bus system using switched differential-algebraic predictors. The bank carries the estimated state through restoration and retains the preceding intervention as a candidate explanation of the remaining response. Nominal signatures describe separation under the assumed plant and noise model. With thresholds calibrated separately to a target of one normal alarm per hour, the bank correctly attributed the family and source of 99/108 interventions before physical restoration, compared with 72/108 for a continuous generalized likelihood ratio observer and 22/108 for ExtraTrees. The bank gained 25.0 percentage points over the continuous observer (95% interval [13.89, 37.04] across 18 physical parents) and emitted 8 unmatched alarms, compared with 85 and 0, respectively. No method emitted a normal alarm in 3.92 monitored hours. At fixed thresholds in a separate control, resetting the estimated physical state reduced timely attribution from 108/108 to 75/108. Independent per-PMU noise at 0.071% added RMS total vector error degraded load attribution. The bank–comparator contrast does not isolate history retention. The results describe sequential, nonoverlapping interventions in a known catalog on one simulator family.

INDEX TERMS change detection and isolation, differential-algebraic equations, event attribution, measurement integrity, phasor measurement units, physics-informed inference, power system dynamics.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, physical state continues to evolve. A second intervention may begin before that response decays. With eight PMUs, the observed change can combine recovery from the earlier event with forcing from a new source (Fig. 1(a)).

The inference problem is to decide whether a new source is needed to explain the incoming reports. Because the preceding response is still evolving, the observation law depends on the earlier intervention. Classical change detection and isolation treats a single change of an otherwise stationary law [1]–[3], and the generalized-likelihood approach of Willsky and Jones detects jumps in a fixed linear system [4]. Power-system outage detectors compare network hypotheses under quasi-static or linearized models [5]–[7], and trajectory libraries diagnose faults from sparse PMUs [8]. These precedents motivate retaining a restored intervention as an explicit competitor when attributing later observations in a switched DAE network.

We develop a bank of switched-DAE predictors that carries the estimated state through restoration. Each history specifies a source, onset and restoration time; load severity is integrated analytically. Separate active and cause weights allow an intervention to end while its response remains an explanation of later reports (Section IV).

We then examine two consequences of this construction. Nominal signature energy describes source separation under the assumed plant and noise model, and a reset analysis quantifies the innovation bias caused by discarding physical state (Section V). A held-out comparison measures timely attribution and extra alarms against a continuous physical GLR and ExtraTrees after calibrating each procedure to the same normal-alarm target (one normal alarm/h; Section VII-A). This comparison evaluates complete procedures. Isolating the benefit of explicit recovery histories requires controls that distinguish state continuity from the uncertainty lost when histories are compressed.

The evaluation covers a known catalog of sequential load

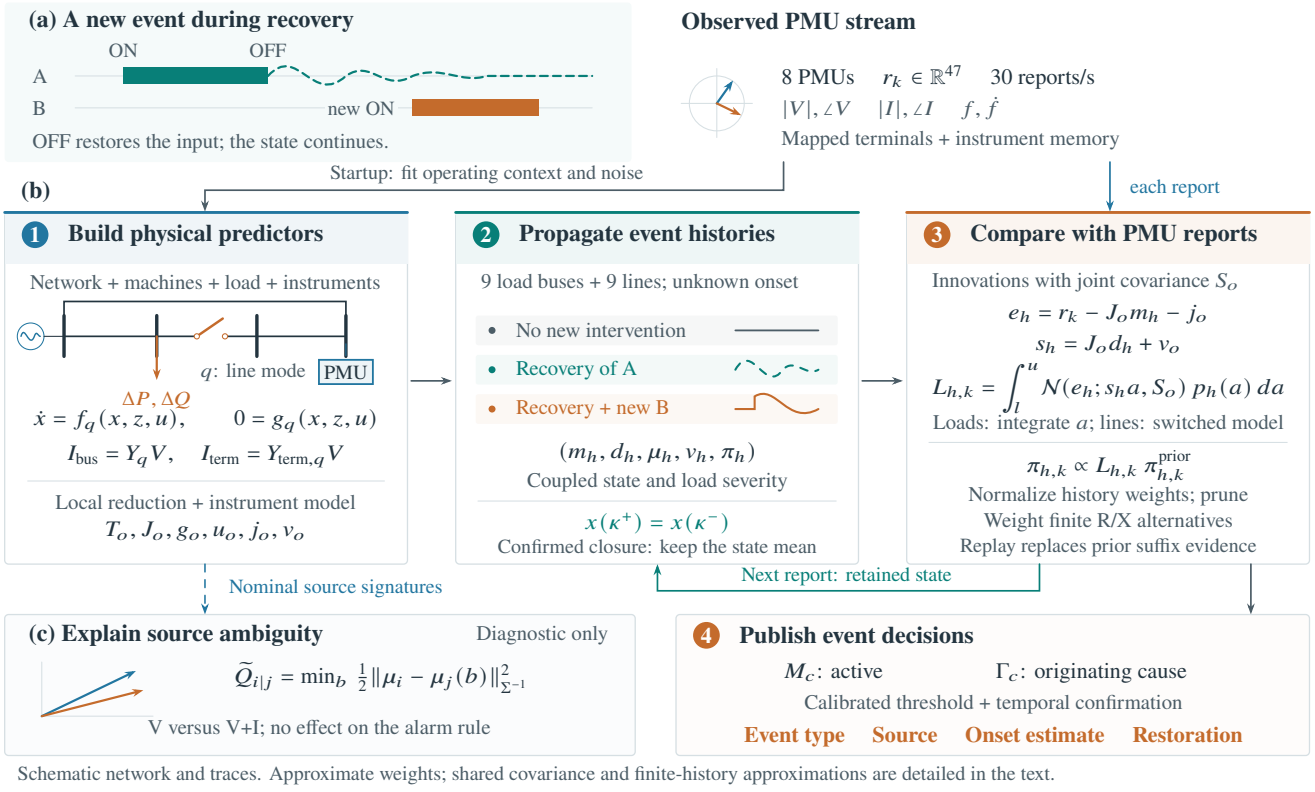


FIGURE 1. Recovery-aware event attribution from sparse PMUs. (a) A new intervention can begin while an earlier response persists. (b) Network-derived predictors compare recovery and new-event histories using the same report; integrated severity evidence updates their weights before detection, source diagnosis and restoration are published. The state continues through physical restoration; confirmed episode closure compresses histories to the retained state mean. (c) A separate nominal diagnostic measures source ambiguity; it does not gate publication. The electrical motif and traces are schematic.

and line interventions. The supplement records auxiliary analyses and reproducibility details.

II. RELATED WORK

A. CHANGE DETECTION AND ISOLATION IN DYNAMIC SYSTEMS

Sequential detection with a prior on the change time leads to the Shiryaev statistic, whose posterior-probability threshold is optimal for a geometric change-time prior [9]. Detection *and isolation* among several post-change laws was formalized by Nikiforov [1] and Lai [2]; asymptotically, the delay for a given error level is inversely proportional to the Kullback–Leibler information separating the true post-change law from its competitors [3]. For linear state-space systems, Willsky and Jones introduced signatures of jumps propagated through a Kalman filter and estimated their magnitude by generalized likelihood ratios [4]. Separating a constant bias from the state, as Friedland did for recursive filtering [10], gives the affine state–severity representation used in our load histories. Interacting and multiple-hypothesis filters propagate mode-conditioned states and mode-sequence histories with pruning and merging [11], [12]. Our method inherits these mechanisms. Its post-change models, however, are switched DAE operators of a power network, and a restored intervention remains a competing cause of later observations.

B. PHYSICAL INFERENCE OF GRID EVENTS

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [13]–[15]. Functional observability and descriptor-model diagnosability distinguish recovering a state quantity from identifying the input that produced it [16]–[19]. Phasor-angle hypotheses support outage identification [5], Bayesian outage inference with uncertain states [20], and quickest line-outage detection [6], [7]. Yang *et al.* add a time-varying AC likelihood chart [21] and particle-filter generator states [22] for persistent line outages, without restoration or loads. Anguluri *et al.* recover sparse forcing locations and inputs of linear dynamics by group LASSO [23], and sensor placement determines which sources are distinguishable [24]–[26]. Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window [8] and derive error/delay guarantees for an adaptive matrix CuSum [27]. Trajectory sensitivities and Gaussian evidence have been combined for parameter inference in power-system DAEs [28], [29]. PMU-based nodal-inertia estimation extends this neighboring line of work to spatially resolved dynamic parameters under ambient and transient conditions [30]. State and parameter estimates do not by themselves identify the intervention that produced them. Here the DAE-derived signatures provide a profiled separation diagnostic at a supplied onset, which we compare

retrospectively with attribution outcomes; it is not a general detectability guarantee. Le *et al.* recently proposed a physics-informed graph neural operator that maps sparse, intermittent PMU windows to unmeasured rotor angle and speed trajectories, with swing equations in the decoder and a learned correction for model mismatch [31]. That preprint addresses dynamic state reconstruction. Our method identifies which intervention explains observations while an earlier response is still evolving.

C. LEARNED DIAGNOSIS FROM SPARSE MEASUREMENTS

Learned and knowledge-driven representations identify PMU disturbances and assets [32]–[34]. For sparse-PMU transient-stability assessment, Wang *et al.* combine dominant-bus selection with a bidirectional recurrent assessor and an online check of model applicability under system changes [35]. This is a useful precedent for monitoring whether a learned assessment remains valid, but its output is a stability assessment rather than a source attribution for successive physical events. Barati's gated-recurrent-unit model predicts disruptions from historical synchrophasor streams, a forecasting task distinct from assigning an observed intervention to a physical source during recovery [36]. SHRED reconstructs IEEE-39 dynamic states from a few PMUs [37], and Dey *et al.* localize distribution topology changes with hardware-in-the-loop validation [38]. Graph neural networks encode the grid topology in the learned map, for event identification [39], physics-informed state estimation [40] and localization after reconstructing unmeasured trajectories [41]. These applications span state reconstruction and event diagnosis. In the present attribution task, the physical bank derives candidate responses from the network equations. Its signature analysis describes separation under the assumed plant and noise model. ExtraTrees provides a learned comparison on the same 18-source catalog.

D. MEASUREMENT INTEGRITY

Bad-data detection in state estimation tests residuals against the redundancy of a network model [42]; analytical-redundancy parity relations generalize this to dynamic systems [43]. For synchrophasors, robust estimators separate outliers from excursions [44], utility data show missing, locked, quantized and time-shifted reports [45], and attack identification characterizes which corrupted sensor sets remain distinguishable from system behavior [46]. Recent P-class synchrophasor-estimation work optimizes front-end accuracy and latency under IEC/IEEE measurement requirements [47], [48]; a review of PLL-based synchronization also relates front-end phase tracking to PMU applications in distributed generation [49]. End-to-end PMU tests under fault transients quantify how estimator errors affect a downstream application [50]. Multisnapshot state estimation models time-correlated instrument-transformer errors together with line-parameter uncertainty [51]; IEC 61850-9-2 co-simulation has also been used to assess OpenPMU and virtual-PMU state estimation [52]. These contributions address measurement generation, measurement uncertainty or state estimation. Our bank starts from reported phasor channels and does not validate a PMU estimator or a hardware-in-the-loop interface. The two parity relations used in the supplement require no state estimator and are connected to the information available for physical attribution when a PMU is lost.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the reference record released with the SGSMA 2026 competition. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The causal attribution test addresses load changes and line openings with restoration; missing and corrupted data are diagnosed by the integrity relations of the supplementary integrity analysis. Faults and generation changes provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the definition of absolute frequency. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The causal stream is exported from one balanced complex phasor per PMU: phase A supplies the voltage and current magnitudes and angles. The CSV exporter constructs B and C by equal magnitudes and angle offsets of -120° and $+120^\circ$. Retaining A therefore loses no independent phase information in these synthetic records; it is not measured positive-sequence extraction or a model of unbalanced faults. The vector is $y_k = (|V_A|_{\mathcal{P}}, \angle V_{A\mathcal{P}}, |I_A|_{\mathcal{P}}, \angle I_{A\mathcal{P}\setminus\{39\}}, f_{\mathcal{P}}, \dot{f}_{\mathcal{P}})_k$. The supplementary measurement table gives all eight current-terminal assignments, polarities, gains and phase offsets. Products of voltage and current measured at different terminals are power proxies.

TABLE 1. Closest inference formulations. Only the quickest-change detector of Chen *et al.* and the two ExtraTrees adapters were executed on the common evaluation data.

Route	Mechanism	Evaluated target	Role in this study
Willsky–Jones GLR; Nikiforov/Lai	Filter signatures; detection–isolation statistics	One jump or one change in a fixed system	Information analysis (Section V)
IMM / M3H	Mode-conditioned filters, history pruning and merging	Hybrid switching-state estimation	History-bank mechanism
Chen / Rovatsos	Network likelihoods, sequential tests	Line outages	Chen executed on persistent openings
Yang	Time-varying AC likelihood chart; particle-filter states with EWMA	Persistent line outages, ten PMUs	Closest dynamic precedent, not executed
Warner high-fidelity	Simulated V/I paths, onset search	Short-circuit line candidates	Closest trajectory precedent
DAE / stochastic-state estimation	Physical state and parameter models	States, inputs or parameters	Components
Wang et al. sparse-PMU TSA	Bus selection; recurrent assessor; validity check	Stability assessment under grid changes	Adjacent; no source attribution
Le et al. GNO (arXiv 2026)	Graph attention; swing-equation decoder	Machine-state trajectories from sparse windows	Adjacent state estimation; not event attribution
Learned PMU diagnosis	Features or learned representations	Event families and locations	Two ExtraTrees adapters executed
This work	Switched-DAE histories with recovery	Successive catalog load/line events	Attribution, alarm cost, nominal separation

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$\begin{aligned} h_k &= H_q(x_k, z_k, u_k), \\ \eta_k &= F_M(\eta_{k-1}, h_k; \theta_M), \\ y_k &= G_M(\eta_k, h_k; \theta_M) + b + \varepsilon_k. \end{aligned} \tag{3}$$

Figure 4 maps these physical and measurement operations to the report consumed by inference. Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. H_q selects electrical phasors; η_k retains the causal phase history used for frequency/ROCOF and the measurement-filter state. The observation thus depends on an augmented state. Bus frequency comes from phase evolution, not a machine's rotor speed. Instrumental noise is generated by the noise model described in Section VI.

An intervention history is the discrete tuple $h = (e, c, \tau, \kappa, \mathcal{I})$: family, source, onset, restoration time ($\kappa = \infty$ for a sustained event) and, for a load, a severity interval $\mathcal{I}$. The severity $a \in \mathcal{I}$ is a continuous latent variable conditioned on h and integrated in the evidence; a line opening has no severity. A load changes its local P/Q setpoints by the common factor $1 + a$; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Parity diagnosis (supplement)
Unknown	Unresolved diagnosis	Not a new physical family

Switch-aligned reports use the consistent post-switch algebraic solution. The local reduction requires an invertible algebraic Jacobian after fixing the angle reference; the catalog excludes islanding line alternatives. Successful solves and local residual checks validate the admitted cases, not global regularity for arbitrary switching topologies.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after a 12-s warmup. This startup estimate assumes an intervention-free prefix.

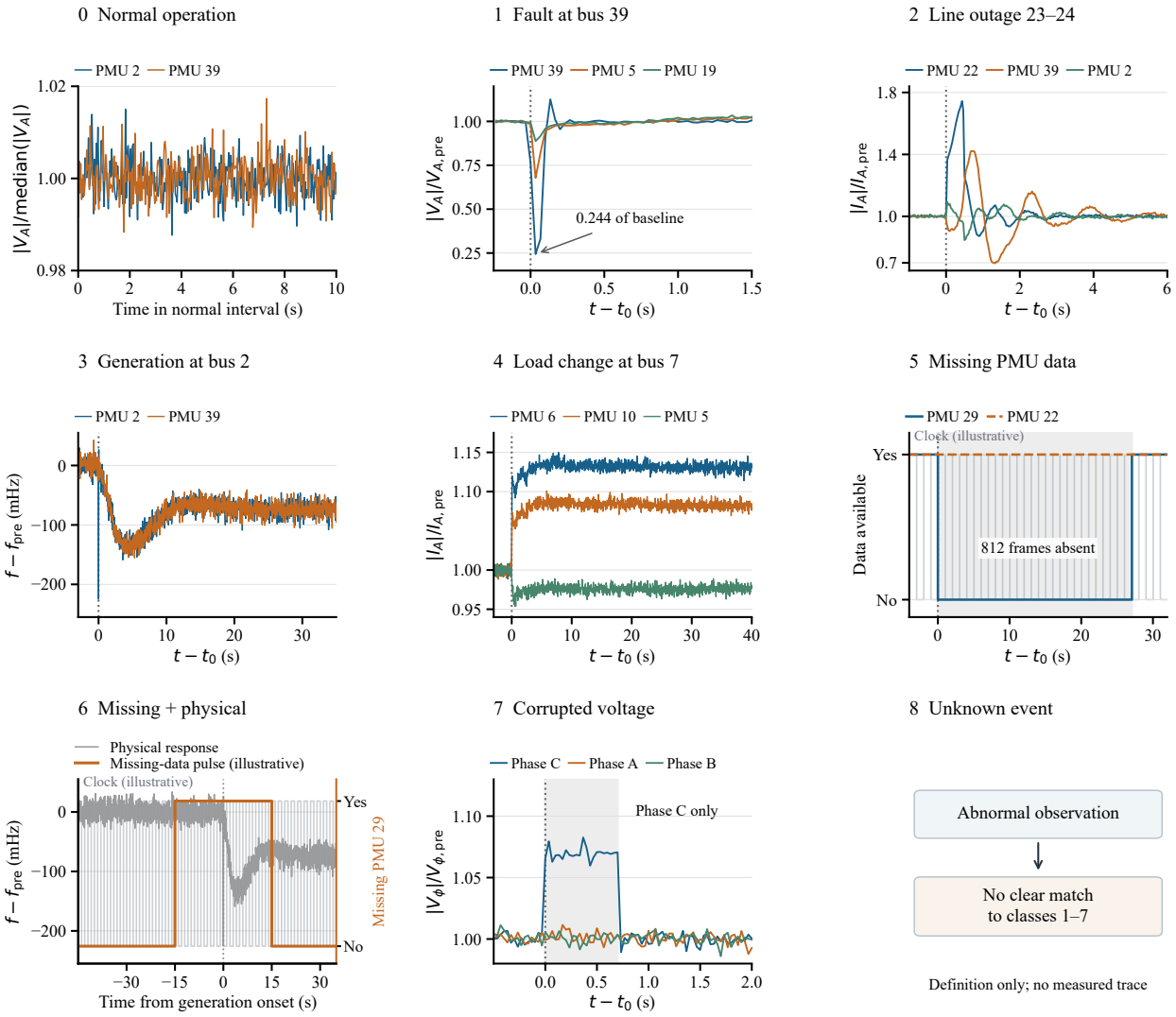


FIGURE 2. **Representative signatures from the archived RAW0001 reference simulation of the SGSMA 2026 competition, arranged by label 0–8. Class 0 shows a normal interval; classes 1–5 and 7 retain the original trace values. In class 6, the gray PMU 2 frequency response accompanies a generation change, with an illustrative orange PMU 29 missing-data pulse from −15 to +15 s (Yes: missing; No: present); this window is not the recorded outage interval. The gray clock trains in panels 5 and 6 are illustrative, not measured report timing. Class 8 is a definition schematic, not an observed trace or a validated bank output. Event traces are aligned to source labels for illustration only. Only load and line mechanisms enter the main causal evaluation.**

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

At each report, the bank compares predictions from candidate intervention histories (Fig. 1(b)). Reduced state-space models carry the physical and instrumental state forward; their dynamics and measurement maps come from the network DAE. This avoids solving a full nonlinear simulation for every candidate. Conditional filters account for uncertain state and persistent model discrepancy, and the resulting evidence updates the weights of the competing histories.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

Each network hypothesis is linearized at the operating point $(\bar{x}_\ell, \bar{u}_\ell)$ determined by its prefix-fitted load and dispatch factors. For topology q, the constructor holds $\bar{x}_\ell$ fixed and solves $g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0$. On a regular index-one chart with

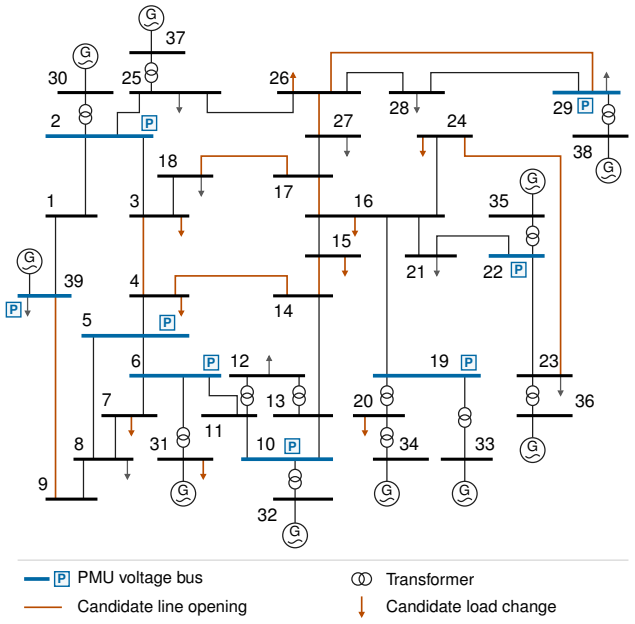


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.**

a fixed angle reference, the derivatives at that consistent point give, suppressing ℓ,

$$\begin{aligned}\delta z &= -g_z^{-1}(g_x\delta x + g_u\delta u),\\ \delta\dot{x} &= c_q + A_q^{red}\delta x + B_q^{red}\delta u,\\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

Here $c_q = f_q(\bar{x}, \bar{z}_q, \bar{u})$ can be nonzero after a line opening even though the differential state is continuous. For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

The electrical offset is $d_q^{el} = H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u})$. Thus both the changed algebraic solution and the changed terminal map enter the prediction. These operators propagate an unmeasured-source perturbation to the selected voltage and oriented-current channels. Branch stamps in g_q and H_q encode network connectivity; eliminating unmeasured algebraic variables generally produces dense reduced operators. A line switch changes both the voltage response and the terminal operator in H_q. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

Exponential discretization integrates c_q over each mode interval; composition handles a switch within a report. After instrumental augmentation, the resulting offsets enter $b_{\ell,q}$ and $d_{\ell,q}$ in

$$\begin{aligned}\xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k,\\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining observation noise after instrument whitening. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture. The supplementary section "Implemented Predictor and Conditional Uncertainty" gives the affine construction and its code correspondence.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

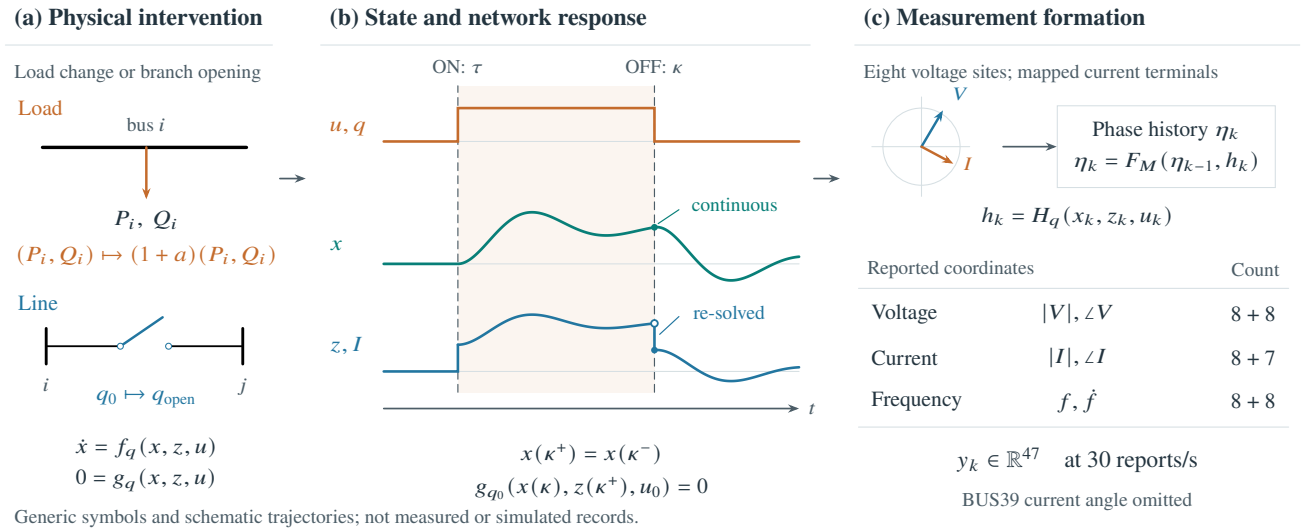


FIGURE 4. Illustrative chain from load/line intervention to causal PMU report. At OFF, the operator returns to baseline while x and phase-history η persist; algebraic voltages and currents are re-solved. Eight PMUs yield 47 coordinates at 30 reports/s; BUS39 current angle is omitted. Symbols and traces are schematic. Phase B/C are reconstructed from A; raw waveforms and harmonics are not inference inputs.

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} &= \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ &\quad + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

As in separate-bias filtering [10], the conditional state mean is represented as $m_h + d_h a$; d_h is the Willsky–Jones signature of a unit severity propagated through the filter [4]. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With the shared covariance P_ρ for the current discrepancy regime ρ,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \tag{13}$$

The amplitude distribution is updated by the same integral. Its truncated variance $V_{a,h}$ differs from the Gaussian parameter v_h. The affine conditional representation retains state–severity dependence:

$$\begin{aligned} \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ \mathrm{Var}(\xi \mid h) &= P_\rho + d_h V_{a,h} d_h^\top. \end{aligned} \tag{14}$$

Conditioning on past reports is implicit. The shared P_ρ solves the nominal-operator Riccati equation of regime ρ; S_o and K_o are then formed for each switched operator o. This is a steady-state approximation, not an exact covariance recursion along each switched history. At closure, the filter retains the augmented-state mixture mean and records the trace of the discarded amplitude and between-history covariance; it does not propagate that covariance. The supplement separates these approximations from the exact local amplitude integral and states the covariance needed for a full mixture.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight is updated with the one-step predictive density

$$w_{\ell,k} = \frac{w_{\ell,k-1}\, p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1}\, p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}, \tag{15}$$

$$p_\ell(y_k \mid \mathcal{F}_{k-1}) = \sum_{h'} \pi_{\ell,h',k-1} \sum_{h \in \mathcal{T}(h')} p_\ell(h \mid h')\, L_{\ell,h,k}. \tag{16}$$

The transition set $\mathcal{T}(h')$ contains continuation, restoration at either half-report phase for an eligible active history, and a birth of every catalog cell and severity interval at either phase from the no-intervention history; the transition probabilities sum to one, which the implementation asserts at every report. The density is evaluated before any state is corrected and before retrospective proposals are selected. Onsets proposed

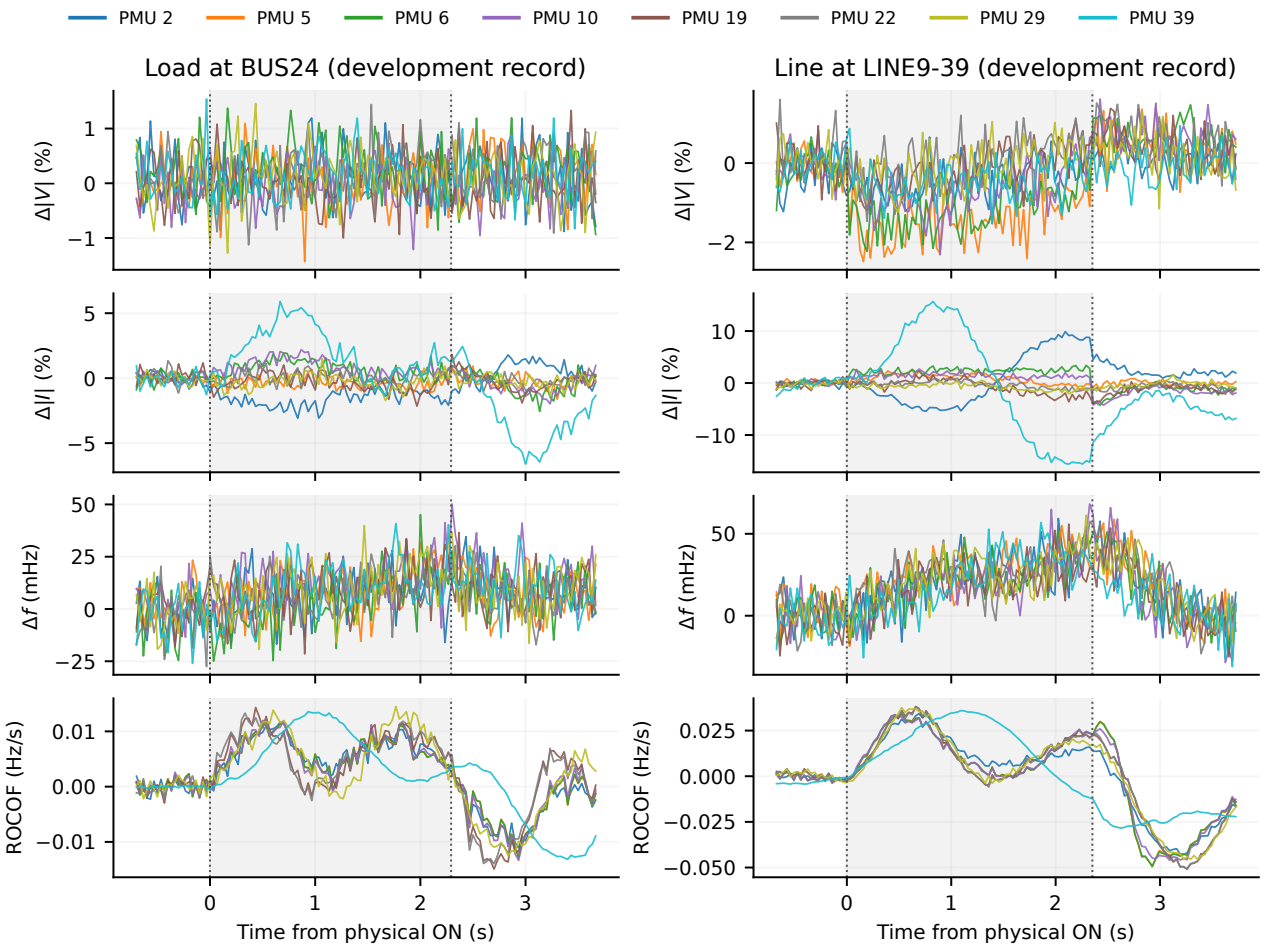


FIGURE 5. **Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.**

later for earlier reports change the inner history posterior through suffix replay but do not re-weight w_ℓ; this is the one point where the outer and inner recursions differ from a single joint posterior. Startup-fitting reports are not scored again. The mixture covers a finite set of network alternatives rather than a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t,\kappa)\delta x_h(\kappa), \\ \delta y_h^{net}(t) &\simeq C_0(t)\Phi_0(t,\kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{17}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a predicted trajectory.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{18}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

The implementation uses M for active-event publication and Γ for source diagnosis. Their update weights are approximate because of pruning and shared steady-state covariance approximations; they are not calibrated posterior probabilities. A descriptive algebraic decomposition of ΔM into within-model evolution and network-model reweighting

is given in the supplement; it does not enter the estimator's decision rule.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

An alarm is published when the active mass $\sum_c M_{c,k}$ exceeds 0.99 for six consecutive reports. The rule has the structure of a Shiryaev change-time threshold under the geometric prior [9], but pruning and covariance approximations make 0.99 an operational threshold on approximate mass, not a calibrated posterior-probability guarantee. Six-report confirmation suppresses brief excursions. OFF is published when the restoration mass exceeds 0.99 for 15 reports. The diagnosis reports the family with the largest cause mass and its highest-mass source; episode-replacement rules handle a different source that begins before the previous episode closes. Every message records its emission time and the last report used, so a diagnosis can be correct but too late to precede physical OFF.

These are the development publication settings. The common-budget comparison in Section VII-A selects thresholds separately on calibration records. The nominal comparator is the same procedure with only the nominal network hypothesis.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1 - \exp[-(1/30)/30]$, fixed during method development before the independent schedules. They are deliberately conservative: test interventions last 2–5 s and are separated by 3–8 s, much shorter than the prior means of 60 and 30 s. A low birth hazard raises the evidence needed for an alarm by $-\log \pi_c$; Section V compares this margin with nominal signature energy. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. Each interval is a separate history with its own weight and truncated amplitude distribution; its birth weight is the prior probability of that interval. The physical operators are shared across these four histories. For a fixed path with all four intervals retained, their weighted evidence equals the single-interval integral. This partition is a bank representation, not numerical quadrature or a piecewise physical approximation. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The supplement maps each operation to its implementation and lists the complete publication configuration.

G. RELATION TO MULTIPLE-HYPOTHESIS FILTERING

For a retained history h, let $p_\ell(h \mid h')$ contain the admitted birth, continuation and restoration transitions. Before pruning, its normalized evidence update has the standard form

$$\pi_{\ell,h,k} \propto L_{\ell,h,k} \sum_{h'} p_\ell(h \mid h') \pi_{\ell,h',k-1}. \tag{19}$$

This is the mode-sequence update of IMM and M3H filtering [11], [12]. Three features are specific to the present problem: the modes are oriented electrical intervention operators derived from the network DAE, histories carry a restoration time so that a restored intervention remains a competing cause, and the output separates active intervention from originating cause through (18). Retrospective births reuse a stored pre-onset state, and confirmed closure compresses histories into the retained augmented-state mean.

Algorithm 1 Causal physical-history attribution

Require: Fixed plant/catalog, configuration, and PMU reports as they arrive
Ensure: Time-stamped detection, source diagnosis, and restoration messages

Fit operating context, reference, and noise from the permitted prefix
Initialize model/history states, weights, and pre-onset state store
for each newly received report y_k after startup **do**
    Map y_k to inference coordinates r_k
    Predict each retained history; carry physical state through OFF
    Save outer-model predictive densities before screening
    Propose eligible births from stored pre-onset states
    for each accepted proposal **do**
        Replay its suffix from the saved state, replacing old suffix scores
    end for
    Score histories before state correction; integrate load severity
    Apply line operators; correct states and update history weights
    Prune the beam; record discarded probability mass
    Update outer-model weights from the saved predictive densities
    Aggregate active-event mass M_k and cause mass Γ_k
    Apply consecutive-report alarm, diagnosis, and OFF rules
    if OFF is confirmed **then**
        Compress closed histories into the retained augmented-state mean
    end if
    Store current states and emit messages with the last report used
end for

H. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

No event truth is available to the algorithm. Retrospective replay changes the inner history weights but does not re-weight the outer model densities. The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. Decisions use only reports already received. Computation time is reported separately in Section VII.

V. NOMINAL SIGNATURES AND STATE CONTINUITY

The local operators give a nominal signature for each candidate. We use these signatures to explain separation and the effect of resetting the physical state under the assumed model. The numerical comparisons with completed runs are retrospective.

A. SIGNATURES AND INFORMATION

Let intervention i start at report 0 with severity a and let the plant follow the local switched model (8) without noise. Run the conditional filter of candidate history j, started at the same onset, on these reports. Its innovation $e_j(k)$ and severity sensitivity $s_j(k)$ follow the recursion (13). The filter of the true history at the true severity has zero deterministic innovation in this construction. We measure nominal separation by a profiled signature energy,

$$\Lambda_{ij}(k) = \tfrac{1}{2} \min_{b\in[l,u]} \sum_{r=0}^{k} \left\| e_j(r) - s_j(r)\, b \right\|^2_{S_j^{-1}}, \tag{20}$$

where a line candidate has no severity to minimize over. The quantity Λ_{i0} compares against the no-intervention path; $\min_{j\neq i} \Lambda_{ij}$ measures separation from the nearest alternative at the supplied onset. These are deterministic diagnostics. For fixed Gaussian paths with means m_i, m_j and joint report covariances Σ_i, Σ_j, the expected log likelihood ratio under i is

$$\begin{aligned} D_{ij} = \tfrac{1}{2}\big[&(m_i - m_j)^\top \Sigma_j^{-1}(m_i - m_j) \\ &+ \operatorname{tr}(\Sigma_j^{-1}\Sigma_i) - d + \log(|\Sigma_j|/|\Sigma_i|)\big], \end{aligned} \tag{21}$$

where d is the stacked report dimension. The covariance terms cancel when the covariances coincide. Equation (20) profiles severity on deterministic signatures using the candidate innovation covariance; it is not, in general, the full Gaussian KL divergence above. Optimizing a deterministic signature, profiling severity from noisy observations and integrating it under a prior are different operations.

For an idealized comparison of one onset and source against the null,

$$\lambda^\star = \log 99 - \log \pi_c. \tag{22}$$

This is the score required for odds of 99 to 1 if π_c is used as the approximate prior odds. For a load with $\Lambda_{i0} = a^2 G_i/2$ and covariance scaled by α^2, the corresponding reference amplitude at horizon T is

$$a_{\mathrm{ref}}(\alpha, T) = \alpha \sqrt{2\lambda^\star / G_i(T)}. \tag{23}$$

This does not specify a detection probability or a threshold guarantee for the bank: unknown onset, competing paths, marginal likelihoods and consecutive-report confirmation all affect publication.

B. THE COST OF FORGETTING THE PHYSICAL STATE

Consider a noise-free, matched-model continuation with no new intervention after an inferred closure at report k_c. Suppose the conditional mean at closure equals the augmented state. Carrying this state gives zero deterministic innovation. If the physical coordinates are set to zero while the instrument and discrepancy states are kept, the no-intervention filter receives the deterministic bias

$$\bar{e}_0(k) = C_0 \Phi_0(k, k_c)\, \delta x(k_c) - \sum_{r=k_c}^{k-1} C_0 \Phi_0(k, r{+}1) K_0\, \bar{e}_0(r), \tag{24}$$

where $\delta x(k_c)$ is the zeroed physical part of the augmented state, $\Phi_0(k,r) = T_0^{k-r}$ and $C_0 = J_0$ are the no-intervention operators of (13), and K_0 is its gain. This is the free recovery response (17) seen through a filter that tries to absorb it. Any birth hypothesis j that explains part of this bias has a profiled energy gain $\Gamma_j^{sp}(k) = \Lambda_0^{reset}(k) - \Lambda_j^{reset}(k)$, where both energies use the same reports from k_c to k. A gain above $\lambda^\star$ crosses the idealized score reference; it does not guarantee a published alarm. Under the same construction with carry, the null energy is zero, so every alternative has gain $\Gamma_j^{sp,carry}(k) \leq 0$. The positive part of the best gain is therefore zero. Reset bias depends on the excited modes, their decay, and their projection onto the observed channels.

C. RETROSPECTIVE DIAGNOSTICS

We evaluated the signature and reset diagnostics in the nominal quiet covariance regime. The true-path deterministic residual was below 10^{-8} nats. Fig. 6 summarizes separation under those conditions.

For a 10% load step, the one-second nominal separation from the null ranges from 4603 to 6.9×10^6 nats; for lines it is at least 4.9×10^6 nats. These large margins are consistent with the short delays under matched conditions. The reference amplitude under uniform noise scaling is highest at bus 31. Neither diagnostic accounts for independent per-PMU noise or a shifted plant. The reset diagnostic (Section VII-E1) evaluates the deterministic effect of discarding a previously estimated state.

A separate channel diagnostic holds source amplitudes and a common temporal covariance fixed. At 1 s, adding terminal currents to voltage increases the minimum pairwise separation by a median factor of 1.35 across the 18 sources (range 1.03–8.41). Allowing rival loads to fit severity over the admitted interval gives a median factor of 1.53. The supplement derives both diagnostics and shows how fitting severity can reduce source separation. Neither result measures attribution accuracy.

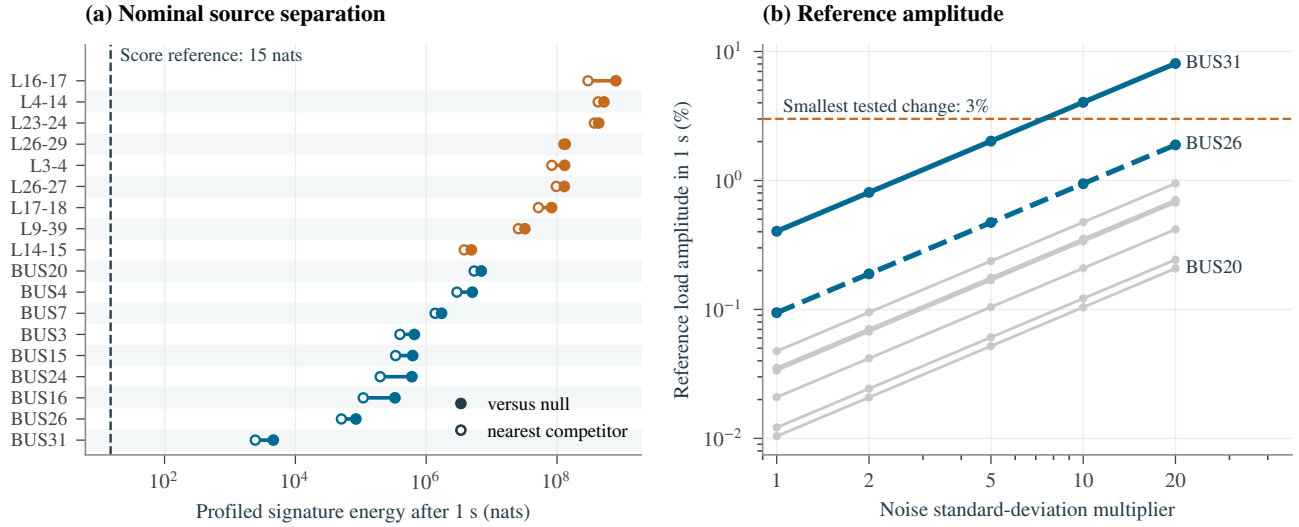


FIGURE 6. **Retrospective signature diagnostics using the nominal switched-DAE operators. (a) Profiled signature energy after 1 s against the no-intervention history (filled circles) and against the nearest competing source (open circles), for a 10% load step or a line opening; the dashed line marks the idealized score reference $\lambda^\star$. (b) Reference load amplitude at 1 s, (23), versus a uniform multiplier of the calibrated noise; gray lines are the other seven load buses.**

VI. EXPERIMENTAL DESIGN

The primary comparison measures timely source attribution and extra alarms for the history bank, a continuous GLR and ExtraTrees after calibrating each to the same normal-alarm budget. A separate evaluation varies network parameters and noise, while component controls examine state reset and severity scoring. Table 4 distinguishes these populations; repeated conditions of a parent sequence are paired observations.

The plant is an IEEE-39 model in Julia/PowerDynamics with Sauer–Pai machines, IEEE Type I exciters, TGOV1 governors and constant-impedance ZIP loads. Each physical parent is integrated continuously, preserving differential states through ON/OFF. The PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001. At a switch-aligned report the export uses the post-switch algebraic value. The bus-31 candidate is the existing 9.2-MW/4.6-Mvar load at the generator bus that serves as slack only in the initial power flow; the dynamic model has no slack bus, so its load change is absorbed by machine dynamics like any other.

Instrument noise follows a stochastic model fitted to normal segments of the competition reference record: per-family vector autoregressions with a strong common factor across PMUs. Voltage-magnitude noise is about 0.4% of nominal, but the noise of angle *differences* between PMUs is only 10^{-4} to 10^{-2} degrees, because most angle noise is common to all units. This small differential noise is what makes megawatt-scale changes observable at remote PMUs (Section V); it is a property of the reference data, smaller than the idiosyncratic error expected from field PMUs [48]. Section VII-F therefore tests larger and independent per-PMU noise.

The two development campaigns each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

TABLE 3. **Candidate assets fixed during method development, before the independent schedules. This is a known catalog rather than a representative sample of all network assets.**

Family	Candidate physical sources
Load buses	3, 4, 7, 15, 16, 20, 24, 26, 31
Lines	3–4, 4–14, 9–39, 14–15, 16–17, 17–18, 23–24, 26–27, 26–29

The five-model bank and its nominal version were fixed before the independent schedules were drawn. Both ExtraTrees models were trained specifically for the causal load/line catalog, using 32 earlier physical parents for training and eight for development validation. Both training sets include normal operation and all 18 candidate sources. The reference uses 846 causal features from 1-, 5- and 10-s windows; the short-window variant adds 611 features from 3-, 8- and 15-report windows. Each model has 400 trees with the same predeclared hyperparameters. They inherit feature families from the SGSMA project, but their fitted weights are distinct from the original hackathon submission. These models and their adapters were fixed before the present independent schedules; no retraining uses the current evaluation records. All methods receive the same 30-fps observations. The supplementary reproduction table records the training lineage and model hashes.

The learned adapter publishes a change after six consec-

TABLE 4. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Common-budget comparison	18 event parents, 108 interventions; 24 normal parents	Timely family/source, extra alarms and paired differences	Complete procedures; thresholds selected on separate calibration data
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Supplementary signature test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Supplementary recovery geometry	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; separate from stream accuracy

utive identical argmax labels; its tree votes are uncalibrated scores. In its own development validation the reference model already attributed no load before OFF and the short-window variant four, while lines were attributed well; both were nevertheless retained unchanged under a predeclared rule. They are therefore reported as fixed adapters at their original operating points, not as the best attainable learned baseline. Learners retrained on the bank's simulations locate at most 49 of 72 held-out events, against 72 for the bank (supplement).

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [6]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared research-plant adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all simulated instrumental effects. We preserve both matched-law implementation checks and model-mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

The independent campaign specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

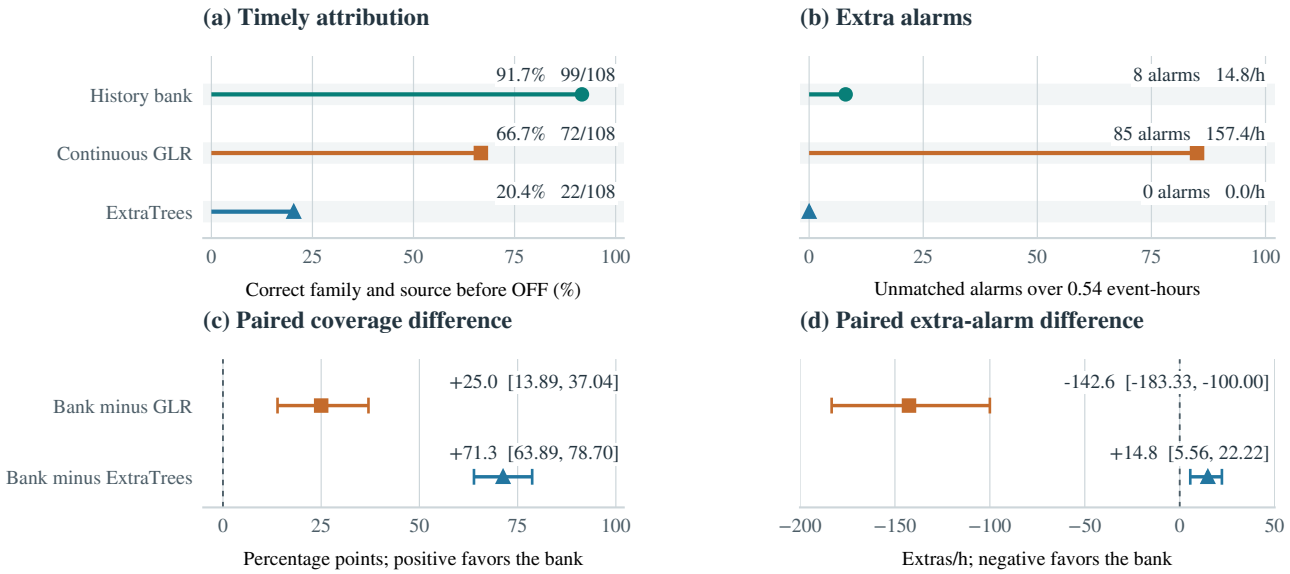
Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time.

VII. RESULTS AND CONTROLLED COMPARISONS

A. COMPARISON AT A COMMON CALIBRATION BUDGET

On a separate cohort, the nominal bank and the continuous generalized likelihood ratio (GLR) observer used the same nominal operators, eight PMUs, source catalog and permitted startup prefix. The GLR carries one committed state through restoration. ExtraTrees was refitted on disjoint simulated parents covering the same catalog. A common publication clock and calibration procedure selected a threshold for each arm under a normal-alarm budget of one per hour. The fixed methods were then scored once on 18 event parents and 24 normal parents. Interventions did not overlap, and the next onset was 3–8 s after physical restoration.

Figure 7 shows a 25.0 percentage point gain over GLR (paired 95% CI [13.89, 37.04]), with fewer extra alarms. The gain over ExtraTrees is 71.3 points (95% CI [63.89, 78.70]), with more extra alarms. The intervals resample entire physical parents within the network/noise strata; events within a parent do not count as independent samples. All three methods



Normal records: 0 alarms in 3.92 h per method; one-sided 95% Poisson upper bound 0.76/h.
Paired 95% intervals: 10,000 resamples of 18 event parents. Each method meets the calibration alarm budget.

FIGURE 7. Coverage and alarm cost on held-out sequences. (a) Correct family and source before OFF, out of 108 interventions. (b) Unmatched alarms over 0.54 monitored event-hours. (c,d) History bank minus each comparator, with 95% intervals from 10,000 paired resamples of 18 event parents. Positive coverage differences and negative alarm-rate differences favor the bank. The bank improves both outcomes over GLR but trades more extras for higher coverage than ExtraTrees. Thresholds were selected on separate calibration data under a target of one normal alarm/h; normal exposure and its Poisson upper bound are shown separately.

had zero normal alarms in 3.92 monitored hours per method (one-sided 95% Poisson upper bound: 0.76/h). The result is conditional on the count model and the aggregate exposure. The two physical methods differ in candidate search, score semantics and commitment, so the comparison does not isolate the effect of explicit recovery histories.

B. INDEPENDENT EVALUATION: TIMELY ATTRIBUTION AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched alarms on event records, and its nominal counterpart 643/648 with 92. These are paired replays of 108 interventions in 18 physical parents, not 648 independent events. On the 72 normal records (42,336 s, 11.8 h of monitored time per method) neither bank raises an alarm, which bounds the normal false-alarm rate below 0.26 per hour at 95% confidence (Poisson). Every matched diagnosis emitted before OFF names the correct family and source; the timely misses are late or absent diagnoses. Table 5 and Fig. 8 give the family split.

The extra alarms concentrate where the network model is wrong: 68 of the 72 occur when the true R/X perturbation lies outside the five-model bank, 68 alarms in 1.08 h of out-of-bank monitored event-record time (about 63 per hour), against four in the other four strata together. Out-of-bank mismatch also produces the only four timely misses. The supplement gives the paired parent-cluster intervals; the physical bank's median alarm delay is about 0.27 s in the matched and in-bank conditions, comparable to the learned adapters (0.19–0.30 s).

TABLE 5. Held-out continuous-stream decisions. Each family has 324 replays of 54 distinct interventions; extras are unmatched alarms on event records. The ExtraTrees adapters keep their original operating points.

Method	Load	Line	Extras
Physical bank	322/324	322/324	72
Nominal bank	322/324	321/324	92
ExtraTrees	15/324	315/324	29
ExtraTrees short	30/324	322/324	16

We compare these delays retrospectively with the nominal signature crossing. For the nominal bank on matched networks, line alarms arrive after a median 0.18 s, at the six-report confirmation floor of 0.20 s less the sub-report onset phase, and doubling the noise leaves them unchanged. Load alarms arrive after 0.35 s, again unchanged by doubled noise but about 4 reports after the nominal signature reference crossing. Line delays are consistent with the publication floor, while load delays exceed that model-based reference. The two ExtraTrees adapters recover lines (315 and 322 of 324) but few loads (15 and 30). The diagnosis-time distributions in the supplement show that this is a coverage failure: every load diagnosis the adapters do emit before OFF is correct, but most load interventions receive none. A per-report audit on four further matched records (12 load and 12 line interventions, 1,245 and 1,143 in-event reports) locates the failure before any publication rule. During every load intervention the ref-

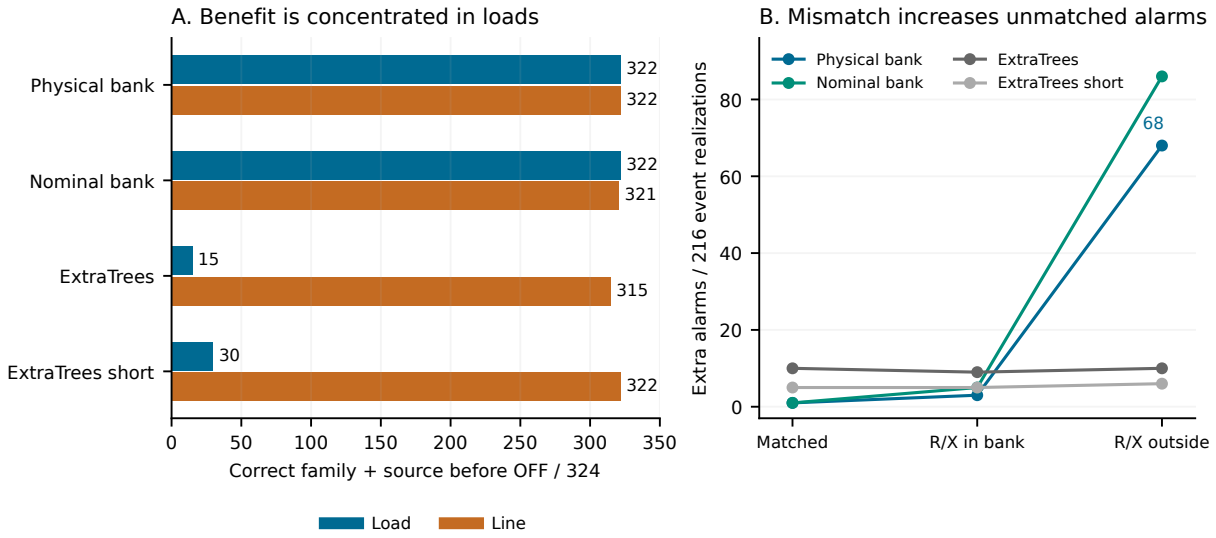


FIGURE 8. **Held-out continuous evaluation. (A) Timely correct family and source by event family, with 324 paired replays per family. (B) Unmatched alarms on event records for the two noise levels of each R/X condition (216 event replays per condition). Each method keeps its own publication rule.**

erence model's per-report argmax is the normal class in all reports, and the median of the largest true-class score is 0.09; during line interventions the true line wins 96% of the reports with median peak score 0.93. The short-window variant ranks the true load first in only 2 of 12 load interventions. The learned representation thus does not separate 3–14% load changes from normal operation, whereas the nominal physical signatures have large separation energies for the same changes (Fig. 6). The same pattern was already present in their development validation (Section VI), so the comparison shows that the physical bank recovers load interventions that these particular adapters miss; it is not a test of the best attainable learned detector at a common alarm budget. In the persistent-line test, both banks and both adapters identify the correct line first on all 162 openings. The adapter of Chen *et al.*'s quickest-change detector identifies 74/162; 66 records stop before the opening and 22 afterward on a wrong line, and it stops on all 72 paired normal records (first stop after 2,522 s of cumulative normal exposure, against 42,336 s monitored without alarm by the other methods). Its white-injection, quasi-static model does not describe the colored, dynamically filtered measurements of this plant, so its threshold is far from its nominal false-alarm rate. We therefore read this result as a test of transfer under model mismatch, not as a ranking of quickest-change detection. After the first identification, the two learned adapters raise 164 and 179 further unmatched alarms on the same persistent records, against 0 for the physical bank. Stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

Table 6 restates the evaluation in the metrics prescribed by the competition guide. Per-report detection F1 is 0.91 for the bank and 0.58–0.64 for the learned adapters, whose precision is higher because they alarm only on the line interventions. None of the methods raises an alarm on the normal records. The bank has no trained weights: its parameters are the network model, the noise model estimated from normal data, and a few prior and publication constants fixed during development. The nominal bank processes one minute of PMU data in 48 s on one CPU thread.

C. A CONTINUOUS SEQUENCE AND A DELAYED DECISION

A development trajectory, shown in the supplement, illustrates the timing criterion. The network keeps moving after OFF, and a later event is evaluated against that predicted response. Under out-of-bank network error, the bus-31 source is identified only after its restoration: correct, but not timely. The active-mass trace, the ΔM decomposition and the development counts are in the supplement.

D. A CONTINUOUS SINGLE-STATE PHYSICAL CONTROL

A development control keeps one committed augmented physical state through restoration instead of competing histories. It uses the same nominal plant and measurement operators, profiles severity over short alternatives of at most 19 reports and applies the same six-report publication clock. On 54 development replays (three parents, three R/X conditions), this single-state generalized likelihood ratio (GLR) detector attains 37/54 timely decisions with 37 extras, against 44/54 and 6 for the nominal history bank and 51/54 and 3 for the five-model bank. The development schedules pack six interventions into 80-s records instead of 120 s, which likely explains why the nominal bank scores lower here than in Table 5. The three procedures also differ in onset search and severity scoring, so this comparison indicates, but does not isolate, the benefit of explicit recovery histories over a single continuous state.

TABLE 6. Independent evaluation in the metric vocabulary of the competition guide (648 interventions, 706 min of normal records, three R/X conditions pooled). Detection is scored per report; Top-k requires the correct family and source among the first k candidates. Run time is per minute of PMU data on one CPU thread.

Method	Precision	Recall	F1	FA/min	Delay (s)	Top-1	Top-3	Run time (s/min)
Physical bank, five R/X models	0.888	0.926	0.907	0	0.20	0.995	0.995	179
Physical bank, nominal model	0.874	0.927	0.900	0	0.27	0.995	0.995	48
Submitted ExtraTrees	0.938	0.424	0.584	0	0.27	0.509	0.509	–
Retrained short-window ExtraTrees	0.944	0.483	0.639	0	0.19	0.543	0.543	–

E. SEPARATING SEVERITY SCORING FROM STATE CONTINUITY

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank, we fixed a subsequent 2×2 study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{25}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{26}$$

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k / v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

1) Component Outcomes

Table 7 and Fig. 9 report coverage and alarm cost for the four arms. With the state carried, marginal and bounded-profile severity scores tie on every reported endpoint. Resetting the 114 estimated physical-state deviations at inferred closure reduces timely attribution from 108/108 to 75/108 and raises the unmatched alarms from 1 to 166 under both scores. The reset leaves instrumental state, covariance regime, prior weights, thresholds and publication rules intact, and the simulated plant never resets.

Fig. 10 compares this outcome with the retrospective reset diagnostic. The profiled energy gain exceeds $\lambda^\star$ after every intervention, by at least two and up to seven orders of magnitude, and the reset arm is harmed after 84 of the 108 interventions: an extra alarm in the following gap or a lost timely attribution of the next one. Harm occurs after 51/54 line events and 33/54 load events. The bus-31 events carry the smallest profiled gain and are harmed after three of seven. The diagnostic shows that reset leaves a residual that event candidates can explain. Its magnitude alone does not determine the observed outcome: episode handling and publication rules also matter. These are within-simulator contrasts at the original thresholds; the two scores are compared at equal thresholds, not at equal alarm rates.

F. SENSITIVITY TO INDEPENDENT PMU NOISE

To examine sensitivity beyond the calibrated noise, we reran the nominal bank with the same equations and publication thresholds on four held-out calibration event records (12 load and 12 line interventions, matched network) and two 600-s normal records (0.33 h). Six noise scenarios were generated with the simulator's own noise stream: the original realization scaled by $\times 1$, $\times 5$, $\times 10$ and $\times 20$, and the original plus independent per-PMU relative-magnitude and angle noise, each with standard deviation $\sigma = 0.0005$ or 0.001 (angles in radians). The added perturbation has approximately 0.071% or 0.141% RMS total vector error (TVE), respectively, before combination with the original noise. Voltage-angle noise is propagated to frequency and ROCOF by the simulator's 21-report estimator. The $\times 1$ scenario reproduces the original predictions message for message. The bank re-estimates its noise scale and observation discrepancy from the noisy 300-report prefix; the operating point is kept at its nominal-noise estimate because derivative matrices exist only there. The implementation refuses to start when the estimated noise scale leaves its calibrated support of $[0.1, 10]$; for this test

TABLE 7. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.

Procedure	Timely	Extras	Normal FA	False restorations	No timely	Delay (s)
Marginal + carry	108/108	1	0/1,764 s	1	0/108	0.267
Profile + carry	108/108	1	0/1,764 s	1	0/108	0.267
Marginal + reset	75/108	166	0/1,764 s	130	33/108	0.337
Profile + reset	75/108	166	0/1,764 s	130	33/108	0.337

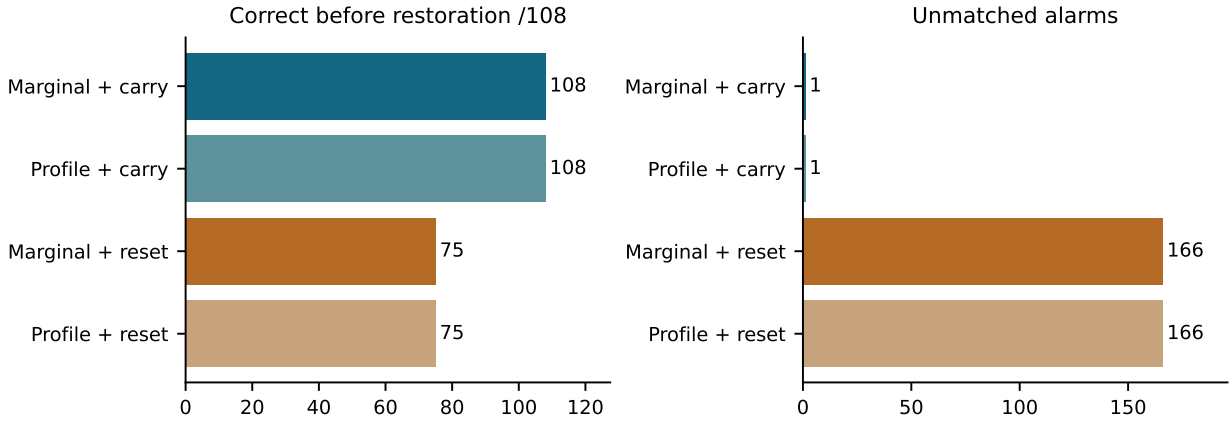


FIGURE 9. Severity-scoring and state-carry study on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the alarm rate may differ between arms.

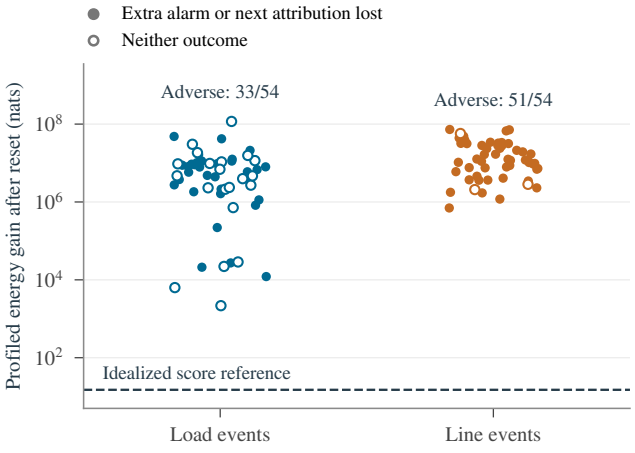


FIGURE 10. Retrospective profiled energy gain after a state reset, (24), for the 108 interventions of the controlled study. Filled: an extra alarm in the following gap or a lost timely attribution of the next intervention; open: neither. The dashed line is the idealized score reference, not the publication threshold. Under matched noise-free continuation, carrying the state gives no positive gain over the null.

only, that bound was widened, and every other line of code is unchanged.

Scaling the calibrated noise by up to twenty preserves all line attributions in this auxiliary cohort. One load intervention, a 4.1% change at bus 26, is missed at $\times 5$ and $\times 10$ but not at $\times 20$. Its amplitude exceeds the nominal reference at all three levels. That disagreement illustrates why a signature-energy crossing cannot guarantee a published attribution. The median load alarm delay changes from 0.35 to 0.36 s. No scenario produces an extra alarm on event records or an alarm on the normal records.

Independent per-PMU noise has a larger effect: the bank attributes 3/12 load interventions at $\sigma = 0.0005$ and 0/12 at $\sigma = 0.001$, while the corresponding line counts are 12/12 and 10/12. A matched-signature diagnostic using only the added independent noise still assigns substantial energy to these load responses (supplement). That restricted diagnostic excludes source search and the bank's covariance approximation; it does not establish achievable attribution accuracy. The bank fits one noise scale per signal family while retaining the reference cross-PMU correlation. Independent noise raises the fitted angle-noise scale and weakens load evidence under that model. Estimating cross-PMU covariance is therefore a candidate remedy that requires a separate controlled evaluation. These results do not establish readiness for field PMUs.

G. COMPUTATIONAL COST

Table 8 records the stages of the evaluation campaign, which ran the five-model bank on one thread. Its history inference takes a median 358.7 s per 120-s record (99.6 ms per report), three times slower than the 30-report/s input, so the five-model bank as implemented is an offline causal replay. The cost is dominated by the number of network hypotheses. Re-running the unchanged nominal bank alone on the same hardware and thread settings (Section VII-F) gives 26.4 ms per report (median over 6 records, maximum 30.0 ms), below the 33.3-ms report interval; the nominal bank attributed 643/648 interventions in the independent evaluation. These

TABLE 8. Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.

Stage	Median (s)	Range (s)	ms/report†
Bank build	80.1	62.0–309.7	22.2
History inference	358.7	218.2–2641.1	99.6
ExtraTrees (both)	9.5	6.7–50.5	2.6
Chen detector	1.3	0.9–10.6	0.4

are amortized processing rates on a desktop CPU; per-report latency percentiles and memory were not instrumented, and the 80-s startup bank construction is a one-time cost. Decision timestamps elsewhere in the paper are report times, not wall-clock latency.

VIII. DISCUSSION

A. WHAT STATE CONTINUITY EXPLAINS

A restored operator can leave a nonzero physical state, so returning the predictor to equilibrium introduces a residual that another candidate can explain. The reset control shows this failure at fixed thresholds; bounded profiling and marginalization tie in that control. The nominal signature diagnostics help explain short delays under the assumed noise model. They do not predict operational error rates. The bank has no online applicability gate to reject decisions after a plant change invalidates its models. Sparse-PMU transient-stability work has reported an explicit self-check for that applicability question [35]. The present signature diagnostics were evaluated retrospectively and do not suppress alarms.

B. COVERAGE, ALARM COST AND SCOPE

In the held-out comparison, the history-bank procedure had higher timely coverage and fewer event-record extra alarms than the continuous GLR. ExtraTrees emits fewer extra alarms but misses more interventions. These are contrasts between complete procedures calibrated to a common normal-alarm target; zero observed normal alarms give upper bounds, not equality of the underlying rates. Under an adequate Markov model, the full augmented-state distribution is sufficient for prediction. Compressing histories to their mean can discard covariance and distinct state alternatives. A history-bank advantage over that compression would therefore concern a practical representation of uncertainty, rather than information unavailable to every state observer. The supplementary distributional analysis separates predictive sufficiency from historical cause attribution. The present comparison does not isolate these effects.

C. TRANSFER AND MEASUREMENT LIMITS

Observations and predictors use the same simulator family. The catalog is fixed, the gaps between interventions are 3–8 s, and no intervention overlaps another. Transfer of the dynamic bank to a distinct plant, shorter gaps and unseen sources are untested. Under independent per-PMU noise at 0.071% added RMS TVE, the bank misses most load changes. This limits transfer beyond the calibrated measurement model. The terminal map retains the provisional bus-5 current terminal and excludes the bus-39 current angle. A dropout with held observations prevents recovery in the auxiliary stress test. The bank did not start on the retrospective competition record; the integrity checks and separate attributor in the supplement do not validate the dynamic bank on that record.

The tested IEEE-39 plant also has no grid-forming inverter controller dynamics. Applying the method to converter-dominated networks would require adding the relevant control and current-limiting states and testing those predictors anew; GFMI surveys describe several such modeling choices [53].

D. APPROXIMATION AND PROCESSING COST

Pruning, steady-state covariances, startup fitting and compression make the attribution weights approximate. Retrospective proposals change inner history weights without updating the outer model weights. The nominal bank processed reports faster than they arrived in a measured replay on one thread; the five-model version did not. This timing is specific to the reported hardware and records.

IX. CONCLUSION

We formulated sequential load and line attribution from sparse PMUs as a comparison of switched physical histories. The predictor retains estimated state at restoration and separates an active intervention from a preceding cause of recovery. On a held-out simulation under a common calibration budget, the nominal bank made timely, correct family-and-source attributions for 99/108 interventions, versus 72/108 for a continuous physical GLR and 22/108 for ExtraTrees. The respective unmatched alarm counts were 8, 85 and 0. A fixed-threshold state-reset control shows the cost of discarding the estimated state. The complete-procedure comparison does not isolate the benefit of retaining recovery histories from the loss of uncertainty under compression. The findings apply to sequential interventions from a known catalog in one simulator family. Transfer to another dynamic plant, shorter gaps and unseen sources remain untested; independent per-PMU noise at 0.071% added RMS TVE severely degrades load attribution.

REFERENCES

[1] I. V. Nikiforov, "A generalized change detection problem," *IEEE Transactions on Information Theory*, vol. 41, no. 1, pp. 171–187, 1995.
[2] T. L. Lai, "Sequential multiple hypothesis testing and efficient fault detection-isolation in stochastic systems," *IEEE Transactions on Information Theory*, vol. 46, no. 2, pp. 595–608, 2000.
[3] A. Tartakovsky, I. Nikiforov, and M. Basseville, *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. Boca Raton, FL: CRC Press, 2014.
[4] A. S. Willsky and H. L. Jones, "A generalized likelihood ratio approach to the detection and estimation of jumps in linear systems," *IEEE Transactions on Automatic Control*, vol. 21, no. 1, pp. 108–112, 1976.

[5] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.
[6] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.
[7] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.
[8] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023.
[9] A. N. Shiryaev, "On optimum methods in quickest detection problems," *Theory of Probability and Its Applications*, vol. 8, no. 1, pp. 22–46, 1963.
[10] B. Friedland, "Treatment of bias in recursive filtering," *IEEE Transactions on Automatic Control*, vol. 14, no. 4, pp. 359–367, 1969.
[11] H. A. P. Blom and Y. Bar-Shalom, "The interacting multiple model algorithm for systems with markovian switching coefficients," *IEEE Transactions on Automatic Control*, vol. 33, no. 8, pp. 780–783, 1988.
[12] Y. Boers and H. Driessen, "A multiple model multiple hypothesis filter for Markovian switching systems," *Automatica*, vol. 41, no. 4, pp. 709–716, 2005.
[13] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.
[14] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.
[15] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.
[16] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.
[17] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.
[18] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.
[19] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.
[20] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 6, pp. 1140–1153, 2014.
[21] X. Yang, N. Chen, and C. Zhai, "A control chart approach to power system line outage detection under transient dynamics," *IEEE Transactions on Power Systems*, vol. 36, no. 1, pp. 127–135, 2021.
[22] ——, "Dynamic power systems line outage detection using particle filter and partially observed states," 2021, version 3.
[23] R. Anguluri, O. Kosut, and L. Sankar, "Localization and estimation of unknown forced inputs: A group LASSO approach," *IEEE Transactions on Control of Network Systems*, vol. 10, no. 4, pp. 1997–2009, 2023.
[24] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.
[25] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.
[26] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.
[27] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.
[28] R. Nagi, X. Huan, and Y. C. Chen, "Bayesian inference of parameters in power system dynamic models using trajectory sensitivities," *IEEE Transactions on Power Systems*, vol. 37, no. 2, pp. 1253–1263, 2022.
[29] A. Alalem Albustami, A. F. Taha, and S. Mahadevan, "DAE-aware bayesian inference for joint generator-network parameter estimation," *Electric Power Systems Research*, vol. 263, p. 113688, 2027.
[30] D. Brahma, A. K. Singh, A. S. Mir, N. Senroy, and B. C. Pal, "An analytical method for nodal inertia estimation of power systems," *Journal of Modern Power Systems and Clean Energy*, vol. 14, no. 3, 2026.
[31] H. H. Le, H. Wang, C. Moya, M. Netto, and G. Lin, "Dynamic state estimation from sparse PMUs via physics-informed graph neural operators," 2026, arXiv:2611.10468, version 1, submitted November 7, 2026.
[32] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.
[33] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
[34] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.
[35] G. Wang, J. Guo, S. Ma, X. Zhang, Q. Guo, S. Fan, and H. Xu, "Data-driven transient stability assessment using sparse PMU sampling and online self-check function," *CSEE Journal of Power and Energy Systems*, vol. 9, no. 3, pp. 910–920, 2023.
[36] M. Barati, "Faster than real-time prediction of disruptions in power grids using PMU: Gated recurrent unit approach," in *2019 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*. IEEE, 2019, pp. 1–5.
[37] A. Pomarico, A. Berizzi, and J. N. Kutz, "A shallow recurrent decoder for dynamic state estimation with a limited number of PMUs in power systems," *arXiv preprint arXiv:2607.00116*, 2026.
[38] A. Dey, S. Chakraborty, and M. V. Salapaka, "Topology change estimation in power distribution networks: A machine learning approach," *IEEE Transactions on Smart Grid*, vol. 17, no. 4, pp. 3445–3462, 2026.
[39] Y. Yuan, Z. Wang, and Y. Wang, "Learning latent interactions for event classification via graph neural networks and PMU data," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 617–629, 2023.
[40] Q.-H. Ngo, B. L. H. Nguyen, T. V. Vu, J. Zhang, and T. Ngo, "Physics-informed graphical neural network for power system state estimation," *Applied Energy*, vol. 358, p. 122602, 2024.
[41] M. Rimkus, P. Kokoszka, D. Duan, X. Wang, and H. Wang, "Graph neural networks for the localization of faults in a partially observed regional transmission system," *Scandinavian Journal of Statistics*, vol. 52, no. 2, pp. 572–594, 2025.
[42] A. Abur and A. Gómez Expósito, *Power System State Estimation: Theory and Implementation*. New York, NY, USA: Marcel Dekker, 2004.
[43] E. Y. Chow and A. S. Willsky, "Analytical redundancy and the design of robust failure detection systems," *IEEE Transactions on Automatic Control*, vol. 29, no. 7, pp. 603–614, 1984.
[44] K. Chatterjee, K. Mahapatra, and N. R. Chaudhuri, "Robust recovery of PMU signals with outlier characterization and stochastic subspace selection," *IEEE Transactions on Smart Grid*, vol. 11, no. 4, pp. 3346–3358, 2020.
[45] S. Biswas, T. Yin, S. A. R. Naqvi, J. Follum, A. C. Varghese, T. Ahmad, and P. V. Etingov, "An open-access repository of synchrophasor data quality examples: Curation and example applications," *IEEE Access*, vol. 13, pp. 66 445–66 457, 2025.
[46] F. Pasqualetti, F. Dörfler, and F. Bullo, "Attack detection and identification in cyber–physical systems," *IEEE Transactions on Automatic Control*, vol. 58, no. 11, pp. 2715–2729, 2013.
[47] M. M. Islam, M. S. Reza, and M. A. Haque, "A low-latency, high-accuracy second-order generalized integrator-based synchrophasor estimation method for p-class PMU," *IEEE Access*, vol. 14, pp. 40 383–40 397, 2026.
[48] IEC and IEEE, "Measuring relays and protection equipment—part 118-1: Synchrophasor for power systems—measurements," IEC/IEEE Std 60255-118-1:2018, 2018.
[49] A. H. Adetayo, S. A. Zulkifli, M. F. Allehyani, H. S. Samkari, and R. Jackson, "A comprehensive review of PLL-based synchronization and PMU applications for distributed generation: Toward autonomous electrical networks," *IEEE Access*, vol. 14, pp. 158 430–158 447, 2026.
[50] T. Becejac and P. Dehghanian, "PMU multilevel end-to-end testing to assess synchrophasor measurements during faults," *IEEE Power and Energy Technology Systems Journal*, vol. 6, no. 1, pp. 71–80, 2019.
[51] G. D'Antona, C. Laurano, P. A. Pegoraro, C. Sitzia, and S. Toscani, "PMU-based state estimation: A multisnapshot approach for mitigating the impact

of systematic errors and line parameters uncertainty," *IEEE Transactions on Instrumentation and Measurement*, vol. 75, 2026, art. no. 9003911.
[52] D. Celeita, M. A. Rios, D. M. Laverty, J. Forero, A. F. M. Jaramillo, and S. McLoone, "IEC 61850-9-2 based module for state estimation in co-simulated power grids," *International Journal of Electrical and Computer Engineering*, vol. 13, no. 3, pp. 2555–2567, 2023.
[53] D. B. Rathnayake, M. Akrami, C. Phurailatpam, S. P. Me, S. Hadavi, G. Jayasinghe, S. Zabihi, and B. Bahrani, "Grid forming inverter modeling, control, and applications," *IEEE Access*, vol. 9, pp. 114 781–114 807, 2021.

JORGE LUIS MAYORGA TABORDA studied electronic engineering at Universidad de los Andes, Bogotá, Colombia. His academic work has addressed distributed control using population dynamics, including applications to urban traffic systems.

YESSICA AFRICANO RODRIGUEZ carried out undergraduate thesis work on photovoltaic integration at Universidad de los Andes, Bogotá, Colombia. Her published research includes co-simulation and stochastic assessment of photovoltaic hosting capacity in distribution systems.



MASOUD BARATI received the Ph.D. degree in electrical engineering from the Illinois Institute of Technology in 2013. He is with the Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA. His research includes mathematical modeling and optimization of electric power systems.



PAYMAN DEHGHANIAN received the Ph.D. degree from Texas A&M University in 2017. He is an Associate Professor with the Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA. His research includes power-system reliability, resilience, and data-informed decision-making.



DAVID FELIPE CELEITA RODRÍGUEZ received the Ph.D. degree in engineering from Universidad de los Andes, Bogotá, Colombia. He is an Associate Professor and Director of Outreach and Engagement with the Faculty of Engineering, Universidad de La Sabana, Chía, Colombia. His research includes power-system protection, fault location, and real-time simulation.

Digital Object Identifier

Supplementary Material for Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]
[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

ABSTRACT This supplement documents the implemented recovery-aware history bank: its local filtering calculations, measurement map, publication rules and simulation contracts. It records the complete confirmatory outcomes, controlled severity-score/state-carry ablation and continuous single-state control. The operating-point signature and recovery-geometry studies keep their separate, known-onset contracts and denominators. All empirical results reuse the fixed evidence; no new campaign or threshold selection is introduced here. Earlier voltage-reconstruction, simultaneous-source and general diagnosability studies are preserved in the separately compiled Historical Technical Notes.

INDEX TERMS phasor measurement units, sequential diagnosis, physical state continuity, conditional filtering, reproducibility.

APPENDIX A IMPLEMENTED PREDICTOR AND CONDITIONAL UNCERTAINTY

This section specifies the local operators and conditional distributions used by the fixed continuous estimator. It distinguishes identities of that local model from approximations introduced by its finite history bank. The numerical checks below exercise these identities on controlled inputs; they do not recalibrate or rerun the reported evaluation.

A. AFFINE OPERATORS AT A CONSISTENT TOPOLOGICAL ANCHOR

For network hypothesis ℓ, let $(\bar{x}_\ell, \bar{u}_\ell)$ be the operating point determined by the prefix-fitted load and dispatch factors. Fix its differential coordinates while solving the algebraic constraints separately for each topology q:

$$g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0. \tag{1}$$

The resulting point need not be a differential equilibrium after a line opening. Assume a nonsingular algebraic Jacobian and a fixed angle gauge. In the expressions below, f is the differential vector field after division by any nonunit differential mass entries. All derivatives are evaluated at the topology-specific point in (1); the network index is suppressed for readability. Define

$$\begin{aligned} c_q &= f_q(\bar{x}, \bar{z}_q, \bar{u}), \\ A_q^c &= f_x - f_z g_z^{-1} g_x, & B_q^c &= f_u - f_z g_z^{-1} g_u, \\ C_q^c &= H_x - H_z g_z^{-1} g_x, & D_q^c &= H_u - H_z g_z^{-1} g_u, \\ d_q^{el} &= H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u}). \end{aligned} \tag{2}$$

For the admitted scalar load direction $\delta u = La$, the continuous approximation is $\dot{\delta x} = A_q^c \delta x + c_q + B_q^c La$ and $\delta y = C_q^c \delta x + d_q^{el} + D_q^c La$. For a line, the topology selects the operator and the amplitude direction is zero. A nonzero pre-event state contributes through δx; d_q^{el} supplies the zero-deviation offset at the fitted anchor. This is a local expansion, so large excursions can still cause prediction error even when the algebraic solve and derivatives are accurate.

For a mode lasting h, compute the affine flow without inverting A_q^c:

$$\mathcal{A}_q = \begin{pmatrix} A_q^c & c_q & B_q^c L \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$
$$\exp(h\mathcal{A}_q) = \begin{pmatrix} F_q(h) & \beta_q(h) & U_q(h) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3}$$

Thus $\beta_q(h) = \int_0^h \exp(A_q^c s) c_q \, ds$. If an admitted switch divides one report into an old-mode interval h_1 and a new-mode interval h_2, then

$$F = F_2 F_1, \qquad \beta = F_2 \beta_1 + \beta_2, \qquad U = F_2 U_1 + U_2. \quad (4)$$

The physical state is continuous across this composition. The final-mode measurement map uses the consistent algebraic right limit. The predictor then appends phase history and the deterministic memory of the instrumental transformation. Scaling these coordinates and applying the same affine augmentation to (4) produces the stored $T_o, J_o, g_o, u_o, j_o, v_o$ operators. In their report ordering,

$$\begin{aligned} r_k &= J_o \xi_k + j_o + v_o a + \epsilon_k, \\ \xi_{k+1} &= T_o \xi_k + g_o + u_o a + \eta_k. \end{aligned} \quad (5)$$

The measurement is scored before correcting the state for the next report. The constant startup discrepancy is applied to the observations; it does not replace the topology-dependent affine terms.

The construction has three steps. A static solve of (1) exports the network forcing, the PMU reference and their reduced Jacobians; the conditional builder forms the topology offsets, the mode flows and the one-report operators; and a final check verifies that the constructed reference matches the prefix-fitted operating point. The reproduction manifest preserves these files and their hashes. No event truth is used to select the startup operating point.

B. FOUR AMPLITUDE INTERVALS REPRESENT ONE PRIOR

Let $I = [-0.2, 0.2]$ be partitioned into four disjoint intervals I_j up to endpoints of zero probability. Write $Z_J(\mu, v) = \int_J \mathcal{N}(a; \mu, v)\, da$ for any interval J. The initial prior has the exact decomposition

$$\begin{aligned} p_0(a) &= \sum_{j=1}^{4} \alpha_{j,0} \, \mathrm{TN}_{I_j}(a; \mu_0, v_0), \\ \alpha_{j,0} &= Z_{I_j}(\mu_0, v_0) / Z_I(\mu_0, v_0). \end{aligned} \quad (6)$$

The implementation assigns one history per source, proposed onset, restoration path, and interval. Each retains its own $(m_{h,j}, d_{h,j}, \mu_{h,j}, v_{h,j}, w_{h,j})$; there is no per-report collapse to a single amplitude distribution. The four interval histories use the same physical operators. They are neither quadrature nodes nor four separately linearized severity models.

For a fixed physical history with observations Y, let $Z_j(Y) = \int_{I_j} p(Y \mid a, h)\, \mathrm{TN}_{I_j}(a; \mu_0, v_0)\, da$. Then

$$p(Y \mid h) = \sum_j \alpha_{j,0} Z_j(Y), \qquad \alpha_j(Y) = \frac{\alpha_{j,0} Z_j(Y)}{p(Y \mid h)}. \quad (7)$$

Applying the local predictive integral sequentially gives the same identity. Identical linear operators and gains preserve the common untruncated Gaussian parameters for interval copies of the same path; their truncated moments and normalized weights differ. Subsequent beam pruning can remove interval histories, so equivalence with an unpartitioned bank is not guaranteed after pruning. The recorded discarded mass describes that approximation and is not a bound on all future decisions.

C. AMPLITUDE DEPENDENCE AND MIXTURE COMPRESSION

Condition on a retained history and the reports used so far. For $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, set $\sigma = \sqrt{v}$, $\alpha = (l-\mu)/\sigma$, $\beta = (u-\mu)/\sigma$, $Z = \Phi(\beta) - \Phi(\alpha)$, and $\lambda = [\phi(\alpha) - \phi(\beta)]/Z$. Its moments are

$$\begin{aligned} \bar{a} &= \mu + \sigma\lambda, \\ V_a &= v \left[1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{Z} - \lambda^2 \right]. \end{aligned} \quad (8)$$

The truncated-moment routine evaluates the normalizer in the log domain and bounds small numerical excursions of the moments. The Gaussian parameters μ, v are therefore distinct from the conditional amplitude mean and variance.

For the affine conditional state model $\xi \mid a, h \sim \mathcal{N}(m_h + d_h a, P_h)$, total expectation and covariance give

$$\begin{aligned} \bar{\xi}_h &= m_h + d_h \bar{a}_h, \\ \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ V_h &= P_h + d_h V_{a,h} d_h^\top. \end{aligned} \quad (9)$$

With normalized history weights ω_h, including the no-new-event history, the complete mixture moments would be

$$\begin{aligned} \bar{\xi} &= \sum_h \omega_h \bar{\xi}_h, \\ P_{\mathrm{mix}} &= \sum_h \omega_h \left[V_h + (\bar{\xi}_h - \bar{\xi})(\bar{\xi}_h - \bar{\xi})^\top \right]. \end{aligned} \quad (10)$$

This identity applies within a network hypothesis and its common state coordinates. It does not justify averaging absolute rotor states between different network equilibria.

The continuous estimator uses the shared regime covariance P_ρ in place of P_h. It updates the affine means and amplitude distributions, and computes marginal diagonal state variances for diagnostics. At closure it carries $\bar{\xi}$, clears explicit histories, and records the trace of amplitude/mean spread omitted from the new single-history representation. It does not carry the complete covariance in (10), nor the uncertainty of the fitted startup context. Instrumental and discrepancy states remain in the carried augmented mean.

For comparison, an exact conditional covariance recursion for the local linear model (5), with independent noises of covariances R, Q, is

$$\begin{aligned} S_h &= J_o P_h J_o^\top + R, \\ K_h &= T_o P_h J_o^\top S_h^{-1}, \\ P_h^+ &= T_o P_h T_o^\top + Q - K_h S_h K_h^\top. \end{aligned} \quad (11)$$

This is a diagnostic reference, not a description of a new executed continuous estimator. The fixed implementation solves the Riccati equation for the nominal operator in each

TABLE 1. Deterministic local-model checks. Maximum absolute errors over 72 scalar updates, five switch positions, and 12 joint Gaussian updates; each row uses its stated reference.

Check	Maximum error
Log evidence: partition	1.78×10^{-15}
Amplitude mean/variance	4.16×10^{-17}
Log evidence: quadrature	3.11×10^{-15}
Affine flow: ODE reference	1.11×10^{-16}
Joint mean/covariance	8.33×10^{-17}

discrepancy regime and uses that shared covariance for the admitted switched operators.

Sequential likelihood factorization conditions on past reports; it does not require marginal independence of the observations. The instrumental VAR/AR filter and the latent discrepancy AR state already model temporal dependence. Remaining covariance misspecification, linearization error, adaptive proposals, and compression can nevertheless affect the resulting weights. Their empirical calibration requires pre-update residual and coverage diagnostics, separately from the algebraic checks below.

D. DETERMINISTIC VERIFICATION AND ITS BOUNDARY

A deterministic check loads the implemented amplitude routine without running a campaign. It compares sequential one-interval and four-interval evidence and moments with direct scalar quadrature, including zero information and amplitudes on interval boundaries. An independent small switched linear system checks affine-flow composition against numerical integration. A linear Gaussian reference also compares the conditional affine-state recursion with a joint state–amplitude Kalman recursion for an untruncated Gaussian amplitude; the truncated case is checked separately by quadrature. All histories are retained in these checks. Source hashes, tolerances, and complete results accompany the generated table. These checks establish local-model identities on the stated inputs. They establish neither calibration of the pruned network bank nor robustness of its nonlinear physical approximation, and add no event outcomes to the evaluation cohorts.

APPENDIX B PREDICTIVE SUFFICIENCY AND DISTRIBUTIONAL FUSION

This note distinguishes a full posterior over an adequate augmented state from its mean. Let S_k include physical, instrument and discrepancy memory, the predictive mode, uncertain fixed parameters, dwell age, and any other summary needed to make the process Markov. Let H_k retain historical cause labels. Assume, in particular, that mode transitions and other predictive effects are represented in S_k, so the next-step kernel is conditionally independent of old cause labels given S_k:

$$p(S_{k+1}, Y_{k+1} \mid S_{0:k}, H_{0:k}, Y_{1:k}) = p(S_{k+1}, Y_{k+1} \mid S_k).$$

Then the marginal filtering distribution $\pi_k(ds) = p(ds \mid Y_{1:k})$ is predictively sufficient. For a measurable report set B, define $K_B(s) = p(Y_{k+1} \in B \mid S_k = s)$. Conditional independence and the tower rule give

$$\begin{aligned} p(Y_{k+1} \in B \mid Y_{1:k}) &= \mathbb{E}[K_B(S_k) \mid Y_{1:k}] \\ &= \int K_B(s)\, \pi_k(ds). \end{aligned}$$

This is a standard filtering identity, not a new result. Adequacy is substantive: omitted memory, mode, dwell age, uncertain parameters, or model mismatch invalidate the factorization unless represented in the state/kernel.

The posterior mean alone is generally insufficient. For example, with $Y_{k+1} = S_k + \epsilon$, $\epsilon \sim \mathcal{N}(0, R)$, the predictive variance is $\mathrm{Var}(S_k \mid y_{1:k}) + R$; a point estimate omits the first term. Under linear-Gaussian transitions and observations, a Gaussian filtering posterior is specified by its mean and covariance; both must be carried for exact prediction. A comparison of historical fusion with mean-only fusion therefore changes both distributional spread and, where applicable, mode or history uncertainty; it does not isolate lost historical labels.

For a mixture with components j, weights w_j, conditional means m_j and covariances P_j, the aggregate moments are

$$\begin{aligned} \bar{m} &= \sum_j w_j m_j, \\ P &= \sum_j w_j \{P_j + (m_j - \bar{m})(m_j - \bar{m})^\top\}. \end{aligned}$$

If a component also carries scalar severity A_j with mean μ_j, variance V_j, and conditional state representation $Z \mid A_j, j$ having mean $m_j + d_j(A_j - \mu_j)$ and covariance P_j, then its joint state/severity moments are

$$\mathrm{Cov}(Z, A) = \sum_j w_j \{d_j V_j + (m_j - \bar{m})(\mu_j - \bar{\mu})\},$$

$$\begin{aligned} \mathrm{Var}(Z) &= \sum_j w_j (P_j + d_j V_j d_j^\top) \\ &\quad + \sum_j w_j (m_j - \bar{m})(m_j - \bar{m})^\top, \\ \mathrm{Var}(A) &= \sum_j w_j \{V_j + (\mu_j - \bar{\mu})^2\}. \end{aligned}$$

Here $\bar{\mu} = \sum_j w_j \mu_j$. The joint moments require a common severity coordinate. Components with no severity, such as line hypotheses, use the state-only identity. Thus within-component severity correlation and between-history spread both contribute. These identities describe moment-preserving compression only; the resulting Gaussian approximation need not preserve a multimodal predictive distribution.

Predictive filtering and retrospective attribution are distinct. A causal filter updates history weights using reports available at each time. A later proposal for an earlier onset is a smoothing/replay operation: it needs the stored pre-onset filtering distribution and must replace, rather than duplicate,

the replayed suffix likelihood. The current filtering posterior alone does not in general recover that earlier distribution. A retrospective cause query requires the joint posterior $p(S_k, H_k \mid Y_{1:k})$; the marginal state posterior can suffice for prediction while being insufficient for historical attribution. This distinction matches the implementation's saved-state replay and its explicitly approximate outer model weights.

Finally, if a reconstructed hidden voltage $\widehat{V}$ is computed solely from already observed PMU data Y and fixed model/configuration θ, then it is a deterministic statistic $g(Y, \theta)$. Conditional on Y and θ, it supplies no additional independent evidence: appending it to the likelihood as an independent measurement would double-count data. It may still be a useful representation for a computation or diagnostic. If its construction uses additional measurements or uncertain parameters, those dependencies must instead be modeled jointly. This is the direct data-processing consequence of deterministic post-processing, not a claim that reconstruction has no computational value.

Finite-state enumerations check prediction before and after marginalizing history labels, joint cause queries, the mixture moments and deterministic post-processing. A Gaussian-mixture example also shows that moment matching can change predictive tails. These checks verify the stated identities and counterexamples; they do not validate the power-system estimator or quantify the accuracy of its compressed representations.

APPENDIX C HOW TO READ AND REPRODUCE THE CURRENT METHOD

This companion begins with the recovery-aware causal method used in the current article. Sections C–E give its execution order, local derivations, worked examples and simulation contracts. The separately compiled Historical Technical Notes preserve full diagnosability proofs, earlier state-reconstruction and batch-estimator studies, and secondary evidence. The earlier studies support particular components; they are not additional tests of the complete causal algorithm.

The article retains the physical model, the defining evidence integral, the recovery mechanism, the decision outputs and decisive results. This supplement supplies the intermediate calculations. The Historical Technical Notes develop functional observability, nuisance-profiled information and source ambiguity, and distinguish the implemented batch estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint estimator $\mathcal{E}_{\mathrm{J}}$. Neither should be silently substituted for the causal physical history bank. The conditional Gaussian updates use standard filtering identities [1]; the derivation below specifies their role and the approximations made by the implemented bank.

A. INPUTS, OUTPUTS AND THE INFORMATION BOUNDARY

The inputs are successive reports from eight PMUs, the fixed physical and instrumental operators, the admitted load/line catalog and the fixed method configuration. The output is a timestamped sequence of detection, family/source and restoration messages, together with retained attribution masses. True ON/OFF times, true sources and generating severities belong to the evaluator. They are not inputs to the causal algorithm.

Three times must remain distinct: the latest observed report, the estimated physical onset, and the public emission time. Looking back to propose an onset is compatible with causality when the message is emitted only after the required reports arrive. It does not permit backdating the public alarm.

B. EXECUTION ORDER FOR ONE STREAM

1) **Load the fixed operators.** Retain voltage reference, oriented terminal currents, load input directions, switch modes and instrument memory. A line opening changes the network and observation operators. It is not represented by fitting a load amplitude.
2) **Fit the startup context.** Use the first 300 reports for operating factors, common reference and noise scale. For each network hypothesis, set the voltage/current discrepancy to the difference between the observed prefix mean and its reference prediction. Keep that offset fixed afterwards. Frequency and ROCOF use their empirical prefix centers. The prefix is not scored again as independent model-selection evidence.
3) **Predict the next report.** Propagate the retained histories and evaluate each network model's predictive density before correcting states or choosing new histories with the current report.
4) **Propose candidate onsets.** The fixed screen uses lags 0–18 reports within a ten-second stored state history. Each proposal starts from a stored state before its proposed onset, including any recovery already present. The bank does not enumerate every possible onset in the stored interval.
5) **Score histories.** For loads, integrate severity with the bounded prior and the local conditional Gaussian likelihood. For lines, score the admitted switched operator. Score a report before using that same report for its conditional state correction.
6) **Update and compress.** Update severity and conditional states, network weights, restoration children and the retained beam. Record discarded mass. At episode closure retain the augmented-state mixture mean; the omitted between-history spread is a documented approximation, not an exact posterior covariance.
7) **Publish and record.** Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed detection, diagnosis and restoration rules. Save emission time and the last report used, in addition to estimated onset and source. Continue propagation through OFF.

C. WHY REPLAYED REPORTS ARE NOT NEW EVIDENCE

Let a retained parent have log score $\ell_{0:k} = \ell_{0:\tau-1} + \ell^{old}_{\tau:k}$. Replacing its suffix by a newly proposed physical history gives

$$\ell^{new}_{0:k} = \ell_{0:k} - \ell^{old}_{\tau:k} + \ell^{new}_{\tau:k}. \tag{12}$$

Transition and prior terms must accompany their corresponding history. The rolling screen supplies a proposal, not an extra independent likelihood. Adding the new suffix without removing the old suffix would count the same reports twice. This bookkeeping statement does not make adaptive screening or beam pruning exact Bayesian inference.

D. FIXED NUMERICAL CHOICES

The input grid is 30 reports/s. The five network hypotheses are nominal, global $\pm 2\%$ and selected-branch $\pm 3\%$ paired R/X perturbations. Each hypothesis is fixed in time; the weights change. They do not identify a continuous impedance posterior.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$. Birth mass is divided equally between load and line families, then uniformly among the admitted sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over intervals bounded by $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can create OFF children at either half-report position, inheriting state and severity. After likelihood normalization, the beam retains the union of the 96 highest-weight recent histories, 64 older histories and at least one per occupied source. The union is renormalized.

The internal observer confirms active-mass evidence at 0.99 for six reports and has a separate 15-report internal OFF clock. The public physical-bank publisher requires an internal pending episode and six reports of source-qualified active support above 0.99; its public OFF requires six reports of the diagnosed cell's joint cause-and-restoration mass above 0.99. The detailed source qualification, diagnosis and replacement rules are given below. These thresholds are implementation settings, not guarantees of probability calibration or false-alarm control. Nominal and mixture use their fixed publication paths, so their comparison does not isolate network averaging alone. The admission-rule ablation changes publication, not the recovery dynamics.

E. SOURCE QUALIFICATION AND PUBLIC EPISODE RULES

The source screen and the onset search are distinct. Source qualification uses a 30-report (one-second) conditional innovation vector v with 47 coordinates per report. Fixed normal-mode gains propagate the stored pre-window state and whiten the suffix. For each prepared response s_j and lag from 0 to 25 reports, let

$$\begin{aligned} g_j &= s_j^\top s_j, \qquad u_j = s_j^\top v, \qquad R = v^\top v, \\ \widehat{a}_j &= \mathrm{clip}(u_j/\max(g_j, 10^{-30}), -0.2, 0.2), \\ G_j &= 2\widehat{a}_j u_j - \widehat{a}_j^2 g_j, \\ F_j &= \frac{\max(G_j, 0)}{\max\{(R - G_j)/1409, 1\}}. \end{aligned} \tag{13}$$

Line columns use fixed amplitude one. The source score is the maximum over its columns; the threshold is the $1 - 0.001/936$ quantile of an $F_{1,1409}$ distribution, approximately 24.0115. This threshold and the variance floor are a fixed proposal/publication rule, not a calibrated global false-alarm probability. The statistic is never multiplied into the Bayesian history likelihood. This one-second qualification screen does not enlarge the separate birth search beyond its 0–18 report lags.

Define $b_c = 1$ for a line, and $b_c = \mathbf{1}\{F_c > 24.0115\}$ for a load. At each monitored report, the publisher updates consecutive-report counters for $M_c b_c \geq 0.99$, $\Gamma_c \geq 0.99$, and $\Gamma_c P(\mathrm{OFF} \mid c) \geq 0.99$. A false condition clears its counter. An internal alarm opens a pending episode. A public alarm then requires either six pending reports with $\sum_c M_c b_c \geq 0.99$ or an already accumulated six-report line-active counter. The latter permits line evidence collected before the internal pending message to count.

A public episode receives one diagnosis when its highest cause-mass cell has six consecutive reports above 0.99, or at the 15-report deadline after the public alarm. The output family maximizes summed cause mass; sources are ranked within that family. The diagnosed source becomes the episode owner. Six reports of that owner's joint cause-and-OFF mass above 0.99 emit a public restoration. These publication operations do not rescale M or reset the physical state.

The internal mixture lifecycle separately replaces an episode after six reports with mass at least 0.99 on active histories whose onset is later than its alarm (using the implementation's quarter-report timing guard). It withdraws an unconfirmed episode after the diagnosis delay when null mass has exceeded 0.99 for 15 reports. Its internal restoration requires cause mass at least 0.99 and conditional retired mass at least 0.99 for 15 reports. These terminal messages are propagated to the public episode if already emitted; an unconfirmed pending episode is recorded as rejected. Finalization marks any remaining public episode right-censored. The nominal comparator retains its own fixed internal lifecycle, so the nominal/mixture comparison is not an isolated model-averaging ablation.

APPENDIX D LOCAL CALCULATIONS AND WORKED EXAMPLES

A. COMPLETING THE BOUNDED-AMPLITUDE INTEGRAL

For one retained severity interval, write $e = sa + \epsilon$, with $\epsilon \sim \mathcal{N}(0, S)$, $S \succ 0$, and $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, $v > 0$. Here μ, v are parameters of the underlying untruncated Gaussian, not the bounded distribution's moments. With $g = s^\top S^{-1} s$ and $q = s^\top S^{-1} e$, expansion gives

$$\begin{aligned} &(e - sa)^\top S^{-1}(e - sa) + (a - \mu)^2/v \\ &\quad = e^\top S^{-1} e + \mu^2/v - \mu'^2/v' + (a - \mu')^2/v', \end{aligned} \tag{14}$$

$$v' = (v^{-1} + g)^{-1}, \qquad \mu' = v'(\mu/v + q). \tag{15}$$

Define $Z(\mu,v)=\Phi((u-\mu)/\sqrt{v})-\Phi((l-\mu)/\sqrt{v})$. Integrating the last Gaussian square over $[l,u]$ yields

$$\begin{aligned}\log L = & -\tfrac{1}{2}[e^\top S^{-1}e+\log|S|+n_y\log(2\pi)]\\ & +\tfrac{1}{2}[\log(v'/v)+\mu'^2/v'-\mu^2/v]\\ & +\log Z(\mu',v')-\log Z(\mu,v).\end{aligned}\tag{16}$$

The posterior on that interval is $\mathrm{TN}_{[l,u]}(\mu',v')$. With $\alpha=(l-\mu')/\sqrt{v'}$, $\beta=(u-\mu')/\sqrt{v'}$, $Z'=\Phi(\beta)-\Phi(\alpha)$ and standard normal density ϕ, its mean is

$$\mathbb{E}[a\mid e]=\mu'+\sqrt{v'}\,\frac{\phi(\alpha)-\phi(\beta)}{Z'}.\tag{17}$$

Thus neither the untruncated mean μ' nor a best-fit severity can simply replace the integral. The implementation combines retained intervals using their weights. The identity is exact for this conditional Gaussian model; it does not remove linearization, shared-covariance or pruning error.

B. NUMERICAL EXAMPLE OF ONE INTERVAL

This dimensionless example illustrates the calculation; it is not an IEEE-39 simulation or a new performance result. Set $S=1$, $s=2$, $e=1$, $\mu=0$, $v=1$ and $[l,u]=[0,1]$. Then $g=4$, $q=2$, $v'=0.2$ and $\mu'=0.4$. Substitution in (16) gives $L\simeq 0.342688$ and $\mathbb{E}[a\mid e]\simeq 0.464941$. The different posterior mean follows from truncation. A separate numerical check compares the closed form against direct numerical integration, including the posterior mean. It checks algebra only and does not modify a fixed experiment.

C. CONDITIONAL STATE CORRECTION

Represent the conditional state mean as $m_h+d_h a$. Let T_o,J_o,g_o,u_o,j_o,v_o denote the stored one-report transition and observation operators for switch o. The code forms $e=r_k-J_o m_h-j_o$ and $s=J_o d_h+v_o$. With shared covariance P_ρ in discrepancy regime ρ,

$$\begin{aligned}S_o &= R+J_oP_\rho J_o^\top, & K_o &= T_oP_\rho J_o^\top S_o^{-1},\\ m_h^+ &= T_om_h+g_o+K_oe, & d_h^+ &= T_od_h+u_o-K_os.\end{aligned}\tag{18}$$

To see the last line, apply the correction to a fixed amplitude:

$$T_o(m_h+d_ha)+g_o+u_oa+K_o(e-sa)=m_h^++d_h^+a.$$

The severity distribution is updated by the same likelihood integral. Regime covariances come from nominal-operator Riccati equations. The physical history bank does not propagate an independent exact covariance for every switched history. At closure the mean is retained and the omitted mixture spread recorded.

D. RESTORATION IS NOT A STATE RESET

Consider the illustrative scalar system $\dot{x}=-x+u$, $y=x$, with $x(0)=0$. Apply $u=a$ for $0\le t<\kappa$ and restore $u=0$ afterwards. Direct integration gives

$$x(t)=\begin{cases}a(1-e^{-t}), & 0\le t<\kappa,\\ a(1-e^{-\kappa})e^{-(t-\kappa)}, & t\ge\kappa.\end{cases}\tag{19}$$

Both expressions agree at κ. For $a=1$ and $\kappa=2$, $x(2)=0.864665$ and $x(3)=0.318092$ even though the intervention is OFF. Resetting the reference to equilibrium would discard that predictable recovery. This example explains state continuity; it does not claim that one exponential describes the multimode IEEE-39 response.

E. ACTIVE MASS, CAUSE MASS AND MODEL REWEIGHTING

For one model, suppose an illustrative cell has mass 0.60 in active histories and 0.30 in restored histories, with 0.10 elsewhere. Its active mass is $M_c=0.60$, while its originating-cause mass is $\Gamma_c=0.90$. Recovery can therefore remain attributable without declaring the input active. These assigned masses illustrate definitions, not calibrated probabilities from an experiment.

For two reports, adding and subtracting $\sum_\ell w_{\ell,k}M_{\ell,c,k-1}$ proves

$$\begin{aligned}\Delta M_c = & \sum_\ell w_{\ell,k}(M_{\ell,c,k}-M_{\ell,c,k-1})\\ & +\sum_\ell (w_{\ell,k}-w_{\ell,k-1})M_{\ell,c,k-1}.\end{aligned}\tag{20}$$

For example, unchanged conditional masses $(0.8,0.2)$ and network weights changing from $(0.5,0.5)$ to $(0.75,0.25)$ move M_c from 0.50 to 0.65. The entire 0.15 change comes from model reweighting. A rising aggregate mass need not mean that each model found stronger within-model evidence.

APPENDIX E SIMULATION SCENARIOS AND REPRODUCTION ORDER

A. PLANT, MEASUREMENT AND PROVENANCE

The simulator is a Julia/PowerDynamics research model of the IEEE 39-bus system, not an officially validated equivalent of the competition reference record. Its contract fixes Julia 1.11.9, PowerDynamics 5.0.0, NetworkDynamics 1.3.0 and IEEE39 0.1.0. Initial states, effective parameter vectors, terminal map, reporting operator and stochastic model come from that snapshot. Nominal inventory files do not replace the effective operating point. The load law is the fixed impedance law.

The voltage buses are $\{2,5,6,10,19,22,29,39\}$. Each current is an oriented branch-terminal measurement, not a nodal injection. BUS5 retains the provisional historical 5–6 branch, terminal 5, hypothesis; producer CT identity is unconfirmed and BUS39 remains partially validated. Execution uses full-precision configuration files rather than rounded tables. The competition reference record provided retrospective calibration context only.

Each physical parent is integrated continuously through its interventions. Differential states remain continuous at switching, while algebraic values are recomputed consistently. Reports use $t=k/30$ with endpoints included; the switch-aligned report uses the post-switch algebraic value. Solver steps are recorded separately and observations are not inter-

polated across a switch. The causal instrumental history is preserved through OFF.

B. CONTINUOUS DEVELOPMENT AND CONFIRMATION

The two development campaigns each contain three 80-s event parents with six interventions and one 80-s normal parent. Crossing the four parents with three R/X conditions gives 12 physical trajectories, 18 unique interventions and 54 event realizations per campaign. Both are development evidence in the current article. Repeated conditions are not independent event draws.

The confirmatory evaluation prespecifies 18 continuous parents, 27 persistent-line parents (nine lines, three repetitions), and 12 normal parents. The three R/X conditions produce $3(18+27+12) = 171$ intended physical trajectories; two instrumental-noise scales yield 342 observed records. The continuous parents contain $18 \times 6 = 108$ unique interventions. Their six condition/noise realizations are repeated observations of those parents. The 120-s and 600-s grids contain 3601 and 18001 reports, respectively. Completed results appear in the dedicated confirmation tables.

Continuous evaluation pairs an alarm temporally before inspecting its predicted label. The first unmatched post-ON alarm is eligible up to the next ON, with the inherited 30-s cap. Timely attribution requires correct family and asset emitted before physical OFF. Unmatched alarms, false restorations and open episodes are retained. Persistent-line evaluation instead scores the first stopping decision, including early stops, wrong sources, omissions and censoring. The detector of Chen *et al.* is evaluated only there; it has no load or restoration output. Confidence intervals use 10,000 paired parent-cluster resamples, not independent PMU samples.

C. SEPARATE SIGNATURE AND RECOVERY CONTROLS

The context-signature study uses a single load at buses $\{3, 7, 16, 24\}$, a 10-s record, known ON at 3 s and OFF at 6 s, and one scalar operating-context direction. It contains 57 dictionary/check trajectories and 85 test trajectories (80 interventions and five controls), with three common noise seeds producing 255 observed test records. The context and severity are estimated from measurements, while onset and family are supplied. The generating source is used to assess the response approximation and severity error. Those conditional metrics do not measure unknown-onset causal attribution.

The recovery-geometry control uses 38 trajectories. Prior load histories at buses 7 and 24, followed by delays of 0.5 and 2 s, give four recovery contexts. Three source-pair comparisons within each context give 12 correlated comparisons. The prior input is active from 2 to 3 s; the subsequent response is examined over 1.5 s. The supplied history, onset, family and two-candidate universe define an oracle control, with one uncertain prior-severity direction. It does not evaluate the causal physical method, unknown onset, event cardinality or operational false alarms.

D. REPRODUCTION WITHOUT CHANGING THE EXPERIMENT

1) Read each campaign's fixed plan and manifest. Select the intended population before interpreting any denominator. Keep current causal, signature, recovery-control and historical batch studies separate.
2) Resolve the plant state, parameter and measurement files from the versioned snapshot. Every new simulation is a new run, and a new hypothesis a new candidate; earlier preparation steps are never rerun inside a fixed baseline.
3) Preserve ON/OFF continuity, report endpoints, noise seeds and the boundary between observations and reserved event truth. Run inference before releasing truth to the evaluator.
4) Retain solver/inference failures in intended denominators. Aggregate at the declared parent level and preserve the original comparators and publication rules. Do not retune from confirmation results.
5) Rebuild the two documents from the versioned tables and figures. The default build does not regenerate scientific artifacts or import new campaign results; refreshing artifacts is a separate operation.

The reproducibility package keeps this walkthrough, the campaign evidence and the earlier mathematical development, compiled separately as Historical Technical Notes. A successful PDF build verifies document consistency; it is not a new scientific validation.

APPENDIX F EXECUTED MEASUREMENT MAP

A. MEASUREMENT CONTRACT

The executed channel map is taken from the versioned research-plant snapshot. The eight voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. The internal voltage phasor is in per unit with unit gain and is exported as phase-to-neutral RMS volts using 345 kV/$\sqrt{3}$; voltage angle uses one common affine phase reference plus the stored relative offsets. The reference interpretation is observational: physical common drift and instrument clock drift are not separately identified. Terminal currents are converted to outward branch convention from the simulator's into-terminal observable, then mapped with the row-specific polarity, gain and phase bias. Current bases use $100\,\mathrm{MVA}/(\sqrt{3}\,V_{\mathrm{terminal}})$; the common voltage base is 199.186 kV.

Frequency is computed as $60 + \dot{\theta}/(360)$ from the reported voltage angle, and ROCOF as $\ddot{\theta}/360$ using a causal 21-report quadratic estimate. The generator fit is robust and magnitude weighted; the predictor uses a fixed unweighted quadratic fit over the same history. Four warmup outputs are unavailable; the confirmatory evaluation uses a 10-s startup context and begins monitoring after 12 s. The 47 retained coordinates exclude the BUS39 current-angle coordinate. Noise multipliers 1 and 2 scale the standard deviation under the fixed stream. The stochastic model stores a cross-signal covariance field, but the executed report operator applies noise by signal family without that cross-family factor. These choices describe a

TABLE 2. Eight PMU rows of the executed measurement map. V is a bus voltage in p.u.; I is an oriented branch-terminal phasor. Gains and phase biases are applied to current channels. BUS5 retains the historical 5–6, terminal-5 hypothesis and its producer CT identity is unverified. BUS39 current phase is partially validated.

PMU	V bus	Edge	Branch	Terminal	Side	Polarity	Gain	Bias (°)	I unit	Provenance
2	2	2	2–3	3	dst	1	0.9581	-1.173	A	Terminal hypothesis; CT unverified
5	5	9	5–6	5	src	1	1.0361	0.312	A	BUS5 5–6 historical hypothesis; CT unverified
6	6	11	6–7	6	src	1	1.0077	1.495	A	Terminal hypothesis; CT unverified
10	10	17	10–11	11	dst	-1	1.0041	2.224	A	Terminal hypothesis; CT unverified
19	19	26	16–19	19	dst	1	1.0384	0.481	A	Terminal hypothesis; CT unverified
22	22	34	21–22	22	dst	1	1.0189	0.094	A	Terminal hypothesis; CT unverified
29	29	44	28–29	29	dst	-1	1.0316	-6.170	A	Terminal hypothesis; CT unverified
39	39	1	1–39	39	dst	-1	0.9298	30.053	A	BUS39 partial phase validation

TABLE 3. Fitted ExtraTrees comparators used in the held-out streams. Both were trained for this study and are distinct from the original competition models.

Comparator	Features	Windows	Trees
ExtraTrees	846	1, 5, 10 s	400
ExtraTrees short	1457	1, 5, 10 s; 3, 8, 15 reports	400

simulated sensor configuration, not a verified eight-PMU CT/PT installation.

APPENDIX G LEARNED COMPARATOR LINEAGE AND DECISION CONTRACT

Both models were fitted on 32 physical parents from four earlier development campaigns; validation used 8 parents and 16 observed records from a fifth campaign. All of these were development data for the comparator construction. The fitted label set contains normal operation and all 18 catalog sources. The training target is the active source; after restoration, including physical recovery, its label is normal. This is not a source-excluded transfer test.

Each report selects the largest tree vote, and six consecutive reports of the same label drive the episode interface; there is no separate probability threshold. Tree votes are uncalibrated; the 300-report prefix and the start of monitoring at report 360 are fixed. Model hashes, training metadata and the observed class set are kept in the reproducibility package.

APPENDIX H COMPLETE CONFIRMATORY OUTCOMES AND DIAGNOSTIC AUDIT

The fixed test contains 171 physical trajectories and 342 observed records. Every method has an accounted output for every intended record; the solver-rejection and inference-failure ledgers are empty. The continuous, persistent-line and normal physical-parent counts are 18, 27 and 12. Three network conditions and two noise levels repeat these parents. Normal controls are shared between evaluation lanes, not doubled independent exposure. Tables below are generated directly from the stored summary, records, event matches and paired intervals.

A. TIMELY COVERAGE VERSUS CLASSIFICATION

The additional confusion view in Fig. 1 is a posthoc descriptive audit. It uses the original evaluator's temporal episode association and the first public diagnosis; it does not rematch events using source labels. A diagnosis emitted after physical OFF is counted as unavailable for this view. All diagnoses that are timely and matched in these four procedures have the correct family and exact source. The large load gap therefore reflects whether a diagnosis is available before OFF, not a measured advantage in conditional label accuracy among timely outputs. Unmatched alarms remain in the separate alarm tables. Normal records are not mixed into this event-level matrix.

B. MAINTAINING A SUSTAINED EVENT AFTER THE FIRST STOP

The first-stop score alone hides subsequent fragmentation. Although both physical banks and both learned comparators correctly identify all 162 persistent openings first, the learned comparators later return to an inactive label and issue additional alarms. Table 11 describes those already recorded outputs. It does not change the fixed first-stop outcome or extend Chen's one-stop contract into a recurrent monitor. Retained open episodes are appropriate when an intervention continues beyond the record.

C. MEASUREMENT INTEGRITY IS A SEPARATE EXECUTED COMPONENT

The availability interface correctly identifies all four missing-data intervals at the first received report that reveals the loss. This is a received-report clock, not zero transport or computation latency. Bad-data labels come from the inherited window classifier. Neither missingness nor corruption proves malicious intent. The composition preserves physical-source and affected-PMU identities but does not run a joint physical/cyber Bayesian filter; the physical bank is unavailable on the retrospective replay of the competition reference record under its startup noise-scale contract.

TABLE 4. Continuous outcomes by network and noise stratum. Each cell is timely family/source decisions / intended events; unmatched event-stream alarms. Replays share physical parents.

Network	Noise	Physical bank	Nominal bank	ExtraTrees	ET short
Matched	1	108/108; 1	108/108; 1	56/108; 6	60/108; 5
Matched	2	108/108; 0	108/108; 0	54/108; 4	58/108; 0
In-bank	1	108/108; 2	108/108; 3	56/108; 5	60/108; 5
In-bank	2	108/108; 1	108/108; 2	54/108; 4	57/108; 0
Out-of-bank	1	106/108; 38	105/108; 47	56/108; 5	60/108; 6
Out-of-bank	2	106/108; 30	106/108; 39	54/108; 5	57/108; 0

TABLE 5. Continuous monitoring: unavailable intended records, normal-record alarm incidence, normal alarm counts and available normal exposure, false restorations and open episodes. Counts combine six repeated strata descriptively; normal parents are not independent across strata. Missing outputs remain intended failures.

Method	Unavailable	Normal incidence	Normal alarms	Exposure (h)	False OFF	Open at end
Physical 5-model	0	0/72	0	11.760	58	0
Physical nominal	0	0/72	0	11.760	68	0
ExtraTrees	0	0/72	0	11.760	29	0
ET short	0	0/72	0	11.760	16	0

TABLE 6. Conditional median diagnosis/restoration delays (s), kept within each paired stratum. These medians summarize matched emitted diagnoses/restoration messages with finite delay and must be read alongside omissions, unavailable records and censoring; – denotes no finite estimate. Logical delays do not measure computational throughput.

R/X condition	Noise	Physical 5-model	Physical nominal	ExtraTrees	ET short
Matched	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
Matched	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
In-bank	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
In-bank	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
Out-of-bank	1	0.33/0.18	0.33/0.18	0.23/0.18	0.19/0.18
Out-of-bank	2	0.26/0.18	0.33/0.18	0.30/0.18	0.20/0.18

TABLE 7. Persistent-line comparison on the common observations. Each cell gives correct first source / intended openings; pre-change stops / right-censored openings; mean restricted correct delay (s). Wrong first sources remain in the denominator. Restricted delay retains the horizon penalty for failures and censoring defined by the fixed evaluator.

R/X condition	Noise	Physical 5-model	Physical nominal	Chen *et al.*
Matched	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Matched	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
In-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.04
In-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.05
Out-of-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Out-of-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.59

TABLE 8. Persistent-line coverage and stopping outcomes across six repeated strata, including both fixed ExtraTrees comparators. Normal controls are the same controls used in the continuous table, not additional exposure. Chen *et al.* normal exposure ends at its first stop; the other methods monitor the full available record.

Method	Unavailable	Correct first	Pre-change	Censored	Normal incidence	Exposure (h)
Physical 5-model	0	162/162	0	0	0/72	11.760
Physical nominal	0	162/162	0	0	0/72	11.760
ExtraTrees	0	162/162	0	0	0/72	11.760
ET short	0	162/162	0	0	0/72	11.760
Chen *et al.*	0	74/162	66	0	72/72	0.701

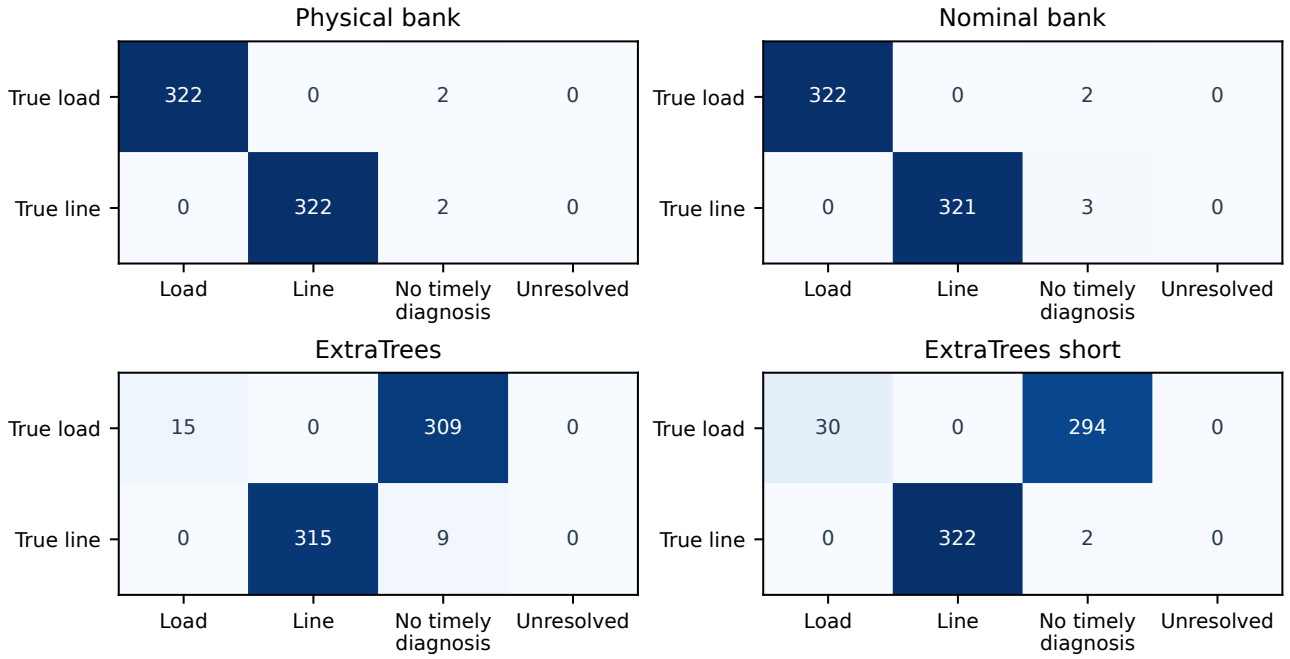


FIGURE 1. **Posthoc confusion audit of the fixed continuous test, with 324 realizations per true family. No timely diagnosis includes omitted and late diagnoses. Counts reuse the registered episode matches; all 432 prediction JSON files used were checked against the stored prediction hashes. These panels do not include unmatched alarms or constitute a new predeclared endpoint.**

TABLE 9. **Timely decisions by source in the continuous test. These are correlated realizations of the registered physical schedules, not source-excluded generalization.**

Family / asset	Physical bank	Nominal bank	ExtraTrees	ET short
Line / LINE14-15	36/36	36/36	27/36	34/36
Line / LINE16-17	36/36	36/36	36/36	36/36
Line / LINE17-18	48/48	48/48	48/48	48/48
Line / LINE23-24	30/30	30/30	30/30	30/30
Line / LINE26-27	24/24	24/24	24/24	24/24
Line / LINE26-29	24/24	23/24	24/24	24/24
Line / LINE3-4	42/42	40/42	42/42	42/42
Line / LINE4-14	46/48	48/48	48/48	48/48
Line / LINE9-39	36/36	36/36	36/36	36/36
Load / BUS15	42/42	42/42	6/42	6/42
Load / BUS16	42/42	42/42	0/42	3/42
Load / BUS20	12/12	12/12	6/12	12/12
Load / BUS24	42/42	42/42	0/42	0/42
Load / BUS26	18/18	18/18	0/18	0/18
Load / BUS3	42/42	42/42	3/42	3/42
Load / BUS31	40/42	40/42	0/42	0/42
Load / BUS4	36/36	36/36	0/36	6/36
Load / BUS7	48/48	48/48	0/48	0/48

TABLE 10. Paired physical-bank minus learned-comparator differences. Timely attribution is in percentage points; extras are per monitored event-record hour, excluding 12 s of startup from each 120-s record. The registered rate differences and 95-percent parent-bootstrap bounds are exactly rescaled by 120/108; original counts, matches and intervals remain stored.

Network	Noise	Method	Reference	Timely difference	Extra-rate difference
Matched	1	Physical bank	ExtraTrees short	44.44 [40.48, 47.92]	-7.41 [-16.67, 0.00]
Matched	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	1	Nominal bank	ExtraTrees short	44.44 [40.74, 48.04]	-7.41 [-14.81, 0.00]
Matched	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	2	Physical bank	ExtraTrees short	46.30 [42.75, 49.12]	0.00 [0.00, 0.00]
Matched	2	Physical bank	ExtraTrees	50.00 [46.30, 53.70]	-7.41 [-14.81, -1.85]
Matched	2	Nominal bank	ExtraTrees short	46.30 [42.86, 49.12]	0.00 [0.00, 0.00]
Matched	2	Nominal bank	ExtraTrees	50.00 [46.38, 53.70]	-7.41 [-14.81, -1.85]
In-bank	1	Physical bank	ExtraTrees short	44.44 [40.62, 47.92]	-5.56 [-14.81, 5.56]
In-bank	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-5.56 [-14.81, 5.56]
In-bank	1	Nominal bank	ExtraTrees short	44.44 [40.62, 47.92]	-3.70 [-12.96, 7.41]
In-bank	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-3.70 [-12.96, 5.56]
In-bank	2	Physical bank	ExtraTrees short	47.22 [43.18, 50.98]	1.85 [0.00, 5.56]
In-bank	2	Physical bank	ExtraTrees	50.00 [46.49, 53.70]	-5.56 [-11.11, 0.00]
In-bank	2	Nominal bank	ExtraTrees short	47.22 [43.33, 51.04]	3.70 [0.00, 9.26]
In-bank	2	Nominal bank	ExtraTrees	50.00 [46.30, 53.70]	-3.70 [-11.11, 3.70]
Out-of-bank	1	Physical bank	ExtraTrees short	42.59 [38.60, 46.43]	59.26 [40.74, 79.63]
Out-of-bank	1	Physical bank	ExtraTrees	46.30 [42.11, 50.00]	61.11 [40.74, 81.48]
Out-of-bank	1	Nominal bank	ExtraTrees short	41.67 [36.11, 46.67]	75.93 [51.85, 100.00]
Out-of-bank	1	Nominal bank	ExtraTrees	45.37 [41.11, 50.00]	77.78 [50.00, 105.56]
Out-of-bank	2	Physical bank	ExtraTrees short	45.37 [41.11, 50.00]	55.56 [38.89, 72.22]
Out-of-bank	2	Physical bank	ExtraTrees	48.15 [43.59, 52.50]	46.30 [27.78, 66.67]
Out-of-bank	2	Nominal bank	ExtraTrees short	45.37 [41.11, 50.00]	72.22 [46.30, 98.15]
Out-of-bank	2	Nominal bank	ExtraTrees	48.15 [43.75, 52.56]	62.96 [35.19, 88.89]

TABLE 11. Descriptive persistence audit after first detection on sustained line openings. All four procedures first identify 162/162 lines. Extras are later unmatched alarms; unmatched OFF counts only restoration messages on unmatched episodes. Open-at-end messages are expected for sustained events and are not automatically errors. False-negative reports include detection delay and later inactive labels. Chen is a first-stop method and is excluded from this recurrent-monitoring diagnostic.

Method	Openings	Extras	Unmatched OFF	Open at end	FN reports
Physical bank	162	0	0	162	810
Nominal bank	162	0	0	162	810
ExtraTrees	162	164	164	16	335374
ExtraTrees short	162	179	178	61	182600

TABLE 12. Retrospective measurement-integrity components over 180 windows. Hits require family and PMU identity. Missingness is observed availability; corrupted-data identification is inherited from the window classifier. These are not counts of attributed cyberattacks.

Component	Positive-window hits	FP	FN
Missing	14/14	0	0
Corrupt	1/2	2	1

APPENDIX I HELD-OUT RECOVERY COMPARISON PROTOCOL

The three arms use one causal reporting clock. The two physical arms share the nominal operators and permitted startup prefix. The continuous GLR retains one committed state and considers transition alternatives over a causal 19-report search; the bank retains conditional histories. ExtraTrees uses causal PMU features and was trained on a disjoint split with all catalog sources represented.

Selection used CAL only: among thresholds satisfying the normal alarm budget, maximize timely joint attribution, then minimize extra alarms and delay. The continuous GLR was rerun for each candidate threshold because commitment changes its state. TEST was evaluated once, after all predictions had been fixed. Timing associations use alarm time, not the true class. Timely attribution requires the right family and source before physical restoration; unavailable cases count in the intended denominator. Event-record extras and normal-record alarms have separate monitored exposures.

A. COMPARISON AT A COMMON CALIBRATION BUDGET

On a separate cohort, the nominal bank and the continuous generalized likelihood ratio (GLR) observer used the same nominal operators, eight PMUs, source catalog and permitted startup prefix. The GLR carries one committed state through restoration. ExtraTrees was refitted on disjoint simulated parents covering the same catalog. A common publication clock and calibration procedure selected a threshold for each arm under a normal-alarm budget of one per hour. The fixed methods were then scored once on 18 event parents and 24 normal parents. Interventions did not overlap, and the next onset was 3–8 s after physical restoration.

Figure 2 shows a 25.0 percentage point gain over GLR (paired 95% CI [13.89, 37.04]), with fewer extra alarms. The gain over ExtraTrees is 71.3 points (95% CI [63.89, 78.70]), with more extra alarms. The intervals resample entire physical parents within the network/noise strata; events within a parent do not count as independent samples. All three methods had zero normal alarms in 3.92 monitored hours per method (one-sided 95% Poisson upper bound: 0.76/h). The result is conditional on the count model and the aggregate exposure. The two physical methods differ in candidate search, score semantics and commitment, so the comparison does not isolate the effect of explicit recovery histories.

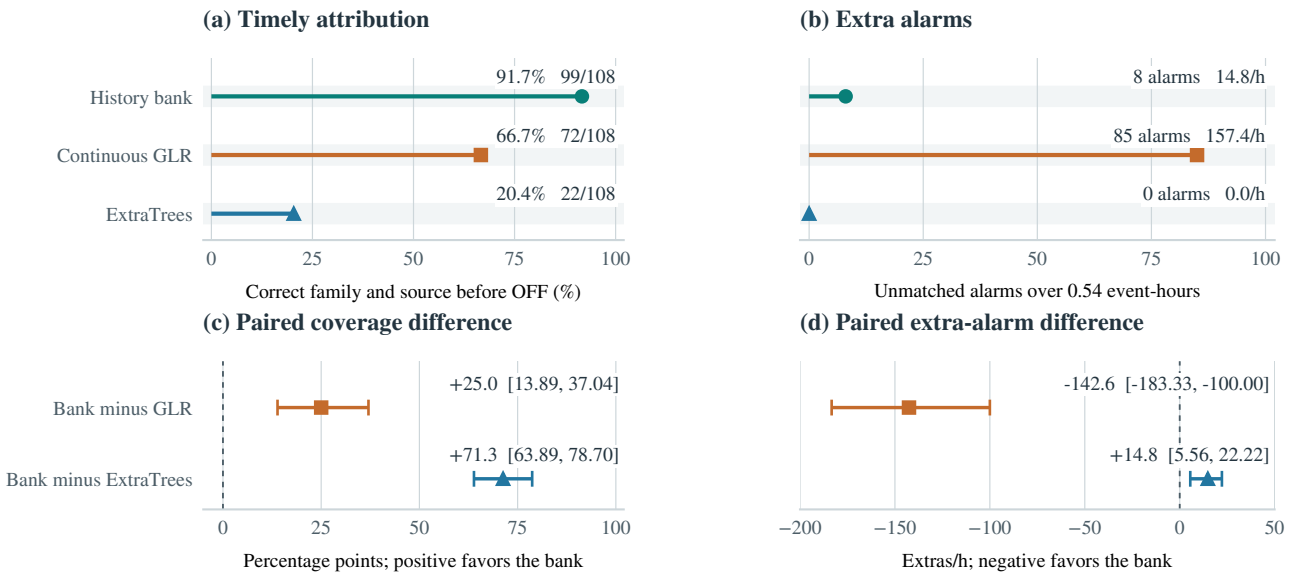
A separate calibration study found no gain from its chosen covariance correction: its selected correction multiplier was zero, and all three calibration arms tied on the evaluated endpoints. The profile and marginal scores likewise tied in the fixed-threshold carry/reset study. Neither finding establishes equivalence beyond its own comparison.

APPENDIX J CONTROLLED COMPONENT ABLATION

The fixed protocol and all output tables are part of the reproducibility package. Each of four arms accounts for all 24 intended records (18 continuous, three normal and three persistent): the evaluation audit records 96/96 available outputs and zero failures. The trial is posthoc and no score threshold is calibrated to these outcomes. Each normal control is a 600-s record with 588 s of evaluated exposure after startup, for 1,764 s total.

a: Exposure convention for the ablation alarm rate.

The ablation table reports extras per event-hour with the complete 120-s event record, including startup, as denominator. Its continuous exposure is $18 \times 120 = 2160$ s. The main article's confirmatory alarm-rate figure instead uses monitored exposure after the 12-s startup. Under that convention the ablation exposure would be $18 \times 108 = 1944$ s and each reported rate difference and interval bound would be



Normal records: 0 alarms in 3.92 h per method; one-sided 95% Poisson upper bound 0.76/h.
Paired 95% intervals: 10,000 resamples of 18 event parents. Each method meets the calibration alarm budget.

FIGURE 2. **Coverage and alarm cost on held-out sequences. (a) Correct family and source before OFF, out of 108 interventions. (b) Unmatched alarms over 0.54 monitored event-hours. (c,d) History bank minus each comparator, with 95% intervals from 10,000 paired resamples of 18 event parents. Positive coverage differences and negative alarm-rate differences favor the bank. The bank improves both outcomes over GLR but trades more extras for higher coverage than ExtraTrees. Thresholds were selected on separate calibration data under a target of one normal alarm/h; normal exposure and its Poisson upper bound are shown separately.**

TABLE 13. **All outcomes, including restoration messages on unmatched episodes and episodes open at the record end.**

method	lane	available	events	timely	extras	false restorations	open eof
Marginal + carry	continuous	18	108	108	1	1	0
Marginal + carry	normal	3	0	0	0	0	0
Marginal + carry	persistent line	3	3	3	0	0	3
Marginal + reset	continuous	18	108	75	166	130	11
Marginal + reset	normal	3	0	0	0	0	0
Marginal + reset	persistent line	3	3	3	0	0	3
Profile + carry	continuous	18	108	108	1	1	0
Profile + carry	normal	3	0	0	0	0	0
Profile + carry	persistent line	3	3	3	0	0	3
Profile + reset	continuous	18	108	75	166	130	11
Profile + reset	normal	3	0	0	0	0	0
Profile + reset	persistent line	3	3	3	0	0	3

multiplied by $120/108$. Counts, event matches and the stored table are unchanged; these two rate conventions must not be compared without this conversion.

APPENDIX K EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.

For a load ON alternative, write the innovation as $e_k - as_k$. The affine conditional state is $m_k + ad_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{21}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{22}$$

$$C = -\frac{1}{2} \sum_k \left(e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{23}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a}U - \tfrac{1}{2}\widehat{a}^2 G - C_{ref}. \tag{24}$$

TABLE 14. Paired descriptive differences with 95-percent physical-parent bootstrap intervals; no multiplicity-adjusted superiority claim.

method	reference	metric	difference	lower95	upper95
Marginal + carry	Profile + carry	timely fraction	0.000	0.000	0.000
Marginal + carry	Profile + carry	extras per event-hour	0.000	0.000	0.000
Marginal + carry	Marginal + reset	timely fraction	0.306	0.241	0.380
Marginal + carry	Marginal + reset	extras per event-hour	-275.000	-325.000	-223.333
Marginal + reset	Profile + reset	timely fraction	0.000	0.000	0.000
Marginal + reset	Profile + reset	extras per event-hour	0.000	0.000	0.000
Profile + carry	Profile + reset	timely fraction	0.306	0.241	0.380
Profile + carry	Profile + reset	extras per event-hour	-275.000	-325.000	-223.333

Here C_{ref} is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c)P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The development comparison contains no such post-startup missing observations.

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

APPENDIX L DEVELOPMENT STREAMS AND ADMISSION CONTROL

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

In the second development campaign, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 16). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the fixed adapters; it neither rules out supervised load detection nor reproduces the historical multi-class hackathon score.

TABLE 15. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

TABLE 16. What the causal component resolves in the second development campaign. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 4 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The

network mixture changes how much mass is assigned to that source, and the mass-increment identity in the main article separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

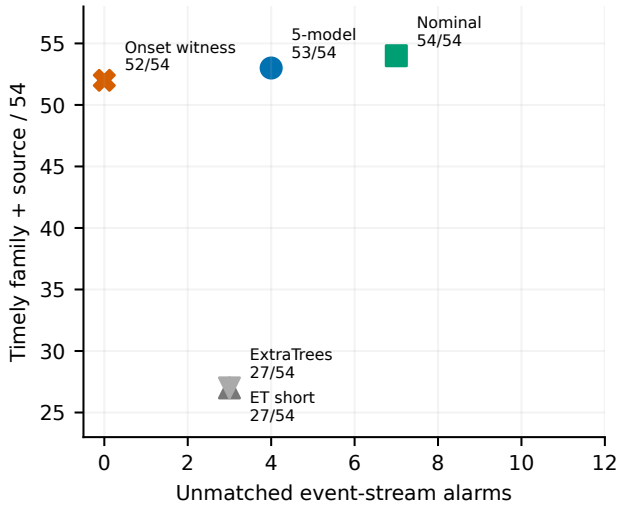


FIGURE 3. **The decision that an accuracy-only table hides. Development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.**

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative result left the five-model and nominal banks unchanged for the independent evaluation.

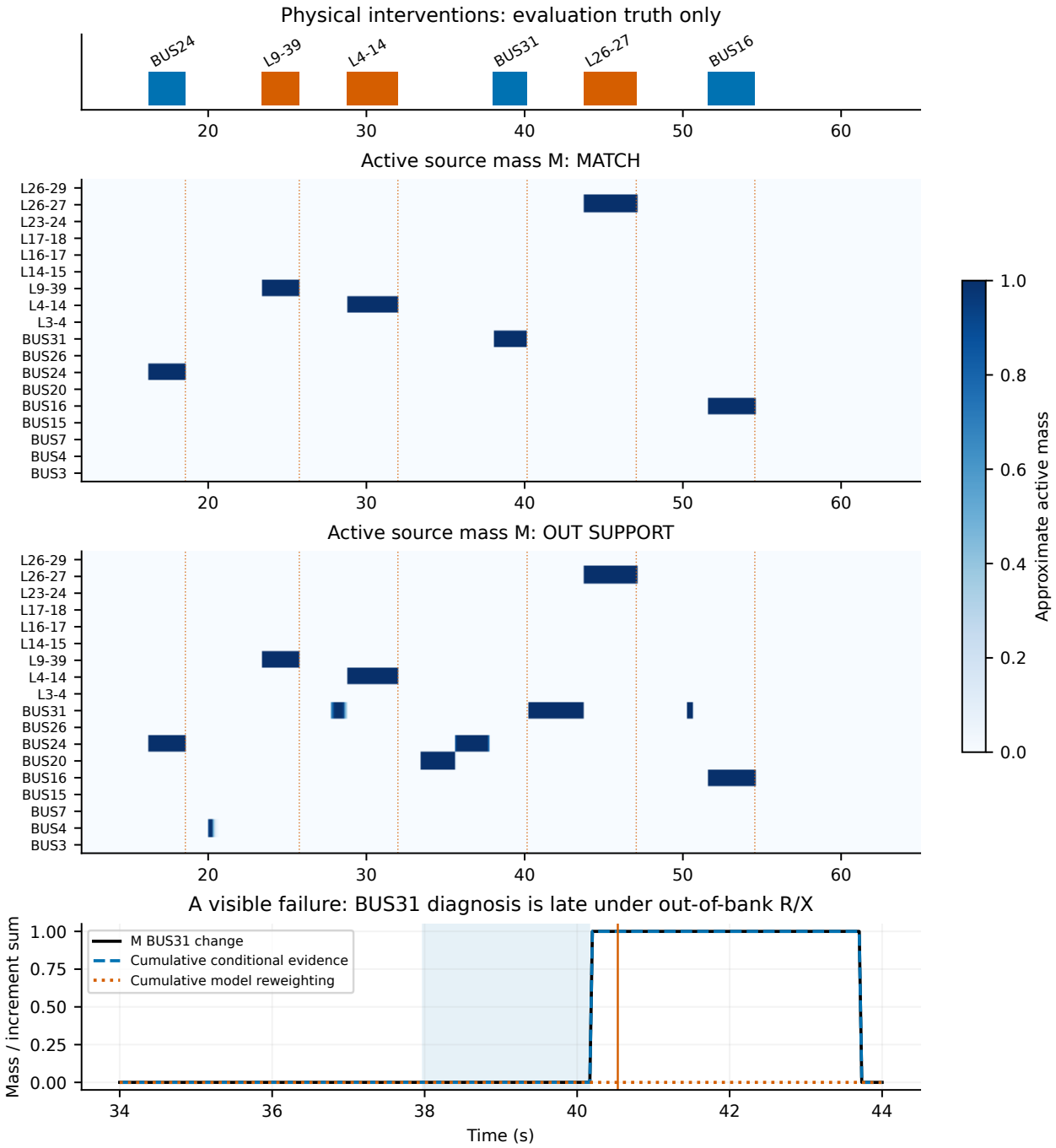


FIGURE 4. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in one development parent. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

APPENDIX M DETECTABILITY COMPUTATION AND NOISE-STRESS PROTOCOL

a: Operators.

The information analysis of the main article uses the nominal bank built by the evaluated implementation for one held-out calibration record of a later campaign whose runtime is byte-identical to the independent evaluation runtime. The independent-evaluation banks had been evicted by the storage-retention policy; the nominal operators differ between records only through the prefix-estimated operating factors, which remain within 2×10^{-5} of unity.

b: Signatures.

For each source i, the noise-free switched predictor is propagated from a zero pre-event deviation with the ON operator at half-phase 0 followed by the stay operator (loads at severity 0.1). Each candidate filter $j \in \{0, \dots, 18\}$ is run on these reports with the quiet-regime gains K_o and inverse innovation covariances S_o^{-1} of the implementation. The cumulative sums $\sum e^\top S^{-1} e$, $\sum s^\top S^{-1} e$ and $\sum s^\top S^{-1} s$ give the signature information of the main article with the competitor's severity profiled on $[-0.2, 0.2]$. As a check, the true-history filter at the true severity leaves a residual energy below 10^{-8}. The idealized score reference uses the birth hazard $1 - \exp[-(1/60)/30]$ shared equally by the 18 cells.

c: Reset cost.

For every intervention of the controlled study, the noise-free truth is propagated through ON, the active interval, OFF and the gap until the next onset. At the inferred closure time of the carry arm, the first 114 coordinates of the true augmented state are zeroed, and the no-intervention and all birth filters are run on the remaining truth in the recovery covariance regime. The largest profiled gain of a birth over the no-intervention history is compared with the idealized score reference $\lambda^\star$, not a publication guarantee. Harm in the reset arm is an unmatched alarm between the OFF of the intervention and the next onset, or a lost timely attribution of the next intervention.

d: Noise stress.

Observations are regenerated from the stored deterministic trajectories with the simulator's own noise stream and seed: $\varepsilon = c + \alpha(v - c)$ for multiplier α, where v is the stream and c its normalization center, exactly as the campaign exporter. The $\alpha = 1$ observations reproduce the original files to 6×10^{-10} and the inference reproduces all 18 published messages of the parity record. The independent-noise scenarios add zero-mean Gaussian perturbations per PMU and report, with relative magnitude and angle (radian) standard deviations of 5×10^{-4} or 10^{-3}; the added voltage-angle noise is passed through the simulator's causal 21-report quadratic frequency and ROCOF estimator. For a reference phasor z, this perturbation gives $\widetilde{z}/z = (1 + \epsilon_m)e^{\mathrm{i}\epsilon_\theta}$, with independent $\epsilon_m, \epsilon_\theta \sim \mathcal{N}(0, \sigma^2)$. Consequently, the RMS TVE of the added perturbation is

$$\sqrt{\mathbb{E}|\widetilde{z}/z - 1|^2} = \sqrt{\sigma^2 + 2(1 - e^{-\sigma^2/2})} \simeq \sqrt{2}\,\sigma.$$

The two settings therefore add approximately 0.071% and 0.141% RMS TVE; these are neither per-report bounds nor the total error including the original noise. The nominal bank is rebuilt from the stored derivative matrices with the noise scale estimated from the noisy prefix. The operating point remains at its $\alpha = 1$ estimate because derivatives exist only there. Estimated angle noise scales of these scenarios are about 48 times the calibrated scale, outside the implemented support $[0.1, 10]$; for this test only the support was widened to $[0.1, 1000]$. Two jobs ran concurrently with another campaign on the same desktop CPU, so the reported processing rates are conservative.

e: Reproduction.

Every number and figure of the analysis is regenerated from the stored operators and outcomes by the reproducibility package, which also contains the noise-stress generator.

TABLE 17. Noise-stress outcomes with fixed publication thresholds and widened startup noise-scale support. σ is the standard deviation of each added relative-magnitude and radian-angle component.

Scenario	Load	Line	Extras	Normal
$\times 1$	12/12	12/12	0	0
$\times 5$	11/12	12/12	0	0
$\times 10$	11/12	12/12	0	0
$\times 20$	12/12	12/12	0	0
$\sigma = 0.0005$	3/12	12/12	0	0
$\sigma = 0.001$	0/12	10/12	0	0

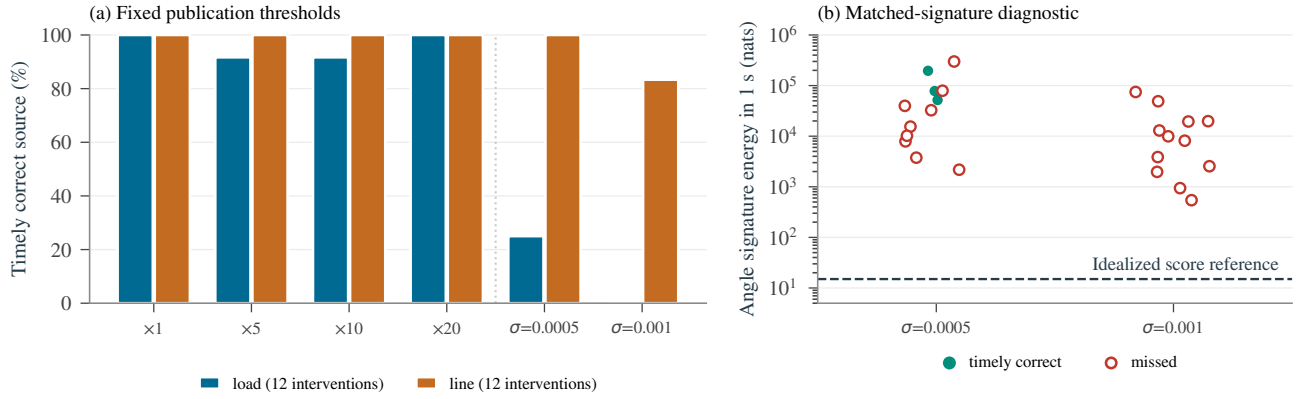


FIGURE 5. Nominal bank under increased PMU noise, with fixed publication rules and widened startup noise-scale support. (A) Timely attribution under uniform noise scaling and added independent noise, with component standard deviation σ (relative magnitude and radian angle). (B) One-second matched-signature energy using only the added independent angle noise, against the observed outcome. The dashed line is an idealized score reference, not the bank's publication threshold.

APPENDIX N WHAT TERMINAL CURRENTS ADD TO VOLTAGE SIGNATURES

This diagnostic compares the observation sets available to distinguish fixed source responses. It uses the nominal switched predictor and its implemented terminal map, including the provisional bus-5 assignment. Loads have a fixed 10% change; lines use their opening operators without an adjustable amplitude. The onset and pre-event state are supplied. These are model-based separations, not attribution outcomes of the sequential detector.

a: A common temporal noise reference.

To compare nested observation sets, all source means share one reference covariance. The quiet no-intervention predictor defines

$$r_k = J_0 m_k + e_k, \tag{25}$$

$$m_{k+1} = T_0 m_k + K_0 e_k, \qquad e_k \sim \mathcal{N}(0, S_0), \tag{26}$$

with independent innovations and fixed initial predictive state. Here m_k is the predictor's conditional mean, rather than a new physical plant state. Stacking reports gives $\Sigma = L(I \otimes S_0)L^\top$, where L has identity diagonal blocks and $L_{kr} = J_0 T_0^{k-r-1} K_0$ for $k > r$. Thus selecting voltage and current coordinates retains the induced cross-time covariance. It is not equivalent to selecting coordinates from innovations produced by a filter that already used all channels.

b: Conditional separation.

Let d be the difference between two fixed mean trajectories. Partition the selected report coordinates into voltage V and current I and set

$$S_{I|V} = \Sigma_{II} - \Sigma_{IV}\Sigma_{VV}^{-1}\Sigma_{VI}, \tag{27}$$

$$d_{I|V} = d_I - \Sigma_{IV}\Sigma_{VV}^{-1} d_V. \tag{28}$$

For a positive-definite common covariance,

$$Q_{VI} = Q_V + Q_{I|V}, \tag{29}$$

$$Q_V = \tfrac{1}{2} d_V^\top \Sigma_{VV}^{-1} d_V, \tag{30}$$

$$Q_{I|V} = \tfrac{1}{2} d_{I|V}^\top S_{I|V}^{-1} d_{I|V}. \tag{31}$$

The second term is nonnegative and measures the contribution of current coordinates after conditioning on voltage for this pair. It is a standard Gaussian identity. It does not establish the contribution of currents after separately fitting severity or other nuisance parameters, and it does not decompose the actual bank's model-dependent likelihoods. The nearest competing source may also change with the observation set.

c: Executed calculation.

The operators and quiet covariance come from one nominal calibration bank built with a 300-report PMU prefix. No test outcomes were used. We evaluated all 153 unordered source pairs at horizons of 3, 15 and 30 reports (0.1, 0.5 and 1 s at 30 reports/s); the stored table includes both directions. These are deterministic comparisons, not independent event replicates. At 1 s the ratio of minimum pairwise separation with V+I to that with V has median 1.35 and range 1.03–8.41 across sources. For each source's nearest V+I competitor, the conditional current term accounts for a median 18.0% of joint separation, with range 2.6%–76.6%. The two summaries answer different questions: the first permits the nearest competitor to change; the second holds the compared pair fixed.

The implementation reconstructs the raw-report covariance recursively and checks it against an explicit innovation transformation. Their relative discrepancy is below 6×10^{-16} at three reports. Full-channel quadratic forms agree with the null-innovation calculation to 3.1×10^{-15} relative error over all pairs and horizons; the conditional decomposition agrees to 1.4×10^{-11}. All nested-channel distances are nondecreasing. The source package contains the compact operator snapshot, channel maps, checksums, calculation and figure-generation scripts.

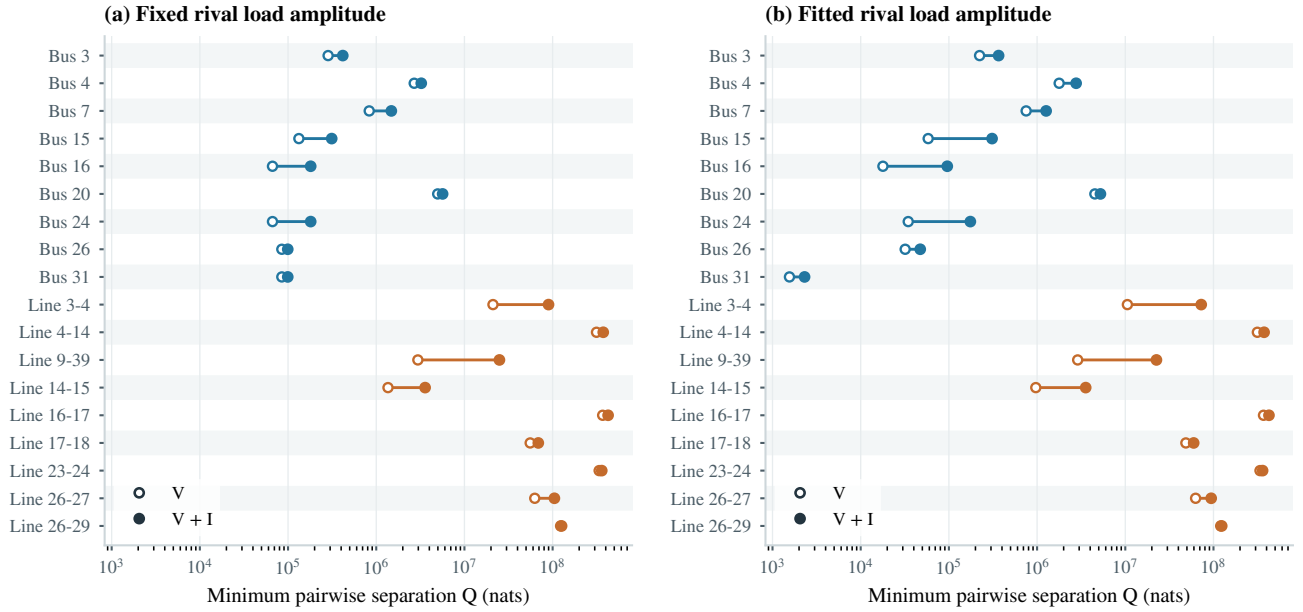


FIGURE 6. **Terminal-current information under a common nominal covariance at 1 s. Blue denotes load changes and orange line openings. (a) Minimum fixed-amplitude pairwise separation using voltage alone (open circles) or voltage and current (filled circles); the minimizing competitor may differ. (b) Rival load amplitudes are fitted over $[-0.2, 0.2]$, while line openings retain their fixed operators. Onset, initial state and true source amplitudes are fixed in both panels; these values do not measure detector performance.**

d: Allowing a rival to fit severity.

For a load rival with mean $\mu_j(b) = o_j + s_j b$, set $d = \mu_i - o_j$. Under the same common covariance, its best bounded fit is

$$b^* = \mathrm{clip}_{[-0.2,0.2]} \frac{s_j^\top \Sigma^{-1} d}{s_j^\top \Sigma^{-1} s_j}, \qquad s_j \neq 0. \tag{32}$$

The profiled separation is $\widetilde{Q}_{i|j} = \frac{1}{2}\|d - s_j b^*\|^2_{\Sigma^{-1}}$. Line rivals retain their complete opening trajectories, with no fitted amplitude. This is deterministic profiling, not integration of the truncated severity prior used by the detector.

For each common feasible amplitude b, the conditional decomposition gives $Q_{VI}(b) = Q_V(b) + Q_{I|V}(b) \geq Q_V(b)$. Hence $\min_b Q_{VI}(b) \geq \min_b Q_V(b)$. This does not imply that the two separately minimized terms add to the joint minimum: both must use the same amplitude. The inequality also holds after minimizing over the same rival catalog, although the nearest rival may change.

We repeated all source comparisons with bounded rival load severity, keeping true load changes at 10% and the same covariance and horizons. Every analytic load fit agreed with an independent scalar numerical optimizer to relative error below 10^{-12}. At 1 s, the ratio of minimum V+I to V separation has median 1.53 and range 1.03–7.82. With V+I, 4 sources change their nearest rival after fitting severity. The bus-31 source retains only 2.3% of its fixed-amplitude minimum separation: its nearest rival changes from bus 26 to bus 7. Thus a large distance between fixed responses can conceal ambiguity between source and magnitude. These nominal comparisons do not establish how often that ambiguity causes an attribution error.

e: Interpretation.

The reference covariance is fixed across sources; candidate-specific covariance terms, uncertain onset, severity marginalization, pruning, publication thresholds and false alarms are outside this diagnostic. The full set contains 16 voltage, 15 current and 16 frequency/ROCOF coordinates. Those derivative coordinates follow the implemented measurement model; their incremental separation is not a statement about independent physical sensors. An empirical channel ablation would require rebuilding each filter with its own observed coordinates and calibrating its publication rule before a new independent test.

A. RECOVERY UNCERTAINTY AND APPARENT SOURCE INFORMATION

A local Gaussian model explains why preserving a recovery mean alone can overstate confidence in a new source. Suppose, conditionally on an earlier history,

$$r = Ox + sa + \epsilon, \qquad x \sim \mathcal{N}(0, P), \quad \epsilon \sim \mathcal{N}(0, R),$$

where x and ϵ are independent and P, R are positive definite. Here R excludes the initial-state uncertainty represented by P. Marginalizing that state gives $\Sigma(P) = R + OPO^\top$. For a fixed local source direction s, Woodbury's identity gives

$$\begin{aligned}\mathcal{J}(P) &= s^\top \Sigma(P)^{-1} s \\ &= s^\top R^{-1} s - b^\top (P^{-1} + O^\top R^{-1} O)^{-1} b.\end{aligned} \tag{33}$$

Here $b = O^\top R^{-1} s$. The nonnegative subtracted term measures how much of the apparent source information can be explained by uncertain recovery coordinates. If $P_2 \succeq P_1$, then

$\mathcal{J}(P_2) \leq \mathcal{J}(P_1)$ for this fixed direction and noise law. This standard Gaussian identity explains a possible confidence error; it does not prove that a larger covariance improves attribution. Likelihood normalization, competing sources and false alarms still matter.

This decomposition is a separate explanatory model. The preceding channel diagnostic already uses a covariance induced by the complete null innovation model. Adding the bank's state covariance to it again would count the same predictive uncertainty twice. Likewise, an uncertainty already marginalized in Σ must not also be fitted as a free recovery direction without redefining the statistical model. The implemented state–severity correlations are specified in Section A-C; independence is not assumed for those stored posterior variables.

Port-Hamiltonian formulations provide a physical interpretation of network interconnection and energy balance [2]. Their reduced swing-model energy is not automatically an energy function for the present lossy DAE with exciters and governors. Similarly, the input-change versus topology-change distinction is exact only within its stated model: a finite impedance-load change can alter algebraic admittance and the local linearization. The implemented load direction is an affine approximation of that change. Energy residuals and reconstructed hidden voltages are therefore not appended as independent likelihood factors to the PMU reports that produced them.

APPENDIX O MEASUREMENT-INTEGRITY DIAGNOSIS

a: Statistics.

For each PMU and report, the three phase phasors of voltage and of current are transformed to symmetrical components (X_0, X_1, X_2). For each voltage or current set, the parity statistic is $q = 2(|X_2|^2+|X_0|^2)/(\overline{|X_2|^2 + |X_0|^2})$, where the overbar is the mean over the first 90 reports (3 s) of the record. With known scale and independent Gaussian per-phase noise, the normalized statistic has an idealized $\mathrm{Gamma}(2, 1)$ tail, $P(q > x) = (1 + x)e^{-x}$. The scale is estimated from the prefix, however, and the test uses the maximum of the voltage and current statistics. Thus the prespecified threshold $x = 23.9$ is a nominal 10^{-9} single-statistic tail reference, not an exact report-level false-alarm probability. Under the ideal marginal model, a union bound gives at most 2×10^{-9} per PMU-report for the maximum. The variance-inflated, post hoc operating point doubles the threshold to 47.9. A report is flagged if either statistic exceeds the threshold. The corrupted phase is the phase whose phasor has the largest projection on X_0.

b: Network-coupling bound.

The admittance matrix of the 39-bus network, including line charging and transformer taps, is augmented with the base-case loads as constant impedances and the generator transient reactances of the New England system on 100 MVA [3]. The bound c_p of the main article is the smallest, over the fault buses at which PMU p is the most affected PMU, of the largest sag ratio at another PMU. Table 18 lists the values. For the three-phase fault at bus 39 of the competition record, the model predicts sag ratios of 0.10–0.29 at the other PMUs. At the deepest fault report the observed ratios are 0.12–0.43, so the bound is conservative there. At the first fault report the largest ratio is only 0.11, 0.37 times c_{39}, because the phasor estimates of the distant PMUs lag the faulted bus. This motivates the margin of one quarter. A balanced change is tested when it exceeds 10% of the PMU's startup level relative to the median of the previous 30 reports. It is an integrity event when the largest change elsewhere is below $c_p/4$ times that change. While a coupling-consistent change is present at any PMU, and for 2 s afterward, sequence parity is suspended.

TABLE 18. Network-coupling bound c_p, the fault bus that attains it and the PMU most coupled to it.

PMU bus	c_p	Fault bus	Most coupled PMU
2	0.39	30	29
5	0.93	5	6
6	0.94	6	5
10	0.71	32	6
19	0.32	34	22
22	0.34	22	19
29	0.16	29	2
39	0.29	39	5

c: Unlabeled incidents.

At the variance-inflated operating point all detections outside labeled episodes are listed here. Single-report current-magnitude spikes occur in phase B of PMU 2 at 11.57 s (2,485 A) and 12.33 s (53 A), against a median of about 585 A, and in phase A of PMU 39 at 5,378.77 s (702 A) and 5,379.03 s (710 A), against 245 A. A balanced 12% jump appears at PMU 29 in the frame immediately before the dropout at 536.57 s. From 2,776.90 to 2,779.10 s, after the reconnection that ends at 2,776.07 s, the three voltage magnitudes of PMU 29 collapse to zero and include negative values. The other PMUs change by less than 0.7% during this interval. Two single-report flags coincide with switching instants: PMU 22 0.47 s after the line opening, and PMU 2 at the onset of the generation change at bus 2.

d: Simulated integrity stress.

The four held-out event records and two normal records of the noise-stress study were used with the calibrated noise ($\times 1$), whose outputs reproduce the stored campaign predictions. Corruptions rotate over the eight PMUs: three in the quiet tail of each event record (75, 90 and 105 s), one 1 s after the third intervention onset, and twelve in each normal record (every 50 s from 30 s). Dropouts rotate over PMUs 29, 5, 19 and 39 in the event records, and over PMUs 6, 10, 22 and 29 in the normal records, each lasting 27 s. The bank, its startup estimates and its operators are those of the stored campaign; the 300-report startup is unchanged in every scenario.

A. INDEPENDENT-EVALUATION FIGURES

B. MEASUREMENT INTEGRITY

The competition data also contain integrity events: frames that never arrive and frames whose values are corrupted. Write the received report of PMU p as

$$y_{p,k} = M_{p,k}\big(h_p(x_k) + v_{p,k} + \beta_{p,k}\big), \tag{34}$$

where $M_{p,k} \in \{0,1\}$ is availability and $\beta_{p,k}$ a corruption that does not enter the network equations. Availability is observed through the frame flag, so a missing PMU is identified exactly. What it removes is information, and the same quantity governs corruption.

Proposition 1 (Integrity isolation): Let the physical-source and no-intervention hypotheses have the same Gaussian covariance, block diagonal across PMUs. Let corruption of PMU p be an unrestricted additive mean shift over the window. The minimum Gaussian Kullback–Leibler divergence from physical source i to the no-intervention hypothesis with PMU p corrupted is $\Lambda_{i0}^{(-p)}$, the mean-separation information for i computed without PMU p's channels, using the same fixed covariance and severity profile. This is also the information for i when PMU p is missing.

Proof. With common covariance, the covariance terms in the Gaussian KL divergence cancel, leaving one half of the covariance-weighted squared mean difference. Minimizing over an unrestricted β_p removes the component supported on PMU p. For block-diagonal covariance, the residual quadratic form is exactly the mean-separation information in the remaining PMUs. □

For a common joint Gaussian covariance Σ and a free shift on the corrupted coordinates, minimizing the quadratic separation over that shift gives $\frac{1}{2} r_c^\top \Sigma_{cc}^{-1} r_c$, where c denotes the remaining coordinates. This uses the marginal covariance of the clean observations; equivalently, Σ_{cc}^{-1} is the Schur complement of the corrupted block in the joint *precision* matrix. Conditioning on corrupted values as though they were clean would answer a different question. This identity does not provide a general integrity detector: it states what information remains under a free additive corruption at one PMU. The two monitors below use narrower physical checks. *Sequence parity.* For balanced interventions (load, line, generation, three-phase fault) the balanced network produces no negative- or zero-sequence voltage or current. These sequence components are therefore ideal-model parity relations in the sense of analytical redundancy [4]: they vanish in this balanced observation model and sensitive to corruption of a single phase. A step ϵ in one phase magnitude gives $|X_2| = |X_0| = \epsilon|X|/3$. The corrupted phase is the one whose phasor is most aligned with X_0. With known scale and independent Gaussian per-phase noise, $q_k = (|X_2|^2 + |X_0|^2)/\sigma^2$ is $\mathrm{Gamma}(2,1)$-distributed. In the implementation the scale is estimated from a 90-report prefix and the statistic is the maximum of voltage and current values, so the resulting threshold is only a nominal single-statistic tail reference, not an exact report-level false-alarm calibration. In the simulated causal cohorts, phases B and C are reconstructed from phase A under the balanced-waveform assumption; they do not provide independent sequence observations. The sequence-parity evaluation below therefore applies only to the separately exported three-phase competition record. *Network coupling.* A modeled single bolted fault that changes the positive-sequence voltage of PMU p by Δ_p changes the other PMUs in proportion to the network impedance matrix $Z = Y^{-1}$. Here Y includes constant-impedance loads and generator transient reactances [3], and a bolted fault at bus f gives $\Delta_j/\Delta_p = |Z_{jf}/Z_{pf}|$. Hence

$$\max_{j\neq p} \Delta_j \;\geq\; c_p\, \Delta_p, \qquad c_p = \min_{f\in\mathcal{F}_p} \max_{j\neq p} \left|\frac{Z_{jf}}{Z_{pf}}\right|, \tag{35}$$

with $\mathcal{F}_p$ the buses at which p is the most affected PMU. For the IEEE 39-bus network c_p ranges from 0.16 (PMU 29) to 0.94. A balanced excursion that violates (35) by a wide margin is inconsistent with the modeled set of single bolted faults. It does not rule out other physical disturbances or report-window effects. Both relations are classical bad-data reasoning [5], applied here to report-rate phasors without a state estimator.

TABLE 19. Labeled integrity episodes of the competition record and their diagnosis. Delay is in reports after the first labeled report; "flagged" counts diagnosed reports within the episode; phase is identified (labeled-data phase). Every episode was located at the correct PMU.

Type	PMU	Start (s)	Length (s)	Delay	Flagged	Phase
Corrupted	2	3.3	0.73	0	22/22	C (C)
Corrupted	39	334.4	0.77	0	23/23	A (A)
Corrupted	39	336.5	1.07	0	15/32	A (A)
Missing	29	536.6	27.07	0	812/812	–
Missing	29	614.3	29.10	0	873/873	–
Missing	29	2584.7	191.43	0	5743/5743	–
Missing	29	2935.8	59.07	0	1772/1772	–

C. MEASUREMENT INTEGRITY

Competition record. The competition record contains 90 min of reports from the eight PMUs (161,379 reports per PMU). Its labels mark four dropout events of PMU 29 (9200 reports) and three corrupted-data episodes at PMUs 2 and 39. A separate class 6 label composes a missing PMU with the generation event at bus 2 and is counted as an additional event by the competition scorer; the integrity monitor still sees the same missing frames. We applied the two parity relations of Section O-B and the frame flag. Separate voltage and current noise scales for each PMU were estimated from the first 3 s of the record; there was no training and labels were used only for scoring. Table 19 and Fig. 10 give the results. All 4 dropouts and all 3 corruptions were diagnosed at their first report and located at the correct PMU. The corrupted phase was identified in 3/3 episodes. The third corruption is intermittent, with a median step below 1%, and is flagged in 15 of its 32 reports.

The three-phase fault at bus 39 produces negative-sequence transients at seven PMUs at once. Its four reports with a

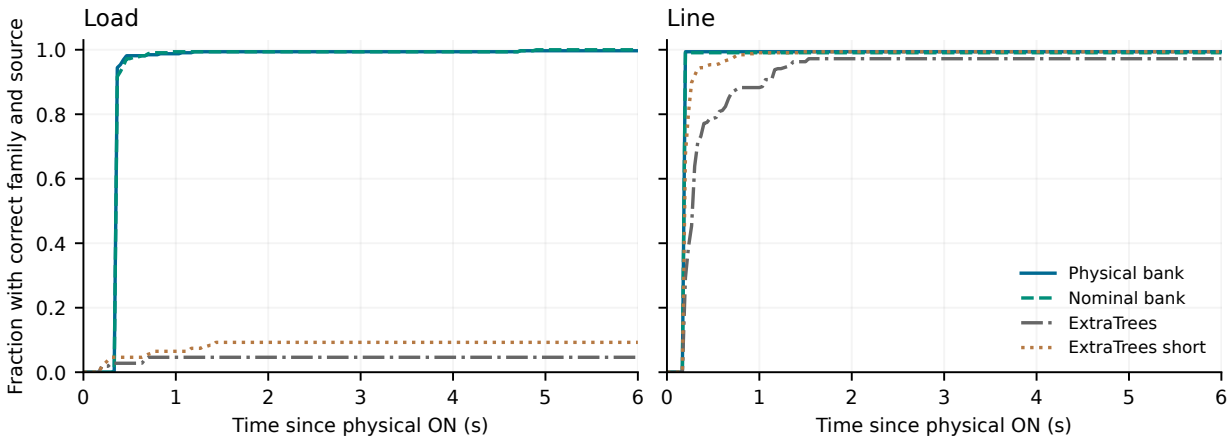


FIGURE 7. **Distribution of correct diagnosis time, using all 324 replays per true family in each denominator. An incorrect or absent diagnosis contributes no mass, and late correct decisions are included, so the curves differ from the before-OFF endpoint.**

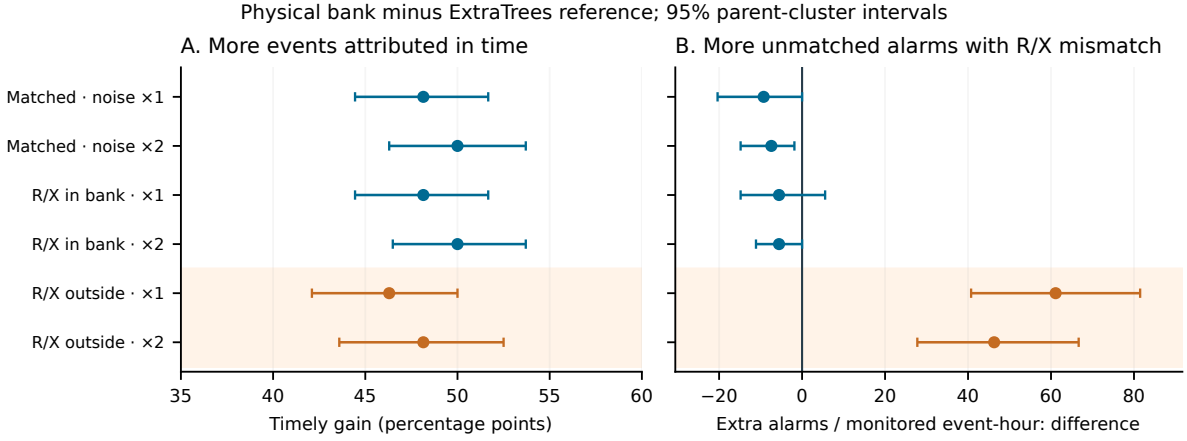


FIGURE 8. **Physical bank minus the ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per monitored event-record hour (the 12-s startup is excluded). Orange rows mark out-of-bank R/X conditions.**

balanced change above 10% lie between 0.37 and 1.5 times the coupling bound, above the quarter of the bound used by the test (Fig. 10B). The lowest ratio occurs at the first report, where the phasor estimation window has reached the faulted bus before the others. The fault is therefore recognized as physical, and parity is suspended while it lasts. The threshold $q > 23.9$ is the nominal 10^{-9} tail point for one idealized sequence statistic. The implemented report statistic is the maximum of voltage and current, so its ideal marginal tail is at most 2×10^{-9} before accounting for scale estimation. Across the eight PMUs this record contains about 1.29 million monitored PMU-reports, for an ideal-model expectation below 0.0026 exceedances; we observe 19 isolated single-report values with $23.9 < q < 47.9$ (about 7.4×10^3 times that reference expectation). This shows that the nominal Gaussian tail model does not calibrate this record. Doubling the assumed noise variance ($q > 47.9$) removes these exceedances and keeps every labeled episode. This operating point was chosen after inspection and is not an independently calibrated FAR. Two single-report flags remain at switching instants: the line opening, and the generation change at the bus of PMU 2. This gives 0.022 false alarms per minute.

The monitor also flags 6 incidents that the labels leave as normal operation. Four are single-report current spikes: 2,485 A and 53 A in phase B of PMU 2, against about 585 A, and 702 A and 710 A in phase A of PMU 39, against 245 A. One is a corrupted frame just before a dropout of PMU 29. The last is a 2.2-s interval after a reconnection in which all three voltage magnitudes of PMU 29 fall to zero, some of them negative, while every other PMU changes by less than 1%. A bolted fault at bus 29 would change another PMU by at least $c_{29} = 0.16$ times the change at PMU 29. The excursions are inconsistent with the tested single-fault model. We report them as unlabelled integrity candidates, separately from false alarms.

Physical bank under integrity events. The physical bank cannot run on the competition record. Its startup test rejects the first 10 s of the record: the ROCOF channels of two PMUs are constant or quantized, and the angle reference drifts by 0.31 deg/s with respect to the calibrated model.

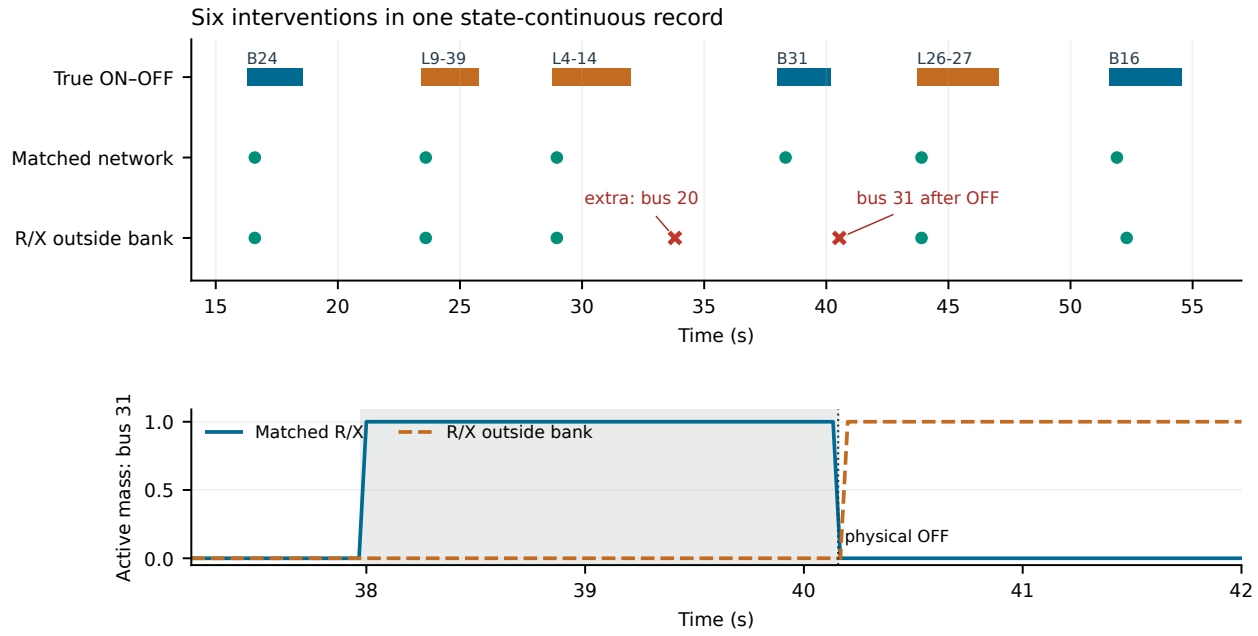


FIGURE 9. **One continuous development trajectory, shown for interpretation. Top: physical ON–OFF intervals and emitted diagnoses under matched and out-of-bank R/X. Bottom: bus-31 active mass with the true intervention shaded. A diagnosis after OFF misses the timely endpoint even when it identifies the source. True timing enters only this retrospective display.**

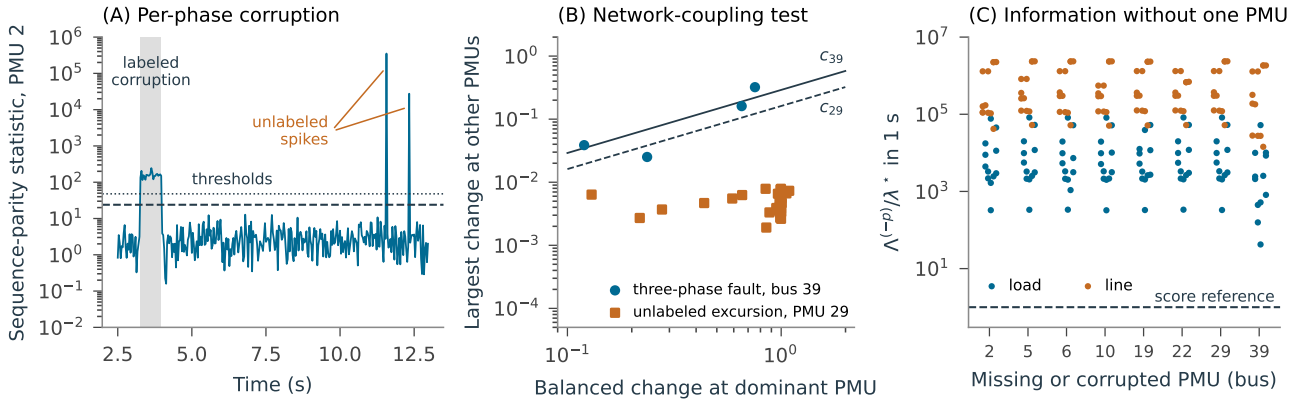


FIGURE 10. **Measurement integrity from network physics. (A) Sequence-parity statistic of PMU 2 on the competition record: a labeled single-phase corruption and two unlabeled single-report current spikes; dashed and dotted lines are the prespecified and variance-inflated thresholds. (B) Balanced voltage changes larger than 10%: reports of the three-phase fault at bus 39 lie near the network-coupling bound c_{39} of (35) and above the quarter of it used by the test, whereas the unlabeled collapse at PMU 29 lies two orders of magnitude below c_{29}. (C) Information for each held-out simulated intervention when one PMU is missing or corrupted (Proposition 1), relative to the idealized score reference $\lambda^\star$, under independent component noise with standard deviation 0.0005 (relative magnitudes and radian angles).**

We therefore inserted the competition's integrity events into the four held-out simulated event records and two normal records of the noise-stress study, and ran the bank with its code unchanged. *Corruption*: we added 40 0.75-s steps of $+2.4\%$ or -2.6% to the positive-sequence voltage magnitude of one PMU, the positive-sequence equivalent of the two labeled corruptions. They rotate over all PMUs, and four fall inside interventions. None of them produced a physical alarm, and the timely attributions were unchanged (12/12 loads, 12/12 lines). *Missing data*: one PMU was removed for 27 s, starting 2 s before an intervention, and replaced by its last valid report, the imputation suggested by the competition guide. Of the 13 interventions that began while a PMU was missing, the bank attributed 1 in time, and the dropouts caused 10 unmatched alarms on the event records. Attribution did not recover after the PMU returned: 0 of the 7 later interventions in these records were attributed in time, against 4 of 4 before the dropout. On the completed normal record the two held intervals produced 2 alarms (the second normal record of this scenario was not run). A held value is a measurement that contradicts the network response, not an absent one; the bank therefore needs the deletion of Proposition 1, an availability-switched observation operator, rather than imputation.

Together these results separate the integrity mechanisms

from the physical ones. Missing data are identified exactly and leave ample nominal signature separation, but the implemented bank cannot use it when fed held values; it needs the availability-switched operator. Corruption is flagged by departures from the assumed measurement and network model. It did not mislead the bank in simulation, and on the competition record it revealed incidents that the labels miss.

D. ALL EVENT TYPES: NETWORK PHYSICS WITHOUT TRAINING

Because the bank cannot start on the competition record (Section O-C), we built a second attributor from the provided network data alone, with no trained parameter, for every event type of the guide. The network is the admittance matrix of the provided power-flow case, with loads as constant impedances and generators as transient reactances behind constant voltages [3]. The integrity layer runs first; unhealthy PMUs are deleted from the fits (Proposition 1), not imputed. The physical layer uses the two time scales of the network equations. *Algebraic jump:* the one-report jump of the PMU voltages and branch currents is fitted by an injection at each of the 39 buses or the removal of each of the 46 branches, removed with its full pre-event current at both terminals; a sag below 0.8 pu is a fault, located by the profile $|Z_{jf}/Z_{ff}|$. *Quasi-steady state:* an episode that does not return within 15 s is fitted 15–25 s after onset by the linearized AC power flow, with generators sharing the imbalance in proportion to dispatch. This separates load changes, generation changes and generator trips, under plausibility limits (a load cannot change by more than its base value, nor a generator by more than its dispatch). A reversed jump, or a return to the pre-episode state, closes an episode. Attributor I detects episodes from the change of the angle differences. Attributor II detects one-report jumps after subtracting the local trend of the neighboring report changes, which separates switching from the smooth electromechanical response. Decisions are taken 0.5 s (switching) or 25 s (persistent) after onset and written back to it, as in the guide's offline scoring; each record takes seconds.

Protocol. No label was used to fit a parameter, but the competition record was not blind. Attributor I took 6 logged iterations on it. Attributor II was designed on the training and calibration records of a simulated cohort and checked on the competition record in two further iterations. Every iteration is stored with its code hash and score; the competition record is thus a validation set. The confirmatory test is the stored test split of that cohort, generated before either attributor existed. Each attributor was fixed and run once, and its predictions were hashed before the schedule was read.

Results. On the competition record (Table 20) both attributors detect, classify and locate all 12 guide-defined event labels of that one record: three corruptions, four dropouts, one three-phase fault, one line outage, one generation change, one load change, and one simultaneous missing-PMU/generation event (class 6). The composite label is counted separately from the coincident generation label, as in the competition scorer. The submitted system located 7 of them. For the generator case, the guide's event label is associated with bus 2. The attributor fits a bus-level effective injection at bus 2 (scale 1.022, with no separate amplitude fit); this is a guide-level Top-1 localization and does not identify a specific generator unit. The load at bus 7 is separated from generation by its reactive-to-active ratio. Six of the eight false-alarm episodes of Attributor II are the impossible measurements of Section O-C, a seventh follows the collapse at PMU 29, and the eighth is an unlabeled change at 3580 s seen at every PMU. Labeled boundaries that extend beyond the observable change keep macro-F1 at 0.88.

TABLE 20. All event types on the competition record (12 guide-defined labels: one fault, one line outage, one generation change, one load change, four dropouts, three corruptions and one simultaneous missing-PMU/generation event). The composite label is scored separately from the coincident generation label. Report-level scores follow the competition guide; false alarms are episodes outside every label.

Method	Class	Top-1	F1	FA/min	Macro-F1
Submitted system	11/12	7/12	0.192	0.18	0.21
Attributor I	12/12	12/12	0.981	0.10	0.90
Attributor II	12/12	12/12	0.979	0.10	0.88

TABLE 21. Stored simulated cohort, evaluated once: 72 load and line pulses (2–5 s, six per record) in 12 records, and 12 normal records (2 h). Top-1 counts events with the right family and exact source. The dynamic bank uses the plant model and an 18-source catalog; the open search ranks all 39 buses and 46 branches.

Method	Detected	Top-1	Lines Top-1	Loads detected	Loads Top-1	Normal alarms
Attributor I, open	13/72	0/72	0/36	4/36	0/36	0
Attributor II, open	48/72	28/72	24/36	12/36	4/36	0
Attributor II, catalog	48/72	40/72	32/36	12/36	8/36	0
Dynamic bank, catalog	72/72	72/72	36/36	36/36	36/36	0

The stored cohort has 72 load and line pulses of 2–5 s, six per record with gaps of 3–8 s (Table 21). Attributor I fails there: its quasi-steady window opens after the pulses end. Attributor II detects all 36 line pulses and locates 24 in an open search over 85 candidates, or 32 within the bank's 18-source catalog. Every remaining line error involves one of three branches whose outage, seen from the eight PMUs, is nearly collinear with an injection at an adjacent bus. It detects only 12 of the 36 load pulses, because a change of 3–14% of one load moves the one-report phasors by less than the startup noise. Longer windows would recover them, but there the response is no longer algebraic. The dynamic bank models that response and attributes all 72 pulses, at 9.2 h of computation against 4 s. No method alarmed in the normal records.

Network physics alone thus attributes every event type that is large enough for its algebraic jump or long enough to settle; short, small pulses need the dynamic model.

APPENDIX P OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of the stream method.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h},$$
$$E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{36}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the fixed common angle drift, the three predicted histories are

$$\mu_i^{\mathrm{N}}(a) = y_0 + aD_i^0 + a^2 Q_i^0,$$
$$\mu_i^{\mathrm{R}}(a) = \mu_i^{\mathrm{N}}(a) + \widehat{b},$$
$$\mu_i^{\mathrm{E}}(a) = \mu_i^{\mathrm{R}}(a) + a\widehat{\eta}E_i. \tag{37}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The research plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three fixed instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its stored instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 12 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the fixed impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the fixed artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

D. HISTORY CONDITIONING IN AN INDEPENDENT REDUCED PLANT

A proposed extension conditions the evidence on the state left by a previous intervention. It relinearizes the signature of the new source at that state, adds the bilinear interaction between the two interventions, or integrates the uncertain

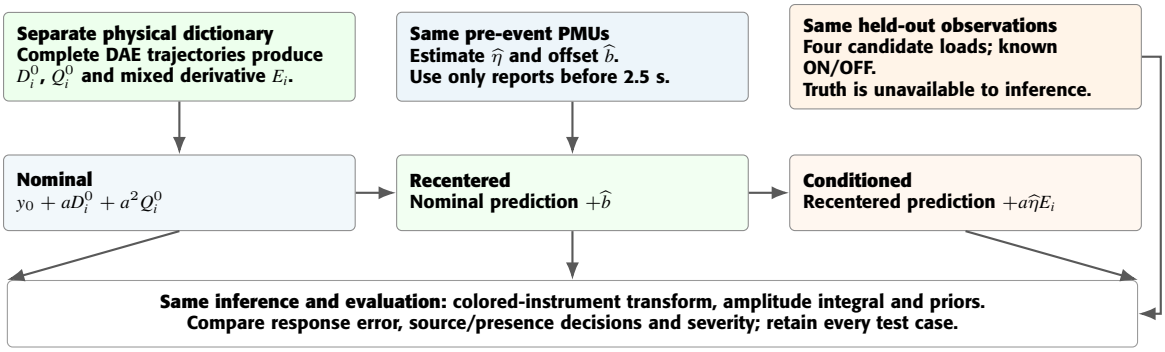


FIGURE 11. Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.

TABLE 22. Controlled signature test, separate from causal detection. Response error is the median whitened incremental error at the generating source/severity over 80 physical cases. Joint correctness includes presence and source on 240 noisy event records; source-only ranking conditions on a load. Severity error is median absolute relative error at the generating source. Normal selections use 15 fixed windows, not normal-hours exposure.

Model	Response error (%)	Joint correct	Source-only correct	Normal event selections	Severity error (%)
Nominal	1.25	114/240	114/240	12/15	3.45
Recentered	1.25	240/240	240/240	0/15	1.2
Conditioned	0.0561	240/240	240/240	0/15	0.329

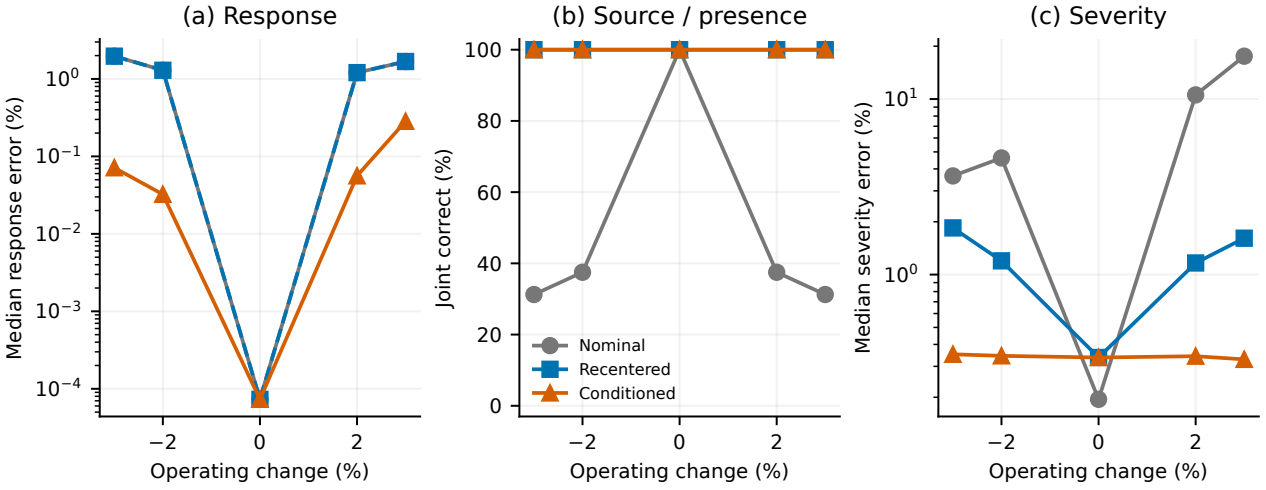


FIGURE 12. Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.

previous severity out of the likelihood. We tested it where it should matter most, with a plant different from the one used for the bank. It has classical machines, a 5% droop governor, constant-impedance loads and the network of the competition case. A load p of 8–16% of its base value is switched on at 2 s and off at 5 s; a weak second load of 2–6% appears at 6.5 s while the network is still recovering. The estimator knows p and the switching times, fits the previous severity by Gauss–Newton and ranks the new source among the eight loads of the catalog. Observation noise is the empirical covariance of the first differences of the competition record, multiplied by 1, 10 or 100. Some cells shorten the history available to the estimator or detune its inertia and damping by 20% and 40%. The protocol, the comparison criterion and the seeds were fixed before the test scenarios were generated, and each cell was run once with 200 scenarios.

Table 23 gives the outcome. In none of the 8 protocol cells does a predictor conditioned on the history make a significantly better source decision than the recentered nominal signature; the accuracies differ by at most 0.5 points. The mechanism is visible in the estimates. The previous severity is recovered to within 0.3 points of the base load in every pro-

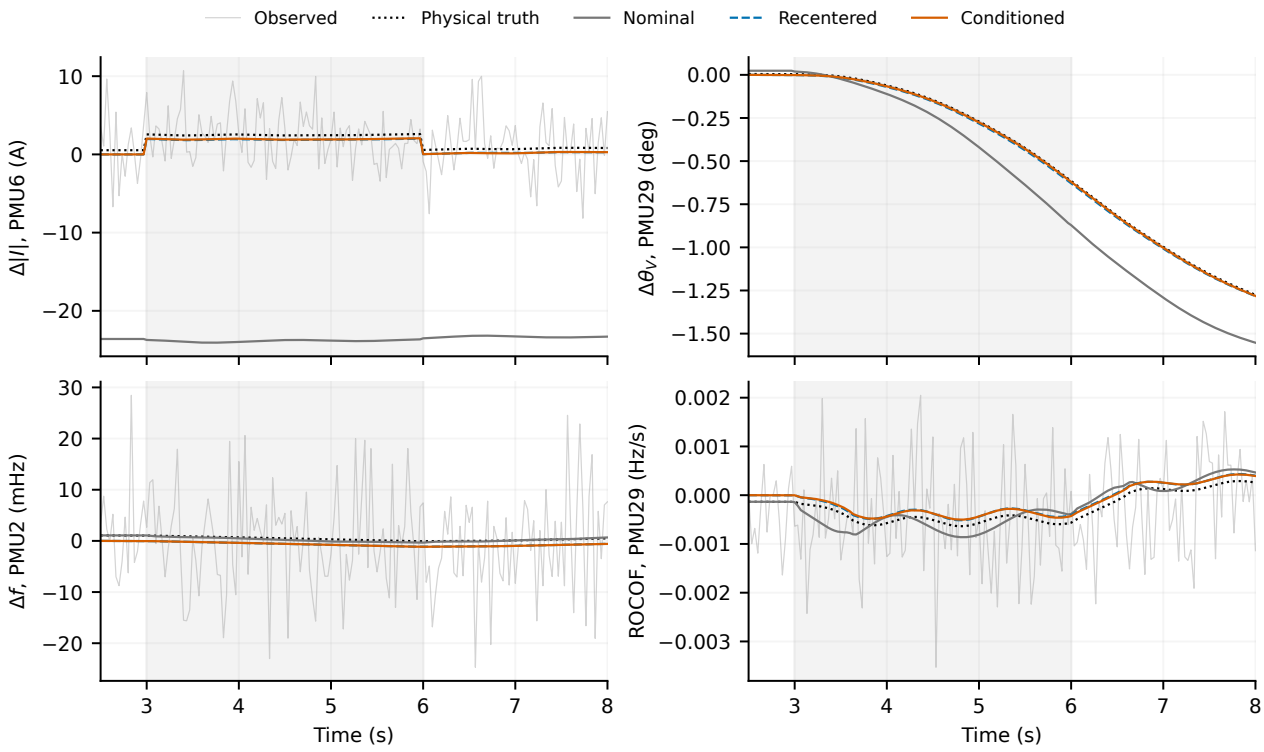


FIGURE 13. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

tocol cell, so the history leaves no uncertainty to marginalize. We then looked for the regime where conditioning should finally pay off. Exploratory runs on separate seeds pushed the previous severity to 30–60%, started the new source 0.2 s after the previous one ended, detuned inertia and damping by 40% and 80%, and cut the history to 0.5–1 s with noise 100 to 1000 times the measured level. Even there the six predictors tied. We fixed two of those regimes (cells I and J) and ran them once on fresh seeds. The previous severity is then recovered only to 0.8–5.4 points, yet the decisions still do not change: the accuracies are equal in cell J and within one point in cell I, where integrating out the previous severity is even worse than the plug-in (by 1.0 point of accuracy and 0.9 nats of log-loss). The reason is structural. The state left by the previous intervention shifts the predicted record of every candidate source by almost the same amount, so it moves all hypotheses together and rarely reorders them; the ranking is set by the shape of the new source's own signature. The nominal signature costs 0.5 ms per decision, against 819 ms for the nonlinear plant and 6.4 s for the quadrature reference. A computable bound on the whitened prediction error certified the decisions of the nominal, bilinear and relinearized predictors in 78–99% of the scenarios. Among the certified decisions, 0 differed from the nonlinear plant. The bound needs the nonlinear prediction it bounds, so it validates the approximation and does not save computation.

On the competition record the same plant ranks the bus-7 load first among 16 load buses (rank 1) and explains 99.5% of the energy of the quasi-steady change after the label-free onset of the attributor. That record cannot separate the predictors, because its only load change arrives about ten minutes after the previous event has recovered. The extension would need a poorly observed previous intervention. A different plant could provide one, but none of the 8 protocol cells or the two extreme cells reached the point where conditioning changes a decision.

E. LEARNED BASELINES TRAINED ON THE BANK'S OWN SIMULATIONS

The ExtraTrees adapters of the main comparison keep the operating points of the earlier system. A fair test of the learned alternative retrains it where the bank was developed. We fitted three learners on the 24 training records of the simulated cohort (72 events) and fixed each alarm threshold on the 24 calibration records, requiring no alarm in normal records. Each was then scored once on the 24 held-out records with the same association rule. Inputs were causal. They were the changes of the 16 phasor channels over the last 0.5 and 1.5 s relative to a reference taken 1.5 to 4 s earlier, whitened with the startup noise of each record. One learner received those changes directly. Two received their projection on the 18 physical signatures of the catalog, which gives them the static physical knowledge of the bank. A multinomial logistic model on the projections serves as a linear control. The held-

TABLE 23. Source accuracy of five predictors in ten stress cells (200 scenarios each; mismatch is the relative error of the inertia, and the damping is detuned twice as much; cells I and J use a previous severity of 30–60%, a new source 0.2 s after the previous one ends, and fresh seeds). Nominal: equilibrium signature on the recentered baseline; plug-in: nonlinear plant at the estimated previous severity; marginalized: plug-in with the previous severity integrated out; quadrature: the same by seven-node quadrature. The last column is the paired accuracy difference marginalized minus plug-in with its 95% bootstrap interval, in percentage points.

Cell	Noise factor	History (s)	Mismatch (%)	Window (s)	Nominal (%)	Plug-in (%)	Marg. (%)	Quad. (%)	Marg.−plug-in (pp)
A	1	6.5	0	1.5	99.5	99.5	99.5	99.5	+0.0 [+0.0, +0.0]
B	100	6.5	0	1.5	98.0	97.5	98.0	98.0	+0.5 [+0.0, +1.5]
C	1	1	20	1.5	95.0	95.0	95.5	95.0	+0.5 [+0.0, +1.5]
D	100	1	20	1.5	93.5	93.5	93.5	93.5	+0.0 [+0.0, +0.0]
E	100	1	20	0.5	80.5	80.5	80.5	80.5	+0.0 [-1.5, +1.5]
F	10	6.5	0	1.5	99.5	99.5	99.5	99.5	+0.0 [+0.0, +0.0]
G	10	1	20	1.5	95.0	95.0	95.5	95.0	+0.5 [+0.0, +1.5]
H	100	0.5	0	1.5	98.0	98.0	98.0	98.0	+0.0 [+0.0, +0.0]
I	100	1	40	1.5	61.5	61.5	60.5	61.5	-1.0 [-2.5, +0.0]
J	1000	0.5	40	0.5	38.0	38.0	38.0	38.0	+0.0 [+0.0, +0.0]

TABLE 24. Learned baselines trained on the simulated cohort of the bank, scored once on its held-out 72 events (36 lines, 36 loads) with the same event-association rule. Thresholds were set on calibration records for zero alarms in normal records. Top-1 counts the events detected with the correct source. The 72 events come from 12 independent records of six events each, so the last column counts the records in which all six were located (family and source), with its exact 95% interval; the events of a record share one noise realization and operating point. Extras are alarms matching no event.

Method	Detected	Top-1	Lines	Loads	Extras	Records (95%)
Dynamic bank	72/72	72	36/36	36/36	7	12/12 (74–100)
Static physical	48/72	40	32/36	8/36	25	0/12 (0–26)
Trees, phasor changes	61/72	41	25/36	16/36	82	1/12 (0–38)
Trees, signatures	58/72	49	32/36	17/36	30	0/12 (0–26)
Logistic, signatures	55/72	32	27/36	5/36	71	0/12 (0–26)

out records had already been opened by the physical methods, so they are not a pristine test for those; they were not used to fit or to tune any learner.

Table 24 gives the outcome. The best learner locates 49 of 72 events at the source, against 72 for the dynamic bank, and the worst locates 32. Giving the learner the physical signatures helps most: it recovers 32 of 36 line outages but only 17 of 36 loads, where the responses are small and electromechanical and the static signatures describe them poorly. The learners also raise 30 to 82 alarms outside the scheduled events, against 7 for the bank. No learner raised an alarm in the normal records, by construction of the threshold. The result does not show that no learner can match the bank. It shows that with the data the bank itself was developed on, three standard learners, two of them given the physics as features, fall well short on loads.

F. TRANSFER TO A DIFFERENT PLANT

The dynamic bank has only been scored on one plant family, so a method could be good at that plant without being good at power systems. We re-simulated the 12 event schedules of the held-out cohort (same sources, instants and amplitudes) with an independent plant. It has classical machines, droop governors, constant-impedance loads and the competition network, and has no excitation system. The new records pass through the same PMU operator and the same instrumental noise, so only the physics differs. The voltages at the monitored buses match the original plant to within 0.1% at rest, while the response to a switching event differs by up to a factor of two. Four normal records of 120 s serve as controls. The exporter advanced one report before assigning the first timestamp, so these comparisons are exploratory and do not support timing or transfer claims. Every method was fixed before these records were read; learner thresholds were set on the original calibration records.

Table 25 gives the outcome. The static physical attributor is not affected by the change of plant: it locates 42 events against 40 on the original cohort. The learners lose up to 6 sources on the new plant and raise 75 to 273 alarms outside the scheduled events, several times more than before. Training on additional noise draws helps the phasor learner on the original cohort but not on the new plant, so this noise augmentation did not resolve the observed mismatch. No method locates all six events of any record, and no method raised an alarm in the normal records. The dynamic bank needs per-record banks built with the plant solver and has not been run on these records. This is the main gap of the evidence: the bank's 72 of 72 is measured on its own plant family, and a run on these records would bound how much of it depends on that match.

G. CLAIMS, EVIDENCE AND THREATS TO VALIDITY

Table 26 lists each headline claim with its evidence, the independent unit behind it and what it leaves open. The units matter. Replays of one physical parent share their trajectory, so the statistical weight of a result is set by the number of parents or records, not of events.

Plant family. The bank was developed and scored on simulations of one plant family. On an independent plant the static attributor keeps its accuracy and the learners lose up to 6 sources, but the bank itself was not run there. Its advantage over the other methods could shrink or grow on another plant, and this is the largest open point of the evidence. *Few independent units.* The held-out cohort has 12 event records

TABLE 25. Transfer to records simulated with a different plant (classical machines, droop governors, no excitation), built from the same 12 event schedules and passed through the same PMU operator and noise. Detected events, events located at the source (Top-1) and alarms matching no event, out of 72 events, for the original cohort and the new plant. The last column gives records with all six events located, out of 12; all methods raised zero alarms on the normal records. The dynamic bank needs per-record banks built with the original plant solver and was not run on these records. The two "more noise" learners were trained on four extra noise draws per training record.

Method	Original cohort			Second plant			Records
	Det.	Top-1	Extras	Det.	Top-1	Extras	(of 12)
Static physical	48	40	25	52	42	48	0
Trees, phasor changes	61	41	82	60	39	170	0
Trees, signatures	58	49	30	68	45	156	0
Trees, phasor changes, more noise	63	48	94	66	42	273	0
Trees, signatures, more noise	55	49	21	59	46	75	0

and the confirmatory simulation 18 parents. A method that is right in all 12 records is bounded below by 74% at 95% confidence, not by the event-level count. *Reuse of the held-out records.* They were opened by the bank, by two static attributors and, once each, by the learners of this section. No learner or threshold was tuned on them, but they are no longer a pristine test for the physical methods. *Competition record.* It is a validation set with logged iterations, and it contains one load change and one line outage among its twelve events. The bank does not start on it, so the competition record does not test the bank. *Event duration.* The simulated interventions last 2–5 s, while the labeled load changes of the record last minutes. The bank is evaluated where recovery dominates, and persistent changes are covered only by the static attributor. *Noise.* Independent per-PMU noise of 0.071% added RMS TVE degrades load attribution. Estimating cross-PMU covariance from the startup prefix is a candidate extension that was not tested. *Baseline strength.* The learners saw 72 training events, and more noise draws did not close the gap. A learner given far more physical variety could do better, so the comparison measures what the available data support.

APPENDIX Q RECOVERY GEOMETRY AND SOURCE SEPARATION

A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new intervention begins. For the exact observation map Ψ and the same pre-intervention history h, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{38}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response s_h at specified severity and context, the distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{39}$$

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [6], [7].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The separately compiled Historical Technical Notes retain the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

B. DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete nonlinear trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied. Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate measurement operator is followed by its fixed innovations transform. For a geometric control, we profile nonnegative competing amplitudes.

TABLE 26. Claims, their evidence and their limits. The last column states what the evidence does not establish.

Claim	Evidence	Independent unit and size	Not established
The bank attributes load and line interventions before restoration	Independent simulation of the bank's plant family	18 parents, 108 interventions, 648 replays; 644/648 timely	Other plant families; the replays of a parent are not independent
Resetting the physical state at restoration costs attribution	Same simulation, paired arms	75 of 108 timely with reset against 108 of 108 with carry	Mechanism beyond the tested conditions
No false alarm in normal operation	Normal records of the same simulation	11.8 h, no alarm; a 95% bound of about 0.25 per hour	Field operation; unmodeled disturbances
Relations of sequence and network coupling diagnose missing and corrupted PMU episodes	Competition record	One record, all labeled episodes	A rate across records
A network-physics attributor locates every labeled event	Competition record, not blind 6 logged iterations on it	12 events in one record	A rate; a blind test
The bank attributes short pulses that the static attributor misses	Held-out simulated cohort	72 events in 12 records; 72 against 48 detected	A different plant (the bank was not run on one)
Learners trained on the bank's simulations fall short	Three learners, scored once	72 events in 12 records; at best 49 located	That no learner can match the bank
Conditioning on the previous intervention does not help	Reduced plant, 8+2 cells	200 scenarios per cell	The plant of the bank; a poorly observed history

Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios exceeds one (Fig. 14). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an improvement of the stream method nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

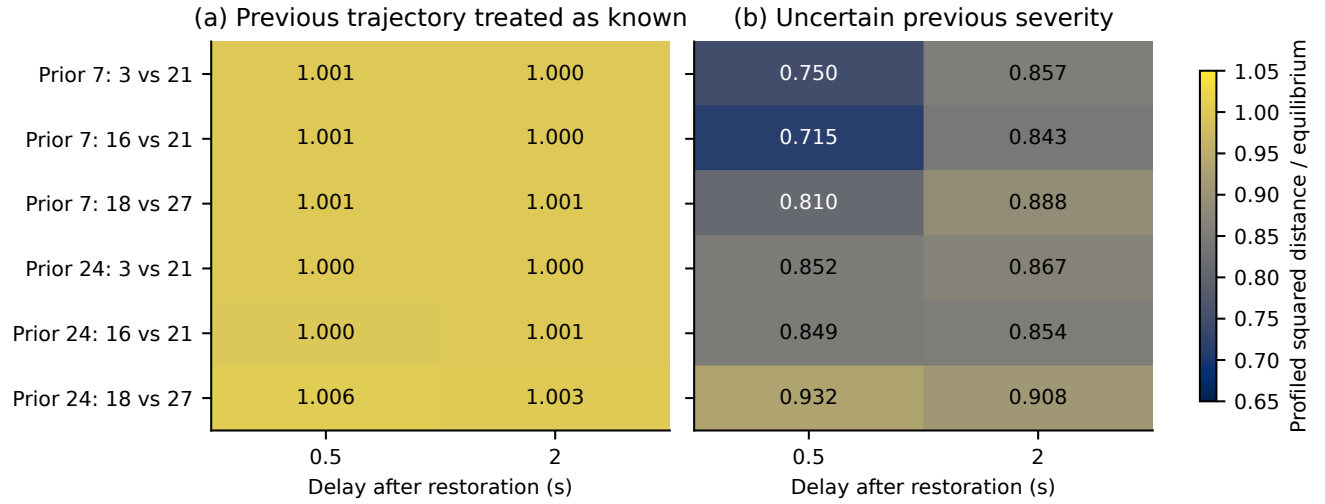


FIGURE 14. **Full nonlinear physical control, not a performance comparison of the stream method. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

REFERENCES

[1] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.
[2] S. Fiaz, D. Zonetti, R. Ortega, J. M. A. Scherpen, and A. J. van der Schaft, "A port-hamiltonian approach to power network modeling and analysis," *European Journal of Control*, vol. 19, no. 6, pp. 477–485, 2013.
[3] M. A. Pai, *Energy Function Analysis for Power System Stability*. Boston, MA, USA: Kluwer Academic Publishers, 1989.
[4] E. Y. Chow and A. S. Willsky, "Analytical redundancy and the design of robust failure detection systems," *IEEE Transactions on Automatic Control*, vol. 29, no. 7, pp. 603–614, 1984.
[5] A. Abur and A. Gómez Expósito, *Power System State Estimation: Theory and Implementation*. New York, NY, USA: Marcel Dekker, 2004.
[6] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.
[7] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.